G-Est Software
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Small-Area Estimation
Area-Level Model with Hierarchical Bayes Estimation

Methodology Specifications

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Table of Contents

Introduction

This document provides a description of the methodology specifications for small-area estimation based on area-level modeling and the Hierarchical Bayes (HB) method. These specifications have provided the framework for the development of software to produce estimates for small areas based on sampling and linking models. The linking model is a linear mixed model with fixed and random effects. The basic concepts behind linear mixed model estimation are described in McCulloch and Searle (2001). The application of linear mixed models to small-area estimation is given in Rao (2003) and Fuller (2009). The Hierarchical Bayes method is described in chapter 10 of Rao (2003).

The programs that have been developed are primarily intended to produce estimates for small-area means and totals but they can be used for other parameters of interest. The methodology that was implemented is described in this document. A description of the programs and their use are given in a separate document titled *G-Est Software - Small-Area Estimation - Area Level Model with Hierarchical Bayes Estimation - Description of Function Parameters and User Guide*.

Terms

This document uses terms in the following list:

small area (or area)	a small domain of interest for estimation
parameter	a characteristic of a variable interest (a mean, total or other) of the small area
direct estimator or design-based estimator	an estimator produced using design-based probability sampling methods
synthetic estimator	a predictive estimator produced from the model
HB estimator	an estimator obtained using Hierarchical Bayes estimation
HB_MM_KV estimation	Hierarchical Bayes estimation using a matched model with known sampling variances
HB_UM_LL_KV estimation	Hierarchical Bayes estimation using an unmatched log linear model with known sampling variances
HB_UM_LCU_KV estimation	Hierarchical Bayes estimation using an unmatched log census undercount model with known sampling variances
HB_MM_EV estimation	Hierarchical Bayes estimation using a matched model with estimated sampling variances
HB_UM_LL_EV estimation	Hierarchical Bayes estimation using an unmatched log linear model with estimated sampling variances
Markov chain Monte Carlo (MCMC) methods	resampling methods (such as the Metropolis-Hastings algorithm) for sampling from a probability distribution by constructing a Markov chain of values that represent pseudo-random selections from the desired distribution

The terms HB_MM_KV, HB_UM_LL_KV, HB_UM_LCU_KV, HB_MM_EV and HB_UM_LL_EV are short forms used to refer to the five types of models available under HB estimation. They are also the names of the corresponding methods of estimation.

Notation

The following table provides a summary of the notation used in this document. More details are given throughout the document text.

notation	definition
A	the set of mutually exclusive groups of areas denoted by $A = A_{EXC} \cup A_{VAL} \cup A_{NUL}$
A_{EXC}	the areas excluded from the small-area model
A_{VAL}	the areas with valid values for the direct estimates and variances
A_{NUL}	the areas with invalid or missing values for the direct estimates and/or variances
m	the number of areas in A_{VAL}
i	the identifier of each area
N_i	the number of units in the population of area i
n_i	the number of sample units in area i
C_i	the known number of units in area i for the log census undercount model of the HB_UM_LCU_KV method
θ_i	the parameter of interest for small area i - a mean ($\theta_i = \overline{Y}_i$), a total ($\theta_i = Y_i$), or other
$\hat{\theta}_i^{DIR}$	a direct (design-based) estimate of θ_i
$\hat{\theta}_i^{HB}$	the HB estimate of θ_i ($\hat{\theta}_i^{HB}$ is $\hat{\theta}_i^{HB\_MM\_KV}$, $\hat{\theta}_i^{HB\_UM\_LL\_KV}$, $\hat{\theta}_i^{HB\_UM\_LCU\_KV}$, $\hat{\theta}_i^{HB\_MM\_EV}$ or $\hat{\theta}_i^{HB\_UM\_LL\_EV}$ corresponding to the different models)
$\hat{\theta}_i^{HB,b}$	the benchmarked HB estimate produced from the initial model estimate $\hat{\theta}_i^{HB}$ (if benchmarking is applied)
$\hat{\theta}_i^{DIR} = \theta_i + e_i$	the sampling model assumed for areas $i \in A_{VAL} \cup A_{NUL}$
$g(\theta_i) = \mathbf{z}_i^T \boldsymbol{\beta} + b_i v_i$	the linking model assumed for areas $i \in A_{VAL} \cup A_{NUL}$ $g(\theta_i) = \begin{cases} \theta_i & \text{for the HB\_MM\_KV and HB\_MM\_EV models} \\ log(\theta_i) & \text{for the HB\_UM\_LL\_KV and HB\_UM\_LL\_EV models} \\ log(\theta_i/(\theta_i + C_i)) & \text{for the HB\_UM\_LCU\_KV model} \end{cases}$

notation	definition
$\mathbf{z}_i^T \boldsymbol{\beta}$	the fixed term or fixed effect for area i
$b_i v_i$	the random effect for area i
$\mathbf{z}_i$	the vector (of dimension p) of known auxiliary values for area i (can include totals, means or other values known for every area)
p	the number of auxiliary variables in $\mathbf{z}_i$
$\boldsymbol{\beta}$	the regression coefficient for the fixed term in the model
$\hat{\boldsymbol{\beta}}^{HB}$	the HB estimate of $\boldsymbol{\beta}$
b_i	the model error coefficient (a known positive constant) for area i
v_i	the model error for area i with $v_i \overset{ind}{\sim} (0, \sigma_v^2)$
e_i	the sampling error for area i where $e_i \overset{ind}{\sim} (0, \psi_i)$
ψ_i	the sampling variance of the direct (design-based) estimator of area i
$\hat{\psi}_i$	a direct (design-based) estimator of the sampling variance of area i
$\hat{\psi}_i^{HB}$	the HB estimate of the sampling variance in area i (used in benchmarking)
σ_v^2	the model variance
$\hat{\sigma}_v^2$	the HB estimate of the model variance
$\hat{\theta}^{DIR}$	the known or estimated value of the parameter over the population in all areas A (this value is used for benchmarking a mean or a total and should not be confused with $\hat{\theta}_i^{DIR}$, the direct estimate in area i)
$v\hat{a}r(\hat{\theta}_i^{HB})$	the HB estimate of the variance of $\hat{\theta}_i^{HB}$ in area i
$v\hat{a}r(\hat{\boldsymbol{\beta}}^{HB})$	the HB estimate of the variance of $\hat{\boldsymbol{\beta}}^{HB}$
$exp(x)$	the exponential function e^x based on the natural logarithm
B	the number of samples in the burn-in period of the MCMC method
G	the total number of generated samples of the MCMC method

notation	**definition**
E_δ	the set of samples used for the estimation of the model parameters $E_\delta = \{ B+\delta, B+2\delta, B+3\delta, \dots ,G \}$ where δ is an integer as follows ... $\delta = 1$ for the HB_MM_KV and HB_MM_EV models $\delta \in [4,5,6,7,8,9,10]$ for the HB_UM_LL_KV and HB_UM_LL_EV models $\delta \in [4,5,6,7,8,9,10]$ for the HB_UM_LCU_KV model
E	the number of samples in E_δ
$N(\boldsymbol{u},\boldsymbol{\Sigma})$	the multivariate (n dimensional) Normal distribution with mean $\boldsymbol{u}$, variance-covariance $\boldsymbol{\Sigma}$ and probability density function $\frac{1}{\sqrt{(2\pi)^n \lvert\boldsymbol{\Sigma}\rvert}} exp\left[-\frac{(\boldsymbol{x}-\boldsymbol{u})' \boldsymbol{\Sigma}^{-1} (\boldsymbol{x}-\boldsymbol{u})}{2} \right]$
$log\, N(u,\sigma^2)$	the Log-Normal distribution with probability density function $\frac{1}{x} \frac{1}{\sqrt{2\pi\sigma^2}} exp\left[-\frac{(log(x)-u)^2}{2\sigma^2} \right]$ for $x > 0$ x is Log-Normal if $log(x)$ has a univariate Normal distribution $N(u,\sigma^2)$
$IG(\alpha,\beta)$	the Inverse Gamma distribution with shape parameter α , scale parameter β and probability density function $\frac{\beta^\alpha}{\Gamma(\alpha)} \frac{e^{-\beta/x}}{x^{\alpha+1}}$ for $x > 0$
χ^2_n	the Chi-Square distribution with n degrees of freedom and probability density function $\frac{1}{2^{n/2}\,\Gamma(n/2)} x^{(\frac{n}{2}-1)} exp(-\frac{x}{2})$ for $x > 0$

Methodology Notes:

- The set of auxiliary variables in $\boldsymbol{z}_i$ should normally include an intercept term with value 1 for all areas.
- The log function used in $log(\theta_i)$ (the log-linear model) and $log(\theta_i/(\theta_i + C_i))$ (log census undercount model) is the natural logarithm to base e .
- The term $\hat{\theta}^{DIR}$ (the value for benchmarking) is not $\hat{\theta}_i^{DIR}$ (the direct estimate in area i).
- The exponential function e^x should not be confused with e_i (the sampling error in area i). The exponential function is also written as $exp(x)$.

Overview

The programs for the area-level model allow you to produce small-area estimates for parameters such as means, totals or other. Five HB methods have been implemented corresponding to the application of different models. They are as follows.

- HB estimation using a matched Fay-Herriot (FH) model with known sampling variances (HB_MM_KV)
- HB estimation using an unmatched log-linear model with known sampling variances (HB_UM_LL_KV)
- HB estimation using an unmatched log census undercount model with known sampling variances (HB_UM_LCU_KV)
- HB estimation using a matched Fay-Herriot (FH) model with estimated sampling variances (HB_MM_EV)
- HB estimation using an unmatched log-linear model with estimated sampling variances (HB_UM_LL_EV)

Each of these models is formulated in terms of a sampling model and a linking model. They are described in this document. The first three models assume the sampling variances are known while the last two models are based on the more typical situation where we have estimated sampling variances. Currently, there is no unmatched log census undercount model for the case where the sampling variances are estimated rather than known.

The basic framework for small-area estimation under each of these methods is as follows. The sampling and linking models contain various parameters, including the main parameters of interest – the small area totals, means or other. The sampling and linking models include specific distributions for some of these of parameters. We assume a prior distribution for each of the other unknown parameters – those with no explicit distribution. We multiply the prior distributions by the density functions of the distributions associated with the sampling and linking models. This gives us a joint likelihood function in terms of the model parameters. We then use this function to derive a full conditional (posterior) distribution for each of the unknown parameters. For some of these, the resulting distribution has a tractable or well-known form. For others, the resulting distribution is a product of density functions with no known form.

Markov chain Monte Carlo (MCMC) methods are used to obtain estimates from the full conditional distribution of each parameter. This is done by applying an iterative technique known as Gibbs sampling to repeatedly sample from the full conditional distributions. We draw many samples, and for each sample, we produce and store the estimates for the unknown parameters. At the end, we calculate the average for each parameter over all samples. These averages provide us with stable values for the estimates of the parameters. We then produce the small-area estimates using these values.

The accuracy and efficiency of the small-area estimates are strongly dependent on the specified model. It is always important to check the validity of the model before accepting the results. Diagnostics are provided to do this. When the model is inappropriate, the methods may produce invalid estimates or estimates which may be more inefficient than the direct estimates.

The model for small-area estimation

The theory developed for HB estimation is an extension of the theory based on the Fay-Herriot (FH) model for small-area estimation. The FH model is represented by the following two equations:

- Sampling model: $\hat{\theta}_i^{DIR} = \theta_i + e_i$
- Linking model: $\theta_i = \boldsymbol{z}_i^T \boldsymbol{\beta} + b_i v_i$

The sampling model expresses the direct estimate $\hat{\theta}_i^{DIR}$ as the unknown parameter value θ_i plus a random error e_i due to sampling. The linking model assumes the parameter value θ_i can be expressed as a fixed effect $\boldsymbol{z}_i^T \boldsymbol{\beta}$ plus a random effect $b_i v_i$. The fixed effect is a linear combination of the auxiliary variables $\boldsymbol{z}_i$. The sampling error and the model error are assumed to have the following properties.

$e_i \overset{ind}{\sim} (0, \psi_i)$	e_i are independent and identically distributed as Normal random variables with mean 0 and unknown variance ψ_i
$v_i \overset{ind}{\sim} (0, \sigma_v^2)$	v_i are independent and identically distributed as Normal random variables with mean 0 and unknown variance σ_v^2

The two equations in the Fay-Herriot model can be combined into a single linear mixed model reflecting the variability due to sampling (the e_i term) and the variability due to the area random effect (the $b_i v_i$ term).

$$\hat{\theta}_i^{DIR} = \boldsymbol{z}_i^T \boldsymbol{\beta} + b_i v_i + e_i$$

This is known as a matched model because we can substitute the value of θ_i from the linking model directly into the sampling model. However, it is not always appropriate to express θ_i as $\boldsymbol{z}_i^T \boldsymbol{\beta} + b_i v_i$. In some instances, it may be better to use some other specification such as $log(\theta_i) = \boldsymbol{z}_i^T \boldsymbol{\beta} + b_i v_i$. In this log-linear model, we do not attempt to solve for θ_i and substitute it into the sampling model. Similar comments apply to the log census undercount model $log(\theta_i / (\theta_i + C_i)) = \boldsymbol{z}_i^T \boldsymbol{\beta} + b_i v_i$. As a result, these two models are known as unmatched linking models. The traditional method of EBLUP estimation cannot be applied to these models. This has led to the development of a different technique for the estimation of the unknown parameters in both the matched and unmatched models. This technique is known as Hierarchical Bayes estimation and the methods available for it are described in this document.

For each parameter, the small areas are partitioned into 3 mutually exclusive groups A_{EXC}, A_{VAL} and A_{NUL}, so $A = A_{EXC} \cup A_{VAL} \cup A_{NUL}$. The group denoted by A_{EXC} are the exclusion areas. Currently, we define these are areas where the direct parameter estimate is not missing and the sampling variance or estimated sampling variance is 0. These areas are not part of the model. No changes are made to the estimates in the areas of this group. We assume the small-area model is applicable to the non-exclusion areas $i \in A_{VAL} \cup A_{NUL}$. The areas in A_{VAL} are the valid areas for which we have proper parameter estimates and variances. These variances are either known or estimated. We apply the model to the areas in A_{VAL} to produce estimates of the various unknown parameters in the model, such as the regression coefficient $\boldsymbol{\beta}$ and the model variance σ_v^2. The group denoted by A_{NUL} consists of areas with invalid or improper values for the parameter or its variance. As a result, we cannot use the input values for these areas in the estimation of the unknowns in the model. However, we can use the model to produce synthetic estimates for these areas. These estimates are the predicted values from the model.

The resulting small-area estimates for each area are as follows:

- For the areas $i \in A_{EXC}$, the small-area estimate is the direct estimate – no change is made.
- For the areas $i \in A_{NUL}$, the small-area estimate is a synthetic estimate.
- For the areas $i \in A_{VAL}$, the small-area estimate is an HB estimate.

Classification of the small areas into area groups

Table 1 below provides a summary of how the area groups are classified for each of the HB methods.

Table 1 Classification of the areas into area groups for each method

method	area group	classification to area group
HB_MM_KV	A_{EXC}	$\hat{\theta}_i^{DIR} \neq . \ \& \ \psi_i = 0$
	A_{VAL}	$\hat{\theta}_i^{DIR} \neq . \ \& \ \psi_i > 0$
	A_{NUL}	otherwise
HB_MM_EV	A_{EXC}	$\hat{\theta}_i^{DIR} \neq . \ \& \ \hat{\psi}_i = 0$
	A_{VAL}	$\hat{\theta}_i^{DIR} \neq . \ \& \ \hat{\psi}_i > 0 \ \& \ n_i \geq 2$
	A_{NUL}	otherwise
HB_UM_LL_KV	A_{EXC}	$\hat{\theta}_i^{DIR} > 0 \ \& \ \psi_i = 0$
	A_{VAL}	$\hat{\theta}_i^{DIR} > 0 \ \& \ \psi_i > 0$
	A_{NUL}	otherwise
HB_UM_LL_EV	A_{EXC}	$\hat{\theta}_i^{DIR} > 0 \ \& \ \hat{\psi}_i = 0$
	A_{VAL}	$\hat{\theta}_i^{DIR} \neq . \ \& \ \hat{\psi}_i > 0 \ \& \ n_i \geq 2$
	A_{NUL}	otherwise
HB_UM_LCU_KV	A_{EXC}	$\hat{\theta}_i^{DIR} > 0 \ \& \ \psi_i = 0$
	A_{VAL}	$\hat{\theta}_i^{DIR} > 0 \ \& \ \psi_i > 0$
	A_{NUL}	otherwise

We use the input estimates and variances to do the classification into the three area groups.

- The population sizes N_i are not required to determine the area groups.
- The sample sizes n_i are required for methods HB_MM_EV and HB_UM_LL_EV. We must have $n_i \geq 2$ for classification to area group A_{VAL} so that $d_i = (n_i - 1) \geq 1$ in the Inverse Gamma function of the full conditional distributions.

The small-area estimation methods

This section describes the process in each of the HB estimation methods. Each method uses a different sampling and linking model with specific distributional assumptions for some of the unknown parameters. For the remaining unknown parameters in the model, we specify Bayesian prior distributions. Using these various distributions, we derive a joint likelihood function of all the parameters in the model. From the joint likelihood function, we obtain a full conditional (posterior) distribution for each of the unknown parameters. For some of these, we know the exact form of the distribution. For others, the distribution does not have a recognizable form. All HB methods involve estimation of the model parameters through repeated sampling of their respective full conditional distributions. This is done using MCMC methods with Gibbs sampling.

When the full conditional distribution is known, we can sample directly from it. When the full conditional distribution is a product of density functions with no known form, a technique known as the Metropolis-Hastings algorithm is used to generate values that reflect the full conditional distribution. This involves the use of a proposal distribution and the application of a proposal step and an acceptance step. In the proposal step, we select a random value from the proposal distribution. In the acceptance step, we either keep the value or retain the previous value in the Gibbs sampler. The probability of keeping or rejecting the value is a function of the posterior distribution and the proposal distribution. More details are given in Gilks, W.R. (1998).

The HB estimation methods for the matched FH model and the log-linear model occur in pairs, reflecting the nature of the input sampling variances. Each pair consists of a method to handle the case where the sampling variances are assumed known and a method for the more common situation where we have estimated sampling variances. The processing steps for each pair of methods are very similar. However, each method has been documented separately. This is done for completeness of presentation. The log census undercount model assumes the sampling variances are known. There is no corresponding method when the sampling variances are estimated.

The methods that have been implemented are described in step form. We use process notes and implementation notes to provide further details on the process and possible issues in the implementation. Methodology notes provide references to the methodology.

HB estimation with matched FH model and known sampling variances

Inference model for the HB_MM_KV method

This method is based on the linking model $g(\theta_i) = \theta_i$ which leads to the Fay-Herriot matched model. This model is specified as

$$\begin{aligned} &\text{Sampling Model:} \quad \hat{\theta}_i^{DIR} = \theta_i + e_i \\ &\text{Linking Model:} \quad \theta_i = \mathbf{z}_i^T \boldsymbol{\beta} + b_i v_i \end{aligned}$$

where

$e_i \overset{id}{\sim} N(0, \psi_i)$	e_i are independently distributed as Normal random variables with mean 0 and known variance ψ_i
$v_i \overset{iid}{\sim} N(0, \sigma_v^2)$	v_i are independent and identically distributed as Normal random variables with mean 0 and unknown variance σ_v^2

The sampling and linking models are written in the Bayesian framework as follows.

$$\begin{aligned} \left[\hat{\theta}_i^{DIR} \,\middle|\, \theta_i \right] &\sim N\left(\theta_i, \psi_i \right) \\ \left[\theta_i \,\middle|\, \boldsymbol{\beta}, \sigma_v^2 \right] &\sim N\left(\mathbf{z}_i^T \boldsymbol{\beta}, b_i^2 \sigma_v^2 \right) \end{aligned}$$

Quantities such as $\mathbf{z}_i$ and b_i are treated as known under the model. For simplicity, they are not explicitly shown as part of the conditional components of each distribution. The parameters denoted by $\hat{\boldsymbol{\theta}}^{DIR} = \left\{ \hat{\theta}_i^{DIR} \right\}$ are estimated from the survey. Since the parameters in $\boldsymbol{\psi} = \{ \psi_i \}$ are assumed known, the only unknown parameters in this model are $\boldsymbol{\theta} = \{\theta_i\}$ over the areas as well as $\boldsymbol{\beta}$ and σ_v^2. The Bayesian model includes a distribution for each θ_i. We need to specify prior distributions $\pi(\cdot)$ for the other parameters. We assume the following prior distributions for $\boldsymbol{\beta}$ and σ_v^2.

$$\begin{aligned} &\pi(\boldsymbol{\beta}) \propto 1 \\ &\pi(\sigma_v^2) \propto (\sigma_v^2)^{-1/2} \end{aligned}$$

The distribution $\pi(\boldsymbol{\beta}) \propto 1$ is known as a flat prior. These two priors lead to a full conditional distribution for σ_v^2 based on the Inverse Gamma distribution. For a description of this distribution, see the Methodology Note after step 2 of this method.

Methodology Notes:

- The Bayesian model is described on page 237 of Rao (2003).
- A discussion on the use of different priors is given by Gelman (2006).

Full conditional distributions for the inference model of the HB_MM_KV method

The full conditional distributions for the unknown parameters are obtained by multiplying the density functions for the distributions of the Bayesian model and the Bayesian priors. The derivations are given in the section titled Derivation of full conditional distributions for the HB models. The resulting full conditional distributions are shown below.

$$\left[\,\theta_i \mid \hat{\boldsymbol{\theta}}^{DIR}, \boldsymbol{\psi}, \boldsymbol{\beta}, \sigma_v^2\,\right] \sim N\left(\gamma_i\,\hat{\theta}_i^{DIR} + (1-\gamma_i)\,\mathbf{z}_i^T\boldsymbol{\beta},\ \gamma_i\,\psi_i\right) \ \text{ for } i \in A_{VAL}$$

$$\text{where } \gamma_i = \frac{b_i^2\,\sigma_v^2}{b_i^2\,\sigma_v^2 + \psi_i}$$

$$\left[\,\boldsymbol{\beta} \mid \hat{\boldsymbol{\theta}}^{DIR}, \boldsymbol{\psi}, \boldsymbol{\theta}, \sigma_v^2\,\right] \sim N\left(\left(\sum_{i\in A_{VAL}} \frac{\mathbf{z}_i}{b_i}\frac{\mathbf{z}_i^T}{b_i}\right)^{-1}\left(\sum_{i\in A_{VAL}} \frac{\mathbf{z}_i}{b_i}\frac{\theta_i}{b_i}\right),\ \sigma_v^2\left(\sum_{i\in A_{VAL}} \frac{\mathbf{z}_i}{b_i}\frac{\mathbf{z}_i^T}{b_i}\right)^{-1}\right)$$

$$\left[\,\sigma_v^2 \,\middle|\, \hat{\boldsymbol{\theta}}^{DIR}, \boldsymbol{\psi}, \boldsymbol{\theta}, \boldsymbol{\beta}\,\right] \sim IG\left(\frac{m-1}{2},\ \frac{1}{2}\sum_{i\in A_{VAL}}\left(\frac{\theta_i - \mathbf{z}_i^T\boldsymbol{\beta}}{b_i}\right)^2\right)$$

Processing steps for the HB_MM_KV method

Step 0 Determine if the HB_MM_KV method can be applied based on the specified inputs. Continue only if it is possible to apply the method with the given specifications.

Step 1 Use Table 1 (on page 10) to partition the small areas into the 3 area groups (A_{EXC} , A_{VAL} and A_{NUL}) for the parameter.

Step 2 If $m < (p+2)$ then we cannot apply the HB_MM_KV method and we stop.
If $m \geq (p+2)$ then continue by computing the following starting values for the Gibbs sampler.

$$\theta_i^{(0)} = \hat{\theta}_i^{DIR}$$

$$\boldsymbol{\beta}^{(0)} = \left(\sum_{i\in A_{VAL}} \frac{\mathbf{z}_i}{b_i}\frac{\mathbf{z}_i^T}{b_i}\right)^{-1}\left(\sum_{i\in A_{VAL}} \frac{\mathbf{z}_i}{b_i}\frac{\theta_i^{(0)}}{b_i}\right)$$

$$\sigma_v^{2(0)} \sim IG\left(\frac{m-1}{2},\ \frac{1}{2}\sum_{i\in A_{VAL}}\left(\frac{\theta_i^{(0)} - \mathbf{z}_i^T\boldsymbol{\beta}^{(0)}}{b_i}\right)^2\right)$$

Process Note:

- To apply this process, we must have $m \geq (p+2)$ so we can compute a proper regression coefficient. This means that m , the number of areas in A_{VAL} , must be at least 2 more than p , the number of auxiliary variables in $\mathbf{z}_i$.

Methodology Notes:

- The value of $\sigma_v^{2(0)}$ is obtained from an Inverse Gamma distribution $IG(\alpha,\beta)$. To do this, we use the following relationship between the Gamma and Inverse Gamma distributions.

  If $x \sim Gamma(\alpha,\beta)$ with pdf $\frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1}e^{-\beta x}$ for $x > 0$

  (x is Gamma with iscale parameter β)

  Then $x^{-1} \sim IG(\alpha,\beta)$ with pdf $\frac{\beta^\alpha}{\Gamma(\alpha)} \frac{e^{-\frac{\beta}{x}}}{x^{\alpha+1}}$ for $x > 0$

  (x^{-1} is Inverse Gamma with scale parameter β)

- Therefore, we obtain $\sigma_v^{2(0)}$ as follows.
  - First, we generate a random value from a Gamma distribution with shape parameter $\alpha = \frac{m-1}{2}$ and iscale (inverse scale) parameter

    $\beta = \frac{1}{2}\sum_{i\in A_{VAL}} \left(\frac{\theta_i^{(0)} - \mathbf{z}_i^T \boldsymbol{\beta}^{(0)}}{b_i}\right)^2$.
  - Then we take the inverse of this random value as the required value of $\sigma_v^{2(0)}$.

- Note that $\sigma_v^{2(0)}$ is a positive scalar value.

Implementation Notes:

- The function RANGAM in SAS can be used to generate a pseudo-random variable from a Gamma distribution with shape parameter *alpha* (α) and iscale (inverse scale) parameter *beta* (β) as follows.

$$x = \frac{1}{\beta} \times \text{RANGAM}(0, alpha)$$

- The required value of $\sigma_v^{2(0)}$ is then obtained as x^{-1} which is equivalent to $\frac{\beta}{\text{RANGAM}(0, alpha)}$.

Step 3 Beginning with the starting values $\theta_i^{(0)}$, $\boldsymbol{\beta}^{(0)}$ and $\sigma_v^{2\,(0)}$, iterate from $g=1$ to $g=G$, computing these quantities at each iteration g in the following order.

1. $\theta_i^{(g)} \sim N\left(\gamma_i^{(g-1)}\hat{\theta}_i^{DIR} + (1-\gamma_i^{(g-1)})\mathbf{z}_i^T \boldsymbol{\beta}^{(g-1)}, \gamma_i^{(g-1)}\psi_i \right)$ for $i \in A_{VAL}$

   where $\gamma_i^{(g-1)} = \dfrac{b_i^2\, \sigma_v^{2\,(g-1)}}{b_i^2\, \sigma_v^{2\,(g-1)} + \psi_i}$

2. $$\boldsymbol{\beta}^{(g)} \sim N\left(\left(\sum_{i\in A_{VAL}} \frac{\mathbf{z}_i}{b_i}\frac{\mathbf{z}_i^T}{b_i}\right)^{-1} \left(\sum_{i\in A_{VAL}} \frac{\mathbf{z}_i}{b_i}\frac{\theta_i^{(g)}}{b_i}\right), \sigma_v^{2\,(g-1)} \left(\sum_{i\in A_{VAL}} \frac{\mathbf{z}_i}{b_i}\frac{\mathbf{z}_i^T}{b_i}\right)^{-1} \right)$$

3. $$\sigma_v^{2\,(g)} \sim IG\left(\frac{m-1}{2}, \frac{1}{2}\sum_{i\in A_{VAL}} \left(\frac{\theta_i^{(g)} - \mathbf{z}_i^T \boldsymbol{\beta}^{(g)}}{b_i}\right)^2 \right)$$

4. For any areas $i \in A_{NUL}$ compute $\theta_i^{(g)} = u_i$ where $u_i \sim N\left(\mathbf{z}_i^T \boldsymbol{\beta}^{(g)}, b_i^2\, \sigma_v^{2\,(g)} \right)$

Methodology Notes:

- The value of $\theta_i^{(g)}$ for $i \in A_{VAL}$ is obtained by generating a random value from a univariate Normal distribution with mean $\gamma_i^{(g-1)}\hat{\theta}_i^{DIR} + (1-\gamma_i^{(g-1)})\mathbf{z}_i^T \boldsymbol{\beta}^{(g-1)}$ and variance $\gamma_i^{(g-1)}\psi_i$. Each $\theta_i^{(g)}$ is a scalar value and we generate m independent values corresponding to $i \in A_{VAL}$.
- The value of $\boldsymbol{\beta}^{(g)}$ is obtained by generating a random value from a multivariate Normal distribution with mean $\left(\sum_{i\in A_{VAL}} \mathbf{z}_i\, \mathbf{z}_i^T / b_i^2\right)^{-1} \left(\sum_{i\in A_{VAL}} \mathbf{z}_i\, \theta_i^{(g)} / b_i^2\right)$ and variance-covariance matrix $\sigma_v^{2(g-1)} \left(\sum_{i\in A_{VAL}} \mathbf{z}_i\, \mathbf{z}_i^T / b_i^2\right)^{-1}$. Note that $\boldsymbol{\beta}^{(g)}$ is a $p \times 1$ vector.
- The value of $\sigma_v^{2(g)}$ is obtained from an Inverse Gamma distribution as described earlier.
- The values $\theta_i^{(g)} = u_i$ where $u_i \sim N\left(\mathbf{z}_i^T \boldsymbol{\beta}^{(g)}, b_i^2\, \sigma_v^{2\,(g)} \right)$ for $i \in A_{NUL}$ are obtained from the linking model. These values are used to produce the HB estimates for $i \in A_{NUL}$.

Implementation Note:

- The quantity $\left(\sum_{i\in A_{VAL}} \mathbf{z}_i\, \mathbf{z}_i^T / b_i^2\right)^{-1}$ is constant over all iterations so we only need to compute it once.

Step 4 We discard the generated values for $g=1,2,\ldots,B$ where $B<G$ and produce the following HB estimates for the small-areas based on the sample values for $g=B+1,B+2,\ldots,G$.

$$\hat{\theta}_i^{HB} = \begin{cases} \hat{\theta}_i^{DIR} & \text{for } i \in A_{EXC} \\ \dfrac{1}{G-B}\sum_{g=B+1}^{G} \theta_i^{(g)} & \text{for } i \in A_{VAL} \\ \dfrac{1}{G-B}\sum_{g=B+1}^{G} \theta_i^{(g)} & \text{for } i \in A_{NUL} \end{cases}$$

$$v\hat{a}r(\hat{\theta}_i^{HB}) = \begin{cases} \psi_i & \text{for } i \in A_{EXC} \\ \dfrac{1}{G-B-1}\sum_{g=B+1}^{G} (\theta_i^{(g)} - \hat{\theta}_i^{HB})^2 & \text{for } i \in A_{VAL} \\ \dfrac{1}{G-B-1}\sum_{g=B+1}^{G} (\theta_i^{(g)} - \hat{\theta}_i^{HB})^2 & \text{for } i \in A_{NUL} \end{cases}$$

We also compute the following estimate of the regression coefficient and its estimated variance.

$$\hat{\boldsymbol{\beta}}^{HB} = \frac{1}{G-B}\sum_{g=B+1}^{G} \boldsymbol{\beta}^{(g)}$$

$$v\hat{a}r(\hat{\boldsymbol{\beta}}^{HB}) = \frac{1}{G-B-1}\sum_{g=B+1}^{G} (\boldsymbol{\beta}^{(g)} - \hat{\boldsymbol{\beta}}^{HB})(\boldsymbol{\beta}^{(g)} - \hat{\boldsymbol{\beta}}^{HB})^T$$

Methodology Note:

- The value of B is the length of the burn-in period. We do not use the values of $\theta_i^{(g)}$, $\boldsymbol{\beta}^{(g)}$ and $\sigma_v^{2(g)}$ in the burn-in period because they are less stable than the values obtained for $g>B$.

Process Notes:

- We can make this process similar to the other methods by defining B, δ and E as inputs and setting G as $G=B+\delta E=B+E$ (with $\delta=1$).
- The default values are $B=2000$, $\delta=1$ and $E=10000$ so $G=12000$.

Implementation Notes:

- We do not need to store the values of $\theta_i^{(g)}$, $\boldsymbol{\beta}^{(g)}$ and $\sigma_v^{2(g)}$ for $g = 1, 2 \ldots B$. We just need to generate them from one iteration to another.
- However, we need to store the values of $\theta_i^{(g)}$ and $\boldsymbol{\beta}^{(g)}$ for $g > B$ to calculate $\hat{\theta}_i^{HB}$, $v\hat{a}r(\hat{\theta}_i^{HB})$, $\hat{\boldsymbol{\beta}}^{HB}$ and $v\hat{a}r(\hat{\boldsymbol{\beta}}^{HB})$.
- If we are benchmarking the estimates $\hat{\theta}_i^{HB}$ in step 5 then we also need to store $\sigma_v^{2(g)}$ for $g > B$ to compute $\hat{\sigma}_v^{2\,HB} = \frac{1}{G-B}\sum_{g=B+1}^{G} \sigma_v^{2(g)}$, the HB estimate of σ_v^2.

Step 5 If benchmarking is required and the parameter corresponds to a mean or a total then we attempt (see Process Note below) to apply the method of difference adjustment to adjust the estimates $\hat{\theta}_i^{HB}$ and produce benchmark estimates $\hat{\theta}_i^{HB,b}$ and variance estimates $v\hat{a}r(\hat{\theta}_i^{HB,b})$ as follows:

$$\hat{\theta}_i^{HB,b} = \begin{cases} \hat{\theta}_i^{HB} & \text{for } i \in A_{EXC} \\ \hat{\theta}_i^{HB} + \alpha_i(\hat{\theta}^* - \sum_{d\in A} \omega_d\, \hat{\theta}_d^{HB}) & \text{for } i \in A_{VAL} \\ \hat{\theta}_i^{HB} & \text{for } i \in A_{NUL} \end{cases}$$

$$\text{with } \alpha_i = \frac{\omega_i\,(\psi_i + b_i^2\,\hat{\sigma}_v^{2\,HB})}{\sum_{i\in A_{VAL}} \omega_i^2\,(\psi_i + b_i^2\,\hat{\sigma}_v^{2\,HB})} \quad \text{for } i \in A_{VAL}$$

$$\text{and } \hat{\sigma}_v^{2\,HB} = \frac{1}{G-B}\sum_{g=B+1}^{G} \sigma_v^{2(g)}$$

$$v\hat{a}r(\hat{\theta}_i^{HB,b}) = \begin{cases} v\hat{a}r(\hat{\theta}_i^{HB}) & \text{for } i \in A_{EXC} \\ v\hat{a}r(\hat{\theta}_i^{HB}) + (\hat{\theta}_i^{HB,b} - \hat{\theta}_i^{HB})^2 & \text{for } i \in A_{VAL} \\ v\hat{a}r(\hat{\theta}_i^{HB}) & \text{for } i \in A_{NUL} \end{cases}$$

where ω_i is defined in the table below

parameter type	**definition of ω_i for $i \in A$**
TOTAL	$\omega_i = 1$
MEAN	$\omega_i = N_i / N$ where $N = \sum_{i\in A} N_i$

Process Notes:

- The term $\hat{\theta}^{*}$ is either $\hat{\theta}^{DIR}$ (a population value provided by the user) or a value calculated from the parameter estimates as $\sum_{i \in A} \omega_i \hat{\theta}_i^{DIR}$ with $\omega_i = 1$ for a total and $\omega_i = N_i / N$ for a mean. Either way $\hat{\theta}^{*}$ is treated as a constant in the above calculation.
- If the user provides $\hat{\theta}^{DIR}$ then $\hat{\theta}^{*} = \hat{\theta}^{DIR}$ and this value is used for benchmarking.
- If the user does not provide $\hat{\theta}^{DIR}$ then we attempt to calculate $\hat{\theta}^{*}$ as follows.
  - If the parameter is a total and $\hat{\theta}_i^{DIR} \neq \cdot$ for all $i \in A$ then we compute $\hat{\theta}^{*}$ as $\sum_{i \in A} \hat{\theta}_i^{DIR}$.
  - If the parameter is a mean and we have $\hat{\theta}_i^{DIR} \neq \cdot$ and N_i known for all $i \in A$ then we compute $\hat{\theta}^{*}$ as $\sum_{i \in A} N_i \hat{\theta}_i^{DIR} \Big/ \sum_{i \in A} N_i$.
  - Otherwise, benchmarking cannot be applied.

Methodology Notes:

- We only adjust the values of $\hat{\theta}_i^{HB}$ for $i \in A_{VAL}$. We do not adjust $\hat{\theta}_i^{HB}$ for $i \in A_{EXC}$ or $i \in A_{NUL}$.
- Note that by definition $\hat{\theta}_i^{HB,b} = \hat{\theta}_i^{HB} = \hat{\theta}_i^{DIR}$ and $v\hat{a}r(\hat{\theta}_i^{HB,b}) = v\hat{a}r(\hat{\theta}_i^{HB}) = \psi_i$ for $i \in A_{EXC}$.
- The reference to the method of benchmarking by difference adjustment is the November 2007 notes from Fuller.
- The formula for the benchmark variance $v\hat{a}r(\hat{\theta}_i^{HB,b})$ for $i \in A_{VAL}$ is given on page 635 of You, Rao and Dick (2004).

HB estimation with unmatched log-linear model and known sampling variances

Inference model for the HB_UM_LL_KV method

This method uses the linking model $log(\theta_i) = \boldsymbol{z}_i^T \boldsymbol{\beta} + b_i v_i$ which leads to an unmatched log-linear model. The model is specified as

$$\begin{aligned} &\text{Sampling Model :} \quad \hat{\theta}_i^{DIR} = \theta_i + e_i \\ &\text{Linking Model :} \quad log(\theta_i) = \boldsymbol{z}_i^T \boldsymbol{\beta} + b_i v_i \end{aligned}$$

where

$e_i \overset{id}{\sim} N(0, \psi_i)$	e_i are independently distributed as Normal random variables with mean 0 and known variance ψ_i
$v_i \overset{iid}{\sim} N(0, \sigma_v^2)$	v_i are independent and identically distributed as Normal random variables with mean 0 and unknown variance σ_v^2

The sampling and linking models are written in the Bayesian framework as follows.

$$\left[\, \hat{\theta}_i^{DIR} \,\middle|\, \theta_i \right] \sim N\left(\theta_i, \psi_i \right)$$
$$\left[\, log(\theta_i) \mid \boldsymbol{\beta}, \sigma_v^2 \right] \sim N\left(\boldsymbol{z}_i^T \boldsymbol{\beta}, b_i^2 \, \sigma_v^2 \right)$$

The unknown parameters in this model are $\boldsymbol{\theta} = \left\{ \theta_i \right\}$ over the areas as well as $\boldsymbol{\beta}$ and σ_v^2 . Prior distributions $\pi(\cdot)$ for the unknown parameters $\boldsymbol{\beta}$ and σ_v^2 are given by the following specifications.

$$\begin{aligned} &\pi(\boldsymbol{\beta}) \propto 1 \\ &\pi(\sigma_v^2) \propto (\sigma_v^2)^{-1/2} \end{aligned}$$

The distribution $\pi(\boldsymbol{\beta}) \propto 1$ is known as a flat prior. These two priors lead to a full conditional distribution for σ_v^2 based on the Inverse Gamma distribution. For a description of this distribution, see the Methodology Note after step 2 of this method.

Methodology Notes:

- This model is referenced by You and Rao (2002).
- The parameter θ_i has a Log-Normal distribution. By definition, θ_i is Log-Normal if $log(\theta_i)$ is Normal.
- The probability density function of θ_i is given by $\dfrac{1}{\theta_i}\dfrac{1}{\sqrt{2\pi b_i^2\sigma_v^2}}\,exp\left[-\dfrac{(log(\theta_i)-\boldsymbol{z}_i^T\boldsymbol{\beta})^2}{2\,b_i^2\,\sigma_v^2}\right]$.
- A discussion on the use of different priors is given by Gelman (2006).

Full conditional distributions for the inference model of the HB_UM_LL_KV method

Using the distributions of the Bayesian model and the Bayesian priors we arrive at the following full conditional distributions for the estimation of the unknown parameters. The way these are derived is shown in the section titled Derivation of full conditional distributions for the HB models.

$$\left[\,\theta_i \,\middle|\, \hat{\boldsymbol{\theta}}^{DIR}, \boldsymbol{\psi}, \boldsymbol{\beta}, \sigma_v^2\,\right] \propto\ log\,N(\,\boldsymbol{z}_i^T\boldsymbol{\beta}, b_i^2\,\sigma_v^2\,)\,exp\left[-\frac{(\hat{\theta}_i^{DIR}-\theta_i)^2}{2\,\psi_i}\right] \quad \text{for } i\in A_{VAL}$$

$$\left[\,\boldsymbol{\beta} \,\middle|\, \hat{\boldsymbol{\theta}}^{DIR}, \boldsymbol{\psi}, \boldsymbol{\theta}, \sigma_v^2\,\right] \sim N\left(\left(\sum_{i\in A_{VAL}}\frac{\boldsymbol{z}}{b_i}\frac{\boldsymbol{z}_i^T}{b_i}\right)^{-1}\left(\sum_{i\in A_{VAL}}\frac{\boldsymbol{z}_i}{b_i}\frac{log(\theta_i)}{b_i}\right),\ \sigma_v^2\left(\sum_{i\in A_{VAL}}\frac{\boldsymbol{z}}{b_i}\frac{\boldsymbol{z}_i^T}{b_i}\right)^{-1}\right)$$

$$\left[\,\sigma_v^2 \,\middle|\, \hat{\boldsymbol{\theta}}^{DIR}, \boldsymbol{\psi}, \boldsymbol{\theta}, \boldsymbol{\beta}\,\right] \sim IG\left(\frac{m-1}{2}, \frac{1}{2}\sum_{i\in A_{VAL}}\left(\frac{log(\theta_i)-\boldsymbol{z}_i^T\boldsymbol{\beta}}{b_i}\right)^2\right)$$

Processing steps for the HB_UM_LL_KV method

Step 0 Determine if the HB_UM_LL_KV method can be applied based on the specified inputs. Continue only if it is possible to apply the method with the given specifications.

Step 1 Use Table 1 (on page 10) to partition the small areas into the 3 area groups (A_{EXC}, A_{VAL} and A_{NUL}) for the parameter.

Step 2 If $m<(p+2)$ then we cannot apply the HB_UM_LL_KV method of estimation and we stop. If $m\geq(p+2)$ continue by computing the following starting values for the Gibbs sampler.

$$\theta_i^{(0)} = \hat{\theta}_i^{DIR} \quad \text{for } i\in A_{VAL}$$

$$\boldsymbol{\beta}^{(0)} = \left(\sum_{i\in A_{VAL}}\frac{\boldsymbol{z}_i}{b_i}\frac{\boldsymbol{z}_i^T}{b_i}\right)^{-1}\left(\sum_{i\in A_{VAL}}\frac{\boldsymbol{z}_i}{b_i}\frac{log\,(\theta_i^{(0)})}{b_i}\right)$$

$$\sigma_v^{2(0)} \sim IG\left(\frac{m-1}{2}, \frac{1}{2}\sum_{i\in A_{VAL}}\left(\frac{log\,(\theta_i^{(0)})-\boldsymbol{z}_i^T\boldsymbol{\beta}^{(0)}}{b_i}\right)^2\right)$$

Process Notes:

- To apply this process, we must have $m \geq (p+2)$ so we can compute a proper regression coefficient. This means that m, the number of areas in A_{VAL}, must be at least 2 more than p, the number of auxiliary variables in $\boldsymbol{z}_i$.
- There is no problem calculating $log(\theta_i^{(0)})$ in $\boldsymbol{\beta}^{(0)}$ and $\sigma_v^{2(0)}$ because $log(\theta_i^{(0)}) = log(\hat{\theta}_i^{DIR})$ and $\hat{\theta}_i^{DIR} > 0$ for $i \in A_{VAL}$.
- We can also calculate $\boldsymbol{\beta}^{(g)}$ and $\sigma_v^{2(g)}$ in subsequent iterations ($g = 1, \ldots G$) because the generated values of $\theta_i^{(g)}$ in step 3 are always positive since they come from a Log-Normal distribution.

Methodology Notes:

- The value of $\sigma_v^{2(0)}$ is obtained from an Inverse Gamma distribution $IG(\alpha, \beta)$. To do this, we use the following relationship between the Gamma and Inverse Gamma distributions.

  If $x \sim Gamma(\alpha, \beta)$ with pdf $\dfrac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$ for $x > 0$

  (Gamma with iscale parameter β)

  Then $x^{-1} \sim IG(\alpha, \beta)$ with pdf $\dfrac{\beta^\alpha}{\Gamma(\alpha)} \dfrac{e^{-\frac{\beta}{x}}}{x^{\alpha+1}}$ for $x > 0$

  (Inverse Gamma with scale parameter β)

- Therefore, we obtain $\sigma_v^{2(0)}$ as follows.
  - First, we generate a random value from a Gamma distribution with shape parameter $\alpha = \dfrac{m-1}{2}$ and iscale (inverse scale) parameter

    $$\beta = \frac{1}{2} \sum_{i \in A_{VAL}} \left(\frac{log(\theta_i^{(0)}) - \boldsymbol{z}_i^T \boldsymbol{\beta}^{(0)}}{b_i} \right)^2 .$$

  - Then we take the inverse of this random value as the required value of $\sigma_v^{2(0)}$.
- Note that $\sigma_v^{2(0)}$ is a positive scalar value.

Implementation Notes:

- The function RANGAM in SAS can be used to generate a pseudo-random variable from a Gamma distribution with shape parameter *alpha* (α) and iscale parameter *beta* (β) as follows.

$$x = \frac{1}{\beta} \times \text{RANGAM}\,(0,\, alpha)$$

- The required value of $\sigma_v^{2(0)}$ is then obtained as x^{-1} which is equivalent to $\dfrac{\beta}{\text{RANGAM}\,(0,\, alpha)}$.

Step 3 Beginning with the starting values $\theta_i^{(0)}$, $\boldsymbol{\beta}^{(0)}$ and $\sigma_v^{2(0)}$, iterate from $g = 1$ to $g = G$, computing these quantities at each iteration g in the following order.

1. $\theta_i^{(g)} \sim log\, N\left(\mathbf{z}_i^T \boldsymbol{\beta}^{(g-1)}, b_i^2\, \sigma_v^{2(g-1)} \right)$ for $i \in A_{VAL}$
2. $\alpha\,(\theta_i^{(g-1)}, \theta_i^{(g)}) = min \left\{ \dfrac{h\,(\theta_i^{(g)})}{h\,(\theta_i^{(g-1)})},\ 1 \right\}$ for $i \in A_{VAL}$

   where $h\,(\theta_i) = exp\left[-\dfrac{(\hat{\theta}_i^{DIR} - \theta_i)^2}{2\,\psi_i} \right]$
3. If $\xi < \alpha\,(\theta_i^{(g-1)}, \theta_i^{(g)})$ then $\theta_i^{(g)} = \theta_i^{(g)}$, else $\theta_i^{(g)} = \theta_i^{(g-1)}$ for $i \in A_{VAL}$

   where $\xi \sim U\left(0, 1\right)$ is a Uniform distribution on the (0,1) interval
4. $\boldsymbol{\beta}^{(g)} \sim N\left(\left(\sum_{i \in A_{VAL}} \dfrac{\mathbf{z}_i}{b_i} \dfrac{\mathbf{z}_i^T}{b_i} \right)^{-1} \left(\sum_{i \in A_{VAL}} \dfrac{\mathbf{z}_i}{b_i} \dfrac{log\,(\theta_i^{(g)})}{b_i} \right),\ \sigma_v^{2(g-1)} \left(\sum_{i \in A_{VAL}} \dfrac{\mathbf{z}_i}{b_i} \dfrac{\mathbf{z}_i^T}{b_i} \right)^{-1} \right)$
5. $\sigma_v^{2(g)} \sim IG\left(\dfrac{m-1}{2},\ \dfrac{1}{2} \sum_{i \in A_{VAL}} \left(\dfrac{log\,(\theta_i^{(g)}) - \mathbf{z}_i^T \boldsymbol{\beta}^{(g)}}{b_i} \right)^2 \right)$
6. For any areas $i \in A_{NUL}$ compute $\theta_i^{(g)} = exp(\,u_i\,)$ where $u_i \sim N\left(\mathbf{z}_i^T \boldsymbol{\beta}^{(g)}, b_i^2\, \sigma_v^{2(g)} \right)$

Methodology Notes:

- Here we use $log\,N\left(\boldsymbol{z}_i^T \boldsymbol{\beta}^{(g-1)}, b_i^2\, \sigma_v^{2(g-1)} \right)$ as our proposal distribution in the Metropolis-Hastings algorithm.
- The value of $\theta_i^{(g)}$ for $i \in A_{VAL}$ is obtained by generating a random value from a univariate Log-Normal distribution with mean $\boldsymbol{z}_i^T \boldsymbol{\beta}$ and variance $b_i^2\, \sigma_v^2$ (proposal step). Each $\theta_i^{(g)}$ is a scalar value and we generate m independent values corresponding to $i \in A_{VAL}$.
- To obtain a value $\theta_i^{(g)}$ from a Log-Normal distribution we first generate a random value u_i from a Normal distribution with mean $\boldsymbol{z}_i^T \boldsymbol{\beta}$ and variance $b_i^2\, \sigma_v^2$ and then take $\theta_i^{(g)} = exp(u_i)$.
- The quantity $\alpha(\theta_i^{(g-1)}, \theta_i^{(g)})$ is known as the acceptance probability. We generate a Uniform random variable ξ on the interval (0,1). If $\xi < \alpha(\theta_i^{(g-1)}, \theta_i^{(g)})$ then we accept the current value of $\theta_i^{(g)}$, else we reset it to the previous value ($\theta_i^{(g)} = \theta_i^{(g-1)}$) (acceptance step).
- The value of $\boldsymbol{\beta}^{(g)}$ is obtained by generating a random value from a multivariate Normal distribution with mean $\left(\sum_{i \in A_{VAL}} \boldsymbol{z}_i \boldsymbol{z}_i^T / b_i^2\right)^{-1} \left(\sum_{i \in A_{VAL}} \boldsymbol{z}_i\, log(\theta_i^{(g)}) / b_i^2\right)$ and variance-covariance matrix $\sigma_v^{2(g-1)} \left(\sum_{i \in A_{VAL}} \boldsymbol{z}_i \boldsymbol{z}_i^T / b_i^2\right)^{-1}$. Note that $\boldsymbol{\beta}^{(g)}$ is a $p \times 1$ vector.
- The value of $\sigma_v^{2(g)}$ is obtained from an Inverse Gamma distribution as described earlier.
- The values $\theta_i^{(g)} = exp(u_i)$ where $u_i \sim N\left(\boldsymbol{z}_i^T \boldsymbol{\beta}^{(g)}, b_i^2\, \sigma_v^{2(g)} \right)$ for $i \in A_{NUL}$ are obtained from the linking model. These values are used to produce the HB estimates for $i \in A_{NUL}$.

Implementation Note:

- The quantity $\left(\sum_{i \in A_{VAL}} \boldsymbol{z}_i \boldsymbol{z}_i^T / b_i^2\right)^{-1}$ is constant over all iterations so we only need to compute it once.

Step 4 We discard the generated values for $g = 1, 2, \ldots, B$ where $B < G$. For the remaining samples, we define the index set $E_\delta = \{ B+\delta, B+2\delta, B+3\delta, \ldots, G \}$ of size $E = (G-B)/\delta$, by selecting every δ^{th} sample, where $4 \le \delta \le 10$. We then use the samples in this set to produce the following HB estimates of the small areas.

$$\hat{\theta}_i^{HB} = \begin{cases} \hat{\theta}_i^{DIR} & \text{for } i \in A_{EXC} \\ \frac{1}{E} \sum_{g \in E_\delta} \theta_i^{(g)} & \text{for } i \in A_{VAL} \\ \frac{1}{E} \sum_{g \in E_\delta} \theta_i^{(g)} & \text{for } i \in A_{NUL} \end{cases}$$

$$v\hat{a}r(\hat{\theta}_i^{HB}) = \begin{cases} \psi_i & \text{for } i \in A_{EXC} \\ \frac{1}{E-1} \sum_{g \in E_\delta} (\theta_i^{(g)} - \hat{\theta}_i^{HB})^2 & \text{for } i \in A_{VAL} \\ \frac{1}{E-1} \sum_{g \in E_\delta} (\theta_i^{(g)} - \hat{\theta}_i^{HB})^2 & \text{for } i \in A_{NUL} \end{cases}$$

We also compute the following estimate of the regression coefficient and its estimated variance.

$$\hat{\boldsymbol{\beta}}^{HB} = \frac{1}{E} \sum_{g \in E_\delta} \boldsymbol{\beta}^{(g)}$$

$$v\hat{a}r(\hat{\boldsymbol{\beta}}^{HB}) = \frac{1}{E-1} \sum_{g \in E_\delta} (\boldsymbol{\beta}^{(g)} - \hat{\boldsymbol{\beta}}^{HB})(\boldsymbol{\beta}^{(g)} - \hat{\boldsymbol{\beta}}^{HB})^T$$

Methodology Note:

- The value of B is the length of the burn-in period. We do not use the values of $\theta_i^{(g)}$, $\boldsymbol{\beta}^{(g)}$ and $\sigma_v^{2(g)}$ in the burn-in period because they are less stable than the values obtained for $g > B$.

Process Notes:

- Since $E = (G - B) / \delta$ must be divisible by δ, it is better to define B, δ and E as inputs and set G as $G = B + \delta E$.
- The default values are $B = 2000$, $\delta = 4$ and $E = 10000$ so $G = 42000$.

Implementation Notes:

- We do not need to store the values of $\theta_i^{(g)}$, $\boldsymbol{\beta}^{(g)}$ and $\sigma_v^{2(g)}$ for $g=1,2\ldots B$. We just need to generate them from one iteration to another.
- However, we need to store the values of $\theta_i^{(g)}$ and $\boldsymbol{\beta}^{(g)}$ for $g>B$ to calculate $\hat{\theta}_i^{HB}$, $v\hat{a}r(\hat{\theta}_i^{HB})$, $\hat{\boldsymbol{\beta}}^{HB}$ and $v\hat{a}r(\hat{\boldsymbol{\beta}}^{HB})$.
- If we are benchmarking the estimates $\hat{\theta}_i^{HB}$ in step 5 then we also need to store $\sigma_v^{2(g)}$ for $g>B$ to compute $\hat{\sigma}_v^{2\,HB}=\frac{1}{E}\sum_{g\in E_\delta}\sigma_v^{2(g)}$, an estimate of σ_v^2 .

Step 5 If benchmarking is required and the parameter corresponds to a mean or a total then we attempt (see Process Note below) to apply the method of difference adjustment to adjust the estimates $\hat{\theta}_i^{HB}$ and produce benchmark estimates $\hat{\theta}_i^{HB,b}$ and variance estimates $v\hat{a}r(\hat{\theta}_i^{HB,b})$ as follows:

$$\hat{\theta}_i^{HB,b}=\begin{cases}\hat{\theta}_i^{HB} & \text{for } i\in A_{EXC}\\ \hat{\theta}_i^{HB}+\alpha_i(\hat{\theta}^*-\sum_{d\in A}\omega_d\,\hat{\theta}_d^{HB}) & \text{for } i\in A_{VAL}\\ \hat{\theta}_i^{HB} & \text{for } i\in A_{NUL}\end{cases}$$

$$\text{with } \alpha_i=\frac{\omega_i\,(\psi_i+b_i^2\,\hat{\sigma}_v^{2\,HB})}{\sum_{i\in A_{VAL}}\omega_i^2\,(\psi_i+b_i^2\,\hat{\sigma}_v^{2\,HB})}\quad \text{for } i\in A_{VAL}$$

$$\text{and } \hat{\sigma}_v^{2\,HB}=\frac{1}{E}\sum_{g\in E_\delta}\sigma_v^{2(g)}$$

$$v\hat{a}r(\hat{\theta}_i^{HB,b})=\begin{cases}v\hat{a}r(\hat{\theta}_i^{HB}) & \text{for } i\in A_{EXC}\\ v\hat{a}r(\hat{\theta}_i^{HB})+(\,\hat{\theta}_i^{HB,b}-\hat{\theta}_i^{HB}\,)^2 & \text{for } i\in A_{VAL}\\ v\hat{a}r(\hat{\theta}_i^{HB}) & \text{for } i\in A_{NUL}\end{cases}$$

where ω_i is defined in the table below

parameter type	**definition of ω_i for $i\in A$**
TOTAL	$\omega_i=1$
MEAN	$\omega_i=N_i/N$ where $N=\sum_{i\in A}N_i$

Process Notes:

- The term $\hat{\theta}^*$ is either $\hat{\theta}^{DIR}$ (a population value provided by the user) or a value calculated from the parameter estimates as $\sum_{i \in A} \omega_i \hat{\theta}_i^{DIR}$ with $\omega_i = 1$ for a total and $\omega_i = N_i / N$ for a mean. Either way $\hat{\theta}^*$ is treated as a constant in the above calculation.
- If the user provides $\hat{\theta}^{DIR}$ then $\hat{\theta}^* = \hat{\theta}^{DIR}$ and this value is used for benchmarking.
- If the user does not provide $\hat{\theta}^{DIR}$ then we attempt to calculate $\hat{\theta}^*$ as follows.
  - If the parameter is a total and $\hat{\theta}_i^{DIR} \neq \cdot$ for all $i \in A$ then we compute $\hat{\theta}^*$ as $\sum_{i \in A} \hat{\theta}_i^{DIR}$.
  - If the parameter is a mean and we have $\hat{\theta}_i^{DIR} \neq \cdot$ and N_i known for all $i \in A$ then we compute $\hat{\theta}^*$ as $\sum_{i \in A} N_i \hat{\theta}_i^{DIR} \Big/ \sum_{i \in A} N_i$.
  - Otherwise, benchmarking cannot be applied.

Methodology Notes:

- We only adjust the values of $\hat{\theta}_i^{HB}$ for $i \in A_{VAL}$. We do not adjust $\hat{\theta}_i^{HB}$ for $i \in A_{EXC}$ or $i \in A_{NUL}$.
- Note that by definition $\hat{\theta}_i^{HB,b} = \hat{\theta}_i^{HB} = \hat{\theta}_i^{DIR}$ and $v\hat{a}r(\hat{\theta}_i^{HB,b}) = v\hat{a}r(\hat{\theta}_i^{HB}) = \psi_i$ for $i \in A_{EXC}$.
- The reference to the method of benchmarking by difference adjustment is the November 2007 notes from Fuller.
- The formula for the benchmark variance $v\hat{a}r(\hat{\theta}_i^{HB,b})$ for $i \in A_{VAL}$ is given on page 635 of You, Rao and Dick (2004).

HB estimation with unmatched log census undercount model and known sampling variances

Inference model for the HB_UM_LCU_KV method

This method uses the linking model $g(\theta_i) = log\left(\theta_i / \theta_i + C_i\right) = \mathbf{z}_i^T \boldsymbol{\beta} + b_i v_i$ which leads to an unmatched model. This model is used for the estimation of the census undercounts of the Canadian Census of Population. The model is specified as

$$\text{Sampling Model}: \quad \hat{\theta}_i^{DIR} = \theta_i + e_i$$
$$\text{Linking Model}: \quad log\left(\frac{\theta_i}{\theta_i + C_i}\right) = \mathbf{z}_i^T \boldsymbol{\beta} + b_i v_i$$

where

$e_i \overset{id}{\sim} N(0, \psi_i)$	e_i are independently distributed as Normal random variables with mean 0 and known variance ψ_i
$v_i \overset{iid}{\sim} N(0, \sigma_v^2)$	v_i are independent and identically distributed as Normal random variables with mean 0 and unknown variance σ_v^2
C_i	are constants representing the known number of individuals from the census

In this model, θ_i represents the number of individuals not counted in the census while C_i is the known census count. As a result, $\theta_i / (\theta_i + C_i)$ is the proportion of individuals undercounted by the census. This model can also be written in the Bayesian framework as follows.

$$\left[\hat{\theta}_i^{DIR} \,\middle|\, \theta_i\right] \sim N\left(\theta_i, \psi_i\right)$$
$$\left[log\left(\frac{\theta_i}{\theta_i + C_i}\right) \middle| \boldsymbol{\beta}, \sigma_v^2\right] \sim N\left(z_i^T \boldsymbol{\beta}, b_i^2 \sigma_v^2\right)$$

The prior distributions $\pi(\cdot)$ for the unknown parameters $\boldsymbol{\beta}$ and σ_v^2 are given by the following specifications.

$$\pi(\boldsymbol{\beta}) \propto 1$$
$$\pi(\sigma_v^2) \propto (\sigma_v^2)^{-1/2}$$

The distribution $\pi(\boldsymbol{\beta}) \propto 1$ is known as a flat prior. These two priors lead to a full conditional distribution for σ_v^2 based on the Inverse Gamma distribution. For a description of this distribution, see the Methodology Note after step 2 of this method.

Methodology Notes:

- This model is described on page 237 of Rao (2003).
- A discussion on the use of different priors is given by Gelman (2006).

Full conditional distributions for the inference model of the HB_UM_LCU_KV method

Using the distributions of the Bayesian model and the Bayesian priors we obtain the following full conditional distributions for the estimation of the unknown parameters. This is shown in the section titled Derivation of full conditional distributions for the HB models.

$$\left[\,\theta_i \,\middle|\, \hat{\boldsymbol{\theta}}^{DIR}, \boldsymbol{\psi}, \boldsymbol{\beta}, \sigma_v^2\,\right] \propto \frac{g'(\theta_i)}{\psi_i} exp\left[-\frac{(\hat{\theta}_i^{DIR}-\theta_i)^2}{2\,\psi_i} - \frac{\left(g(\theta_i)-\mathbf{z}_i^T\boldsymbol{\beta}\right)^2}{2\,\sigma_v^2}\right] \quad \text{for } i \in A_{VAL}$$

$$\text{where } g(\theta_i) = log\left(\frac{\theta_i}{\theta_i + C_i}\right) \text{ and } g'(\theta_i) = \frac{C_i}{\theta_i\,(\theta_i + C_i)}$$

$$\left[\,\boldsymbol{\beta} \,\middle|\, \hat{\boldsymbol{\theta}}^{DIR}, \boldsymbol{\psi}, \boldsymbol{\theta}, \sigma_v^2\,\right] \sim N\left(\left(\sum_{i\in A_{VAL}} \frac{\mathbf{z}_i}{b_i}\frac{\mathbf{z}_i^T}{b_i}\right)^{-1}\left(\sum_{i\in A_{VAL}} \frac{\mathbf{z}_i}{b_i}\frac{g(\theta_i)}{b_i}\right),\ \sigma_v^2\left(\sum_{i\in A_{VAL}} \frac{\mathbf{z}_i}{b_i}\frac{\mathbf{z}_i^T}{b_i}\right)^{-1}\right)$$

$$\left[\,\sigma_v^2 \,\middle|\, \hat{\boldsymbol{\theta}}^{DIR}, \boldsymbol{\psi}, \boldsymbol{\theta}, \boldsymbol{\beta}\,\right] \sim IG\left(\frac{m-1}{2},\ \frac{1}{2}\sum_{i\in A_{VAL}}\left(\frac{g(\theta_i)-\mathbf{z}_i^T\boldsymbol{\beta}}{b_i}\right)^2\right)$$

Processing steps for the HB_UM_LCU_KV method

Step 0 Determine if the HB_UM_LCU_KV method can be applied based on the specified inputs. Continue only if it is possible to apply the method with the given specifications.

Step 1 Use Table 1 (on page 10) to partition the small areas into the 3 area groups (A_{EXC} , A_{VAL} and A_{NUL}) for the parameter.

Step 2 If $m < (p+2)$ then we cannot apply the HB_UM_LCU_KV method of estimation and we stop. If $m \geq (p+2)$ continue by computing the following starting values for the Gibbs sampler.

$$\theta_i^{(0)} = \hat{\theta}_i^{DIR} \quad \text{for } i \in A_{VAL}$$

$$\boldsymbol{\beta}^{(0)} = \left(\sum_{i\in A_{VAL}} \frac{\mathbf{z}_i}{b_i}\frac{\mathbf{z}_i^T}{b_i}\right)^{-1}\left(\sum_{i\in A_{VAL}} \frac{\mathbf{z}_i}{b_i}\frac{g(\theta_i^{(0)})}{b_i}\right)$$

$$\sigma_v^{2(0)} \sim IG\left(\frac{m-1}{2},\ \frac{1}{2}\sum_{i\in A_{VAL}}\left(\frac{g(\theta_i^{(0)})-\mathbf{z}_i^T\boldsymbol{\beta}^{(0)}}{b_i}\right)^2\right)$$

Process Notes:

- To apply this process, we must have $m \geq (p+2)$ so we can compute a proper regression coefficient. This means that m, the number of areas in A_{VAL}, must be at least 2 more than p, the number of auxiliary variables in $\boldsymbol{z}_i$.
- There is no problem calculating $g(\theta_i^{(0)})$ in $\boldsymbol{\beta}^{(0)}$ and $\sigma_v^{2(0)}$ because $g(\theta_i^{(0)}) = log\left(\dfrac{\theta_i^{(0)}}{\theta_i^{(0)} + C_i}\right) = log\left(\dfrac{\hat{\theta}_i^{DIR}}{\hat{\theta}_i^{DIR} + C_i}\right)$ and $\hat{\theta}_i^{DIR} > 0$ for $i \in A_{VAL}$. We also know $C_i > 0$ for $i \in A_{VAL}$.

Methodology Notes:

- The value of $\sigma_v^{2(0)}$ is obtained from an Inverse Gamma distribution $IG(\alpha, \beta)$. To do this, we use the following relationship between the Gamma and Inverse Gamma distributions.

  If $x \sim Gamma(\alpha, \beta)$ with pdf $\dfrac{\beta^{\alpha}}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$ for $x > 0$

  (x is Gamma with iscale parameter β)

  Then $x^{-1} \sim IG(\alpha, \beta)$ with pdf $\dfrac{\beta^{\alpha}}{\Gamma(\alpha)} \dfrac{e^{-\frac{\beta}{x}}}{x^{\alpha+1}}$ for $x > 0$

  (x^{-1} is Inverse Gamma with scale parameter β)

- Therefore, we obtain $\sigma_v^{2(0)}$ as follows.
  - First, we generate a random value from a Gamma distribution with shape parameter $\alpha = \dfrac{m-1}{2}$ and iscale (inverse scale) parameter

    $$\beta = \frac{1}{2} \sum_{i \in A_{VAL}} \left(\frac{g(\theta_i^{(0)}) - \boldsymbol{z}_i^T \boldsymbol{\beta}^{(0)}}{b_i} \right)^2 .$$

  - Then we take the inverse of this random value as the required value of $\sigma_v^{2(0)}$.
- Note that $\sigma_v^{2(0)}$ is a positive scalar value.

Implementation Notes:

- The function RANGAM in SAS can be used to generate a pseudo-random variable from a Gamma distribution with shape parameter *alpha* (α) and iscale (inverse scale) parameter *beta* (β) as follows.

$$x = \frac{1}{\beta} \times \text{RANGAM}\,(0, alpha)$$

- The required value of $\sigma_v^{2(0)}$ is then obtained as x^{-1} which is equivalent to $\frac{\beta}{\text{RANGAM}\,(0, alpha)}$.

Step 3 Beginning with the starting values $\theta_i^{(0)}$, $\boldsymbol{\beta}^{(0)}$ and $\sigma_v^{2\,(0)}$, iterate from $g=1$ to $g=G$, computing these quantities at each iteration g in the following order.

1. $\theta_i^{(g)} \sim N\left(\hat{\theta}_i^{DIR}, \psi_i \right)$ for $i \in A_{VAL}$
2. $\alpha\,(\theta_i^{(g-1)}, \theta_i^{(g)}) = min\left\{ \frac{h\,(\theta_i^{(g)}, \boldsymbol{\beta}^{(g-1)}, \sigma_v^{2(g-1)})}{h\,(\theta_i^{(g-1)}, \boldsymbol{\beta}^{(g-1)}, \sigma_v^{2(g-1)})},\ 1 \right\}$ for $i \in A_{VAL}$

   where $h\,(\theta_i, \boldsymbol{\beta}, \sigma_v^2) = \frac{1}{\theta_i\,(\theta_i + C_i)}\, exp\left[-\frac{1}{2\,\sigma_v^2} \left(\frac{g\,(\theta_i) - \mathbf{z}_i^T \boldsymbol{\beta}}{b_i} \right)^2 \right]$
3. If $\xi < \alpha\,(\theta_i^{(g-1)}, \theta_i^{(g)})$ then $\theta_i^{(g)} = \theta_i^{(g)}$, else $\theta_i^{(g)} = \theta_i^{(g-1)}$ for $i \in A_{VAL}$

   where $\xi \sim U\,(\,0,1\,)$ is a Uniform distribution on the (0,1) interval
4. $\boldsymbol{\beta}^{(g)} \sim N\left(\left(\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i}{b_i} \frac{\mathbf{z}_i^T}{b_i} \right)^{-1} \left(\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i}{b_i} \frac{g\,(\theta_i^{(g)})}{b_i} \right),\ \sigma_v^{2(g-1)} \left(\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i}{b_i} \frac{\mathbf{z}_i^T}{b_i} \right)^{-1} \right)$
5. $\sigma_v^{2(g)} \sim IG\left(\frac{m-1}{2},\ \frac{1}{2} \sum_{i \in A_{VAL}} \left(\frac{g\,(\theta_i^{(g)}) - \mathbf{z}_i^T \boldsymbol{\beta}^{(g)}}{b_i} \right)^2 \right)$
6. For any areas $i \in A_{NUL}$ compute $\theta_i^{(g)} = \frac{C_i}{exp(-u_i\,) - 1}$ where $u_i \sim N\left(\mathbf{z}_i^T \boldsymbol{\beta}^{(g)}, b_i^2\, \sigma_v^{2(g)} \right)$

Process Notes:

- Since $\theta_i^{(g)} \sim N\left(\hat{\theta}_i^{DIR}, \psi_i \right)$ for $i \in A_{VAL}$, it is possible to generate $\theta_i^{(g)} \le 0$. This is a problem for calculating $g(\theta_i^{(g)}) = log\left(\dfrac{\theta_i^{(g)}}{\theta_i^{(g)} + C_i} \right)$ in $h(\theta_i^{(g)}, \boldsymbol{\beta}^{(g-1)}, \sigma_v^{2(g-1)})$, $\boldsymbol{\beta}^{(g)}$ and $\sigma_v^{2(g)}$.
- This can occur when $\hat{\theta}_i^{DIR}$ is close to 0 or when the variance ψ_i is large relative to $\hat{\theta}_i^{DIR}$.
- To avoid this difficulty, we generate successive values until we obtain a positive value for $\theta_i^{(g)}$.

Methodology Notes:

- Here we use $N\left(\hat{\theta}_i^{DIR}, \psi_i \right)$ as our proposal distribution in the Metropolis-Hastings algorithm.
- The value of $\theta_i^{(g)}$ (for $i \in A_{VAL}$) is obtained by generating a random value from a univariate Normal distribution with mean $\hat{\theta}_i^{DIR}$ and variance ψ_i (proposal step).
- Each $\theta_i^{(g)}$ is a scalar value and we generate m independent values corresponding to $i \in A_{VAL}$. If this value is negative for any area then we generate another value from the univariate Normal distribution. We continue to generate values until we obtain a positive value.
- The quantity $\alpha(\theta_i^{(g-1)}, \theta_i^{(g)})$ is known as the acceptance probability. We generate a Uniform random variable ξ on the interval $(0,1)$. If $\xi < \alpha(\theta_i^{(g-1)}, \theta_i^{(g)})$ then we accept the current value of $\theta_i^{(g)}$, else we reset it to the previous value ($\theta_i^{(g)} = \theta_i^{(g-1)}$) (acceptance step).
- The value of $\boldsymbol{\beta}^{(g)}$ is obtained by generating a random value from a multivariate Normal distribution with mean $\left(\sum_{i \in A_{VAL}} \mathbf{z}_i \mathbf{z}_i^T / b_i^2 \right)^{-1} \left(\sum_{i \in A_{VAL}} \mathbf{z}_i \, g(\theta_i^{(g)}) / b_i^2 \right)$ and variance-covariance matrix $\sigma_v^{2(g-1)} \left(\sum_{i \in A_{VAL}} \mathbf{z}_i \mathbf{z}_i^T / b_i^2 \right)^{-1}$. Note that $\boldsymbol{\beta}^{(g)}$ is a $p \times 1$ vector.
- The value of $\sigma_v^{2(g)}$ is obtained from an Inverse Gamma distribution as described earlier.
- The values $\theta_i^{(g)} = \dfrac{C_i}{exp(-u_i) - 1}$ where $u_i \sim N\left(\mathbf{z}_i^T \boldsymbol{\beta}^{(g)}, b_i^2 \, \sigma_v^{2(g)} \right)$ for $i \in A_{NUL}$ are obtained from the linking model. These values are used to produce the HB estimates for $i \in A_{NUL}$.

Implementation Note:

- The quantity $\left(\sum_{i \in A_{VAL}} \mathbf{z}_i \, \mathbf{z}_i^T / b_i^2\right)^{-1}$ is constant over all iterations so we only need to compute it once.

Step 4 We discard the generated values for $g = 1, 2, \ldots, B$ where $B < G$. For the remaining samples, we define the index set $E_\delta = \{ B+\delta, B+2\delta, B+3\delta, \ldots , G \}$ of size $E = (G-B) / \delta$, by selecting every δ^{th} sample, where $4 \le \delta \le 10$. We then use the samples in this set to produce the following HB estimates of the small areas.

$$\hat{\theta}_i^{HB} = \begin{cases} \hat{\theta}_i^{DIR} & \text{for } i \in A_{EXC} \\ \frac{1}{E} \sum_{g \in E_\delta} \theta_i^{(g)} & \text{for } i \in A_{VAL} \\ \frac{1}{E} \sum_{g \in E_\delta} \theta_i^{(g)} & \text{for } i \in A_{NUL} \end{cases}$$

$$v\hat{a}r(\hat{\theta}_i^{HB}) = \begin{cases} \psi_i & \text{for } i \in A_{EXC} \\ \frac{1}{E-1} \sum_{g \in E_\delta} (\, \theta_i^{(g)} - \hat{\theta}_i^{HB} \,)^2 & \text{for } i \in A_{VAL} \\ \frac{1}{E-1} \sum_{g \in E_\delta} (\, \theta_i^{(g)} - \hat{\theta}_i^{HB} \,)^2 & \text{for } i \in A_{NUL} \end{cases}$$

We also compute the following estimate of the regression coefficient and its estimated variance.

$$\hat{\boldsymbol{\beta}}^{HB} = \frac{1}{E} \sum_{g \in E_\delta} \boldsymbol{\beta}^{(g)}$$

$$v\hat{a}r(\hat{\boldsymbol{\beta}}^{HB}) = \frac{1}{E-1} \sum_{g \in E_\delta} (\, \boldsymbol{\beta}^{(g)} - \hat{\boldsymbol{\beta}}^{HB} \,)(\, \boldsymbol{\beta}^{(g)} - \hat{\boldsymbol{\beta}}^{HB} \,)^T$$

Methodology Note:

- The value of B is the length of the burn-in period. We do not use the values of $\theta_i^{(g)}$, $\boldsymbol{\beta}^{(g)}$ and $\sigma_v^{2(g)}$ in the burn-in period because they are less stable than the values obtained for $g > B$.

Process Notes:

- Since $E=(G-B)/\delta$ must be divisible by δ, it is better to define B, δ and E as inputs and set G as $G=B+\delta E$.
- The default values are $B=2000$, $\delta=4$ and $E=10000$ so $G=42000$.

Implementation Notes:

- We do not need to store the values of $\theta_i^{(g)}$, $\boldsymbol{\beta}^{(g)}$ and $\sigma_v^{2(g)}$ for $g=1,2\ldots B$. We just need to generate them from one iteration to another.
- However, we need to store the values of $\theta_i^{(g)}$ and $\boldsymbol{\beta}^{(g)}$ for $g>B$ to calculate $\hat{\theta}_i^{HB}$, $\hat{var}(\hat{\theta}_i^{HB})$, $\hat{\boldsymbol{\beta}}^{HB}$ and $\hat{var}(\hat{\boldsymbol{\beta}}^{HB})$.
- If we are benchmarking the estimates $\hat{\theta}_i^{HB}$ in step 5 then we also need to store $\sigma_v^{2(g)}$ for $g>B$ to compute $\hat{\sigma}_v^{2\,HB}=\frac{1}{E}\sum_{g\in E_\delta}\sigma_v^{2(g)}$, an estimate of σ_v^2.

Step 5 If benchmarking is required and the parameter corresponds to a mean or a total then we attempt (see Process Note below) to apply the method of difference adjustment to adjust the estimates $\hat{\theta}_i^{HB}$ and produce benchmark estimates $\hat{\theta}_i^{HB,b}$ and variance estimates $\hat{var}(\hat{\theta}_i^{HB,b})$ as follows:

$$\hat{\theta}_i^{HB,b}=\begin{cases}\hat{\theta}_i^{HB} & \text{for } i\in A_{EXC}\\ \hat{\theta}_i^{HB}+\alpha_i(\hat{\theta}^*-\sum_{d\in A}\omega_d\,\hat{\theta}_d^{HB}) & \text{for } i\in A_{VAL}\\ \hat{\theta}_i^{HB} & \text{for } i\in A_{NUL}\end{cases}$$

$$\text{with } \alpha_i=\frac{\omega_i\,(\psi_i+b_i^2\,\hat{\sigma}_v^{2\,HB})}{\sum_{i\in A_{VAL}}\omega_i^2\,(\psi_i+b_i^2\,\hat{\sigma}_v^{2\,HB})} \quad \text{for } i\in A_{VAL}$$

$$\text{and } \hat{\sigma}_v^{2\,HB}=\frac{1}{E}\sum_{g\in E_\delta}\sigma_v^{2(g)}$$

$$\hat{var}(\hat{\theta}_i^{HB,b})=\begin{cases}\hat{var}(\hat{\theta}_i^{HB}) & \text{for } i\in A_{EXC}\\ \hat{var}(\hat{\theta}_i^{HB})+(\hat{\theta}_i^{HB,b}-\hat{\theta}_i^{HB})^2 & \text{for } i\in A_{VAL}\\ \hat{var}(\hat{\theta}_i^{HB}) & \text{for } i\in A_{NUL}\end{cases}$$

where ω_i is defined in the table below

parameter type	**definition of ω_i for $i\in A$**
TOTAL	$\omega_i=1$
MEAN	$\omega_i=N_i/N$ where $N=\sum_{i\in A}N_i$

Process Notes:

- The term $\hat{\theta}^*$ is either $\hat{\theta}^{DIR}$ (a population value provided by the user) or a value calculated from the parameter estimates as $\sum_{i \in A} \omega_i \hat{\theta}_i^{DIR}$ with $\omega_i = 1$ for a total and $\omega_i = N_i / N$ for a mean. Either way $\hat{\theta}^*$ is treated as a constant in the above calculation.
- If the user provides $\hat{\theta}^{DIR}$ then $\hat{\theta}^* = \hat{\theta}^{DIR}$ and this value is used for benchmarking.
- If the user does not provide $\hat{\theta}^{DIR}$ then we attempt to calculate $\hat{\theta}^*$ as follows.
  - If the parameter is a total and $\hat{\theta}_i^{DIR} \neq \cdot$ for all $i \in A$ then we compute $\hat{\theta}^*$ as $\sum_{i \in A} \hat{\theta}_i^{DIR}$.
  - If the parameter is a mean and we have $\hat{\theta}_i^{DIR} \neq \cdot$ and N_i known for all $i \in A$ then we compute $\hat{\theta}^*$ as $\sum_{i \in A} N_i \hat{\theta}_i^{DIR} \Big/ \sum_{i \in A} N_i$.
  - Otherwise, benchmarking cannot be applied.

Methodology Notes:

- We only adjust the values of $\hat{\theta}_i^{HB}$ for $i \in A_{VAL}$. We do not adjust $\hat{\theta}_i^{HB}$ for $i \in A_{EXC}$ or $i \in A_{NUL}$.
- Note that by definition $\hat{\theta}_i^{HB,b} = \hat{\theta}_i^{HB} = \hat{\theta}_i^{DIR}$ and $v\hat{a}r(\hat{\theta}_i^{HB,b}) = v\hat{a}r(\hat{\theta}_i^{HB}) = \psi_i$ for $i \in A_{EXC}$.
- The reference to the method of benchmarking by difference adjustment is the November 2007 notes from Fuller.
- The formula for the benchmark variance $v\hat{a}r(\hat{\theta}_i^{HB,b})$ for $i \in A_{VAL}$ is given on page 635 of You, Rao and Dick (2004).

HB estimation with matched FH model and estimated sampling variances

Inference model for the HB_MM_EV method

This method is based on the linking model $g(\theta_i) = \theta_i$ which leads to the Fay-Herriot matched model. This model is specified as

$$\begin{aligned} \text{Sampling Model: } \quad & \hat{\theta}_i^{DIR} = \theta_i + e_i \\ \text{Linking Model: } \quad & \theta_i = \mathbf{z}_i^T \boldsymbol{\beta} + b_i v_i \end{aligned}$$

where

$e_i \overset{id}{\sim} N(0, \psi_i)$	e_i are independently distributed as Normal random variables with mean 0 and unknown variance ψ_i
$v_i \overset{iid}{\sim} N(0, \sigma_v^2)$	v_i are independent and identically distributed as Normal random variables with mean 0 and unknown variance σ_v^2

In this method, we assume the variances $\boldsymbol{\psi} = \{\psi_i\}$ are unknown but we have corresponding estimates denoted by $\hat{\boldsymbol{\psi}} = \{\hat{\psi}_i\}$. The parameter estimates $\hat{\boldsymbol{\theta}}^{DIR} = \left\{\hat{\theta}_i^{DIR}\right\}$ are assumed to be obtained from the survey. We use the following Bayesian model in the development of the theory.

$$\begin{aligned} & \left[\, \hat{\theta}_i^{DIR} \,\middle|\, \theta_i \,\right] \sim N\left(\theta_i, \psi_i\right) \\ & \left[\, \theta_i \,\middle|\, \boldsymbol{\beta}, \sigma_v^2 \,\right] \sim N\left(\mathbf{z}_i^T \boldsymbol{\beta}, b_i^2 \sigma_v^2\right) \\ & \left[\, d_i\, \hat{\psi}_i \,\middle|\, \psi_i \,\right] \sim \psi_i\, \chi_{d_i}^2 \quad \text{where } d_i = (n_i - 1) \end{aligned}$$

As usual, quantities such as $\mathbf{z}_i$ and b_i are treated as known. The unknown parameters in this model are $\boldsymbol{\theta} = \{\theta_i\}$ and $\boldsymbol{\psi} = \{\psi_i\}$ over the areas as well as $\boldsymbol{\beta}$ and σ_v^2. The above Bayesian model includes a distribution for each θ_i. We provide prior distributions for the other parameters. We use the following prior distributions $\pi(\cdot)$ for $\boldsymbol{\beta}$, σ_v^2 and ψ_i.

$$\begin{aligned} & \pi(\boldsymbol{\beta}) \propto 1 \\ & \pi(\sigma_v^2) \propto (\sigma_v^2)^{-1/2} \\ & \pi(\psi_i) \propto (\psi_i)^{-1/2} \end{aligned}$$

The distribution $\pi(\boldsymbol{\beta}) \propto 1$ is known as a flat prior.

Methodology Notes:

- This Bayesian model is described in the specifications provided by Y. You (2014).
- A discussion on the use of different priors is given by Gelman (2006).

Full conditional distributions for the inference model of the HB_MM_EV method

Using the distributions of the Bayesian model and the Bayesian priors we arrive at the following full conditional distributions for the estimation of the unknown parameters. Details are given in the section titled Derivation of full conditional distributions for the HB models.

$$\left[\, \theta_i \mid \hat{\boldsymbol{\theta}}^{DIR}, \hat{\boldsymbol{\psi}}^{DIR}, \boldsymbol{\psi}, \boldsymbol{\beta}, \sigma_v^2 \,\right] \sim N\left(\gamma_i \, \hat{\theta}_i^{DIR} + (1-\gamma_i)\, \mathbf{z}_i^T \boldsymbol{\beta},\ \gamma_i \psi_i \right) \quad \text{for } i \in A_{VAL}$$

$$\text{where } \gamma_i = \frac{b_i^2\, \sigma_v^2}{b_i^2\, \sigma_v^2 + \psi_i}$$

$$\left[\, \boldsymbol{\beta} \mid \hat{\boldsymbol{\theta}}^{DIR}, \hat{\boldsymbol{\psi}}^{DIR}, \boldsymbol{\theta}, \boldsymbol{\psi}, \sigma_v^2 \,\right] \sim N\left(\left(\sum\nolimits_{i \in A_{VAL}} \frac{\mathbf{z}_i}{b_i} \frac{\mathbf{z}_i^T}{b_i} \right)^{-1} \left(\sum\nolimits_{i \in A_{VAL}} \frac{\mathbf{z}_i}{b_i} \frac{\theta_i}{b_i} \right),\ \sigma_v^2 \left(\sum\nolimits_{i \in A_{VAL}} \frac{\mathbf{z}_i}{b_i} \frac{\mathbf{z}_i^T}{b_i} \right)^{-1} \right)$$

$$\left[\, \sigma_v^2 \mid \hat{\boldsymbol{\theta}}^{DIR}, \hat{\boldsymbol{\psi}}^{DIR}, \boldsymbol{\theta}, \boldsymbol{\psi}, \boldsymbol{\beta} \,\right] \sim IG\left(\frac{m-1}{2},\ \frac{1}{2} \sum\nolimits_{i \in A_{VAL}} \left(\frac{\theta_i - \mathbf{z}_i^T \boldsymbol{\beta}}{b_i} \right)^2 \right)$$

$$\left[\, \psi_i \mid \hat{\boldsymbol{\theta}}^{DIR}, \hat{\boldsymbol{\psi}}^{DIR}, \boldsymbol{\theta}, \boldsymbol{\beta}, \sigma_v^2 \,\right] \sim IG\left(\frac{n_i - 1}{2},\ \frac{1}{2}\left((\hat{\theta}_i^{DIR} - \theta_i)^2 + d_i \, \hat{\psi}_i \right) \right) \quad \text{for } i \in A_{VAL}$$

Processing steps for the HB_MM_EV method

Step 0 Determine if the HB_MM_EV method can be applied based on the specified inputs. Continue only if it is possible to apply the method with the given specifications.

Step 1 Use Table 1 (on page 10) to partition the small areas into the 3 area groups (A_{EXC} , A_{VAL} and A_{NUL}) for the parameter.

Step 2 If $m < (p+2)$ then we cannot apply the HB_MM_EV method and we stop.
If $m \geq (p+2)$ continue by computing the following starting values for the Gibbs sampler.

$$\theta_i^{(0)} = \hat{\theta}_i^{DIR} \quad \text{for } i \in A_{VAL}$$

$$\boldsymbol{\beta}^{(0)} = \left(\sum\nolimits_{i \in A_{VAL}} \frac{\mathbf{z}_i}{b_i} \frac{\mathbf{z}_i^T}{b_i} \right)^{-1} \left(\sum\nolimits_{i \in A_{VAL}} \frac{\mathbf{z}_i}{b_i} \frac{\theta_i^{(0)}}{b_i} \right)$$

$$\sigma_v^{2(0)} \sim IG\left(\frac{m-1}{2},\ \frac{1}{2} \sum\nolimits_{i \in A_{VAL}} \left(\frac{\theta_i^{(0)} - \mathbf{z}_i^T \boldsymbol{\beta}^{(0)}}{b_i} \right)^2 \right)$$

$$\psi_i^{(0)} = \hat{\psi}_i \quad \text{for } i \in A_{VAL}$$

Process Note:

- To apply this process, we must have $m \geq (p+2)$ so we can compute a proper regression coefficient. This means that m, the number of areas in A_{VAL}, must be at least 2 more than p, the number of auxiliary variables in $\mathbf{z}_i$.

Methodology Notes:

- The value of $\sigma_v^{2(0)}$ is obtained from an Inverse Gamma distribution $IG(\alpha,\beta)$. To do this, we use the following relationship between the Gamma and Inverse Gamma distributions.

  If $x \sim Gamma(\alpha,\beta)$ with pdf $\dfrac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1}e^{-\beta x}$ for $x>0$

  (x is Gamma with iscale parameter β)

  Then $x^{-1} \sim IG(\alpha,\beta)$ with pdf $\dfrac{\beta^\alpha}{\Gamma(\alpha)} \dfrac{e^{-\frac{\beta}{x}}}{x^{\alpha+1}}$ for $x>0$

  (x^{-1} is Inverse Gamma with scale parameter β)

- Therefore, we obtain $\sigma_v^{2(0)}$ as follows.
  - First, we generate a random value from a Gamma distribution with shape parameter $\alpha = \dfrac{m-1}{2}$ and iscale (inverse scale) parameter
    $\beta = \dfrac{1}{2}\sum_{i\in A_{VAL}} \left(\dfrac{\theta_i^{(0)} - \boldsymbol{z}_i^T \boldsymbol{\beta}^{(0)}}{b_i}\right)^2$.
  - Then we take the inverse of this random value as the required value of $\sigma_v^{2(0)}$.

- Note that $\sigma_v^{2(0)}$ is a positive scalar value.

Implementation Notes:

- The function RANGAM in SAS can be used to generate a pseudo-random variable from a Gamma distribution with shape parameter *alpha* (α) and iscale (inverse scale) parameter *beta* (β) as follows.

$$x = \frac{1}{\beta} \times \text{RANGAM}\,(0, alpha)$$

- The required value of $\sigma_v^{2(0)}$ is then obtained as x^{-1} which is equivalent to $\dfrac{\beta}{\text{RANGAM}\,(0, alpha)}$.

Step 3 Beginning with the starting values $\theta_i^{(0)}$, $\boldsymbol{\beta}^{(0)}$, $\sigma_v^{2\,(0)}$ and $\psi_i^{(0)}$, iterate from $g=1$ to $g=G$, computing these quantities at each iteration g in the following order.

1. $\theta_i^{(g)} \sim N\left(\gamma_i^{(g-1)}\hat{\theta}_i^{DIR} + (1-\gamma_i^{(g-1)})\,\mathbf{z}_i^T\boldsymbol{\beta}^{(g-1)}, \gamma_i^{(g-1)}\,\psi_i^{(g-1)} \right)$ for $i \in A_{VAL}$

   where $\gamma_i^{(g-1)} = \dfrac{b_i^2\,\sigma_v^{2(g-1)}}{b_i^2\,\sigma_v^{2(g-1)} + \psi_i^{(g-1)}}$

2. $\boldsymbol{\beta}^{(g)} \sim N\left(\left(\sum_{i\in A_{VAL}} \dfrac{\mathbf{z}_i}{b_i}\dfrac{\mathbf{z}_i^T}{b_i} \right)^{-1} \left(\sum_{i\in A_{VAL}} \dfrac{\mathbf{z}_i}{b_i}\dfrac{\theta_i^{(g)}}{b_i} \right), \sigma_v^{2(g-1)} \left(\sum_{i\in A_{VAL}} \dfrac{\mathbf{z}_i}{b_i}\dfrac{\mathbf{z}_i^T}{b_i} \right)^{-1} \right)$

3. $\sigma_v^{2(g)} \sim IG\left(\dfrac{m-1}{2}, \dfrac{1}{2}\sum_{i\in A_{VAL}} \left(\dfrac{\theta_i^{(g)} - \mathbf{z}_i^T\boldsymbol{\beta}^{(g)}}{b_i} \right)^2 \right)$

4. $\psi_i^{(g)} \sim IG\left(\dfrac{n_i-1}{2}, \dfrac{1}{2}\left((\hat{\theta}_i^{DIR} - \theta_i^{(g)})^2 + d_i\,\hat{\psi}_i \right) \right)$

5. For any areas $i \in A_{NUL}$ compute $\theta_i^{(g)} = u_i$ where $u_i \sim N\left(\mathbf{z}_i^T\boldsymbol{\beta}^{(g)}, b_i^2\,\sigma_v^{2\,(g)} \right)$

Methodology Notes:

- The value of $\theta_i^{(g)}$ (for $i \in A_{VAL}$) is obtained by generating a random value from a univariate Normal distribution with mean $\gamma_i^{(g-1)}\hat{\theta}_i^{DIR} + (1-\gamma_i^{(g-1)})\,\mathbf{z}_i^T\boldsymbol{\beta}^{(g-1)}$ and variance $\gamma_i^{(g-1)}\,\psi_i^{(g-1)}$. Each $\theta_i^{(g)}$ is a scalar value and we generate m independent values corresponding to $i \in A_{VAL}$.
- The value of $\boldsymbol{\beta}^{(g)}$ is obtained by generating a random value from a multivariate Normal distribution with mean $\left(\sum_{i\in A_{VAL}} \mathbf{z}_i\,\mathbf{z}_i^T / b_i^2\right)^{-1} \left(\sum_{i\in A_{VAL}} \mathbf{z}_i\, g(\theta_i^{(g)}) / b_i^2\right)$ and variance-covariance matrix $\sigma_v^{2(g-1)}\left(\sum_{i\in A_{VAL}} \mathbf{z}_i\,\mathbf{z}_i^T / b_i^2\right)^{-1}$. Note that $\boldsymbol{\beta}^{(g)}$ is a $p\times 1$ vector.
- The values of $\sigma_v^{2(g)}$ and $\psi_i^{(g)}$ are obtained from the Inverse Gamma distribution as described earlier.
- The values $\theta_i^{(g)} = u_i$ where $u_i \sim N\left(\mathbf{z}_i^T\boldsymbol{\beta}^{(g)}, b_i^2\,\sigma_v^{2\,(g)} \right)$ for $i \in A_{NUL}$ are obtained from the linking model. These values are used to produce the HB estimates for $i \in A_{NUL}$.

Implementation Note:

- The quantity $\left(\sum_{i\in A_{VAL}} \mathbf{z}_i\,\mathbf{z}_i^T / b_i^2\right)^{-1}$ is constant over all iterations so we only need to compute it once.

Step 4 We discard the generated values for $g=1,2,\ldots,B$ and produce the following HB estimates for the small-areas based on the sample values for $g=B+1, B+2,\ldots,G$.

$$\hat{\theta}_i^{HB} = \begin{cases} \hat{\theta}_i^{DIR} & \text{for } i \in A_{EXC} \\ \dfrac{1}{G-B} \sum\limits_{g=B+1}^{G} \theta_i^{(g)} & \text{for } i \in A_{VAL} \\ \dfrac{1}{G-B} \sum\limits_{g=B+1}^{G} \theta_i^{(g)} & \text{for } i \in A_{NUL} \end{cases}$$

$$v\hat{a}r(\hat{\theta}_i^{HB}) = \begin{cases} \hat{\psi}_i & \text{for } i \in A_{EXC} \\ \dfrac{1}{G-B-1} \sum\limits_{g=B+1}^{G} (\,\theta_i^{(g)} - \hat{\theta}_i^{HB}\,)^2 & \text{for } i \in A_{VAL} \\ \dfrac{1}{G-B-1} \sum\limits_{g=B+1}^{G} (\,\theta_i^{(g)} - \hat{\theta}_i^{HB}\,)^2 & \text{for } i \in A_{NUL} \end{cases}$$

We also compute the following estimate of the regression coefficient and its estimated variance.

$$\hat{\boldsymbol{\beta}}^{HB} = \frac{1}{G-B} \sum_{g=B+1}^{G} \boldsymbol{\beta}^{(g)}$$

$$v\hat{a}r(\hat{\boldsymbol{\beta}}^{HB}) = \frac{1}{G-B-1} \sum_{g=B+1}^{G} (\,\boldsymbol{\beta}^{(g)} - \hat{\boldsymbol{\beta}}^{HB}\,)(\,\boldsymbol{\beta}^{(g)} - \hat{\boldsymbol{\beta}}^{HB}\,)^T$$

Methodology Note:

- The value of B is the length of the burn-in period. We do not use the values of $\theta_i^{(g)}$, $\boldsymbol{\beta}^{(g)}$, $\sigma_v^{2(g)}$ and $\psi_i^{(g)}$ in the burn-in period because they are less stable than the values obtained for $g > B$.

Process Notes:

- We can make this process similar to the other methods by defining B, δ and E as inputs and setting G as $G = B + \delta E = B + E$ (with $\delta = 1$).
- The default values are $B = 2000$, $\delta = 1$ and $E = 10000$ so $G = 12000$.

Implementation Notes:

- We do not need to store the values of $\theta_i^{(g)}$, $\boldsymbol{\beta}^{(g)}$, $\sigma_v^{2(g)}$ and $\psi_i^{(g)}$ for $g = 1, 2 \ldots B$. We just need to generate them from one iteration to another.
- However, we need to store the values of $\theta_i^{(g)}$ and $\boldsymbol{\beta}^{(g)}$ for $g > B$ to calculate $\hat{\theta}_i^{HB}$, $v\hat{a}r(\hat{\theta}_i^{HB})$, $\hat{\boldsymbol{\beta}}^{HB}$ and $v\hat{a}r(\hat{\boldsymbol{\beta}}^{HB})$.
- If we are benchmarking the estimates $\hat{\theta}_i^{HB}$ in step 5 then we also need to store $\sigma_v^{2(g)}$ and $\psi_i^{(g)}$ for $g > B$ to compute $\hat{\sigma}_v^{2\,HB} = \frac{1}{G-B}\sum_{g=B+1}^{G} \sigma_v^{2(g)}$ and $\hat{\psi}_i^{HB} = \frac{1}{G-B}\sum_{g=B+1}^{G} \psi_i^{(g)}$, the HB estimates of σ_v^2 and ψ_i respectively.

Step 5 If benchmarking is required and the parameter corresponds to a mean or a total then we attempt (see Process Note below) to apply the method of difference adjustment to adjust the estimates $\hat{\theta}_i^{HB}$ and produce benchmark estimates $\hat{\theta}_i^{HB,b}$ and variance estimates $v\hat{a}r(\hat{\theta}_i^{HB,b})$ as follows:

$$\hat{\theta}_i^{HB,b} = \begin{cases} \hat{\theta}_i^{HB} & \text{for } i \in A_{EXC} \\ \hat{\theta}_i^{HB} + \alpha_i(\hat{\theta}^* - \sum_{d \in A} \omega_d\, \hat{\theta}_d^{HB}) & \text{for } i \in A_{VAL} \\ \hat{\theta}_i^{HB} & \text{for } i \in A_{NUL} \end{cases}$$

$$\text{with } \alpha_i = \frac{\omega_i\,(\hat{\psi}_i^{HB} + b_i^2\,\hat{\sigma}_v^{2\,HB})}{\sum_{i \in A_{VAL}} \omega_i^2\,(\hat{\psi}_i^{HB} + b_i^2\,\hat{\sigma}_v^{2\,HB})} \quad \text{for } i \in A_{VAL}$$

$$\hat{\sigma}_v^{2\,HB} = \frac{1}{G-B}\sum_{g=B+1}^{G} \sigma_v^{2(g)}$$

$$\text{and } \hat{\psi}_i^{HB} = \frac{1}{G-B}\sum_{g=B+1}^{G} \psi_i^{(g)}$$

$$v\hat{a}r(\hat{\theta}_i^{HB,b}) = \begin{cases} v\hat{a}r(\hat{\theta}_i^{HB}) & \text{for } i \in A_{EXC} \\ v\hat{a}r(\hat{\theta}_i^{HB}) + (\,\hat{\theta}_i^{HB,b} - \hat{\theta}_i^{HB}\,)^2 & \text{for } i \in A_{VAL} \\ v\hat{a}r(\hat{\theta}_i^{HB}) & \text{for } i \in A_{NUL} \end{cases}$$

where ω_i is defined in the table below

parameter type	**definition of ω_i for $i \in A$**
TOTAL	$\omega_i = 1$
MEAN	$\omega_i = N_i / N$ where $N = \sum_{i \in A} N_i$

Process Notes:

- The term $\hat{\theta}^{*}$ is either $\hat{\theta}^{DIR}$ (a population value provided by the user) or a value calculated from the parameter estimates as $\sum_{i \in A} \omega_i \hat{\theta}_i^{DIR}$ with $\omega_i = 1$ for a total and $\omega_i = N_i / N$ for a mean. Either way $\hat{\theta}^{*}$ is treated as a constant in the above calculation.
- If the user provides $\hat{\theta}^{DIR}$ then $\hat{\theta}^{*} = \hat{\theta}^{DIR}$ and this value is used for benchmarking.
- If the user does not provide $\hat{\theta}^{DIR}$ then we attempt to calculate $\hat{\theta}^{*}$ as follows.
  - If the parameter is a total and $\hat{\theta}_i^{DIR} \neq \cdot$ for all $i \in A$ then we compute $\hat{\theta}^{*}$ as $\sum_{i \in A} \hat{\theta}_i^{DIR}$.
  - If the parameter is a mean and we have $\hat{\theta}_i^{DIR} \neq \cdot$ and N_i known for all $i \in A$ then we compute $\hat{\theta}^{*}$ as $\sum_{i \in A} N_i \hat{\theta}_i^{DIR} \big/ \sum_{i \in A} N_i$.
  - Otherwise, benchmarking cannot be applied.

Methodology Notes:

- We only adjust the values of $\hat{\theta}_i^{HB}$ for $i \in A_{VAL}$. We do not adjust $\hat{\theta}_i^{HB}$ for $i \in A_{EXC}$ or $i \in A_{NUL}$.
- Note that by definition $\hat{\theta}_i^{HB,b} = \hat{\theta}_i^{HB} = \hat{\theta}_i^{DIR}$ and $v\hat{a}r(\hat{\theta}_i^{HB,b}) = v\hat{a}r(\hat{\theta}_i^{HB}) = \psi_i$ for $i \in A_{EXC}$.
- The reference to the method of benchmarking by difference adjustment is the November 2007 notes from Fuller.
- The formula for the benchmark variance $v\hat{a}r(\hat{\theta}_i^{HB,b})$ for $i \in A_{VAL}$ is given on page 635 of You, Rao and Dick (2004).

HB estimation with unmatched log-linear model and estimated sampling variances

Inference model for the HB_UM_LL_EV method

This method uses the linking model $log(\theta_i) = \mathbf{z}_i^T \boldsymbol{\beta} + b_i v_i$ which leads to an unmatched log-linear model. The model is specified as

$$\begin{aligned} &\text{Sampling Model :} \quad \hat{\theta}_i^{DIR} = \theta_i + e_i \\ &\text{Linking Model :} \quad log(\theta_i) = \mathbf{z}_i^T \boldsymbol{\beta} + b_i v_i \end{aligned}$$

where

$e_i \overset{id}{\sim} N(0, \psi_i)$ — e_i are independently distributed as Normal random variables with mean 0 and unknown variance ψ_i

$v_i \overset{iid}{\sim} N(0, \sigma_v^2)$ — v_i are independent and identically distributed as Normal random variables with mean 0 and unknown variance σ_v^2

In this method, we assume the variances ψ_i are unknown but we have corresponding estimates denoted by $\hat{\psi}_i$. The $\left\{ \hat{\theta}_i^{DIR} \right\}$ are assumed to be estimated from the survey. We use the following Bayesian model in the development of the theory.

$$\begin{aligned} &\left[\hat{\theta}_i^{DIR} \,\middle|\, \theta_i \right] \sim N\left(\theta_i, \psi_i \right) \\ &\left[\log(\theta_i) \,|\, \boldsymbol{\beta}, \sigma_v^2 \right] \sim N\left(z_i^T \boldsymbol{\beta}, b_i^2 \, \sigma_v^2 \right) \\ &\left[d_i \, \hat{\psi}_i \,|\, \psi_i \right] \sim \psi_i \, \chi_{d_i}^2 \quad \text{where } d_i = (n_i - 1) \end{aligned}$$

The unknown parameters in this model are $\boldsymbol{\theta} = \left\{ \theta_i \right\}$ and $\boldsymbol{\psi} = \left\{ \psi_i \right\}$ over the areas as well as $\boldsymbol{\beta}$ and σ_v^2. The above Bayesian model provides inference on θ_i through the distribution for $log(\theta_i)$. We provide prior distributions for the other parameters. We use the following prior distributions $\pi(\cdot)$ for ψ_i, $\boldsymbol{\beta}$ and σ_v^2.

$$\begin{aligned} &\pi(\boldsymbol{\beta}) \propto 1 \\ &\pi(\sigma_v^2) \propto (\sigma_v^2)^{-1/2} \\ &\pi(\psi_i) \propto (\psi_i)^{-1/2} \end{aligned}$$

Methodology Notes:

- This Bayesian model is described in the specifications provided by Y. You (2014).
- The parameter θ_i has a Log-Normal distribution. By definition, θ_i is Log-Normal if $log(\theta_i)$ is Normal.
- The probability density function of θ_i is given by $\frac{1}{\theta_i}\frac{1}{\sqrt{2\pi b_i^2 \sigma_v^2}} exp\left[-\frac{(\,log(\theta_i) - \boldsymbol{z}_i^T \boldsymbol{\beta}\,)^2}{2\, b_i^2\, \sigma_v^2}\right]$.
- A discussion on the use of different priors is given by Gelman (2006).

Full conditional distributions for the inference model of the HB_UM_LL_EV method

Using the distributions of the Bayesian model and the Bayesian priors we arrive at the following full conditional distributions for the estimation of the unknown parameters. The way these are derived is shown in the section titled Derivation of full conditional distributions for the HB models.

$$\left[\,\theta_i \mid \hat{\boldsymbol{\theta}}^{DIR}, \hat{\boldsymbol{\psi}}^{DIR}, \boldsymbol{\psi}, \boldsymbol{\beta}, \sigma_v^2\,\right] \propto\ exp\left[-\frac{(\hat{\theta}_i^{DIR} - \theta_i)^2}{2\psi_i}\right]\frac{1}{\theta_i}\,exp\left[-\frac{(\,log(\theta_i) - \boldsymbol{z}_i^T \boldsymbol{\beta}\,)^2}{2\, b_i^2\, \sigma_v^2}\right] \quad \text{for } \ i \in A_{VAL}$$

$$\left[\,\boldsymbol{\beta} \mid \hat{\boldsymbol{\theta}}^{DIR}, \hat{\boldsymbol{\psi}}^{DIR}, \boldsymbol{\theta}, \boldsymbol{\psi}, \sigma_v^2\,\right] \sim N\left(\left(\sum_{i \in A_{VAL}} \frac{\boldsymbol{z}}{b_i}\frac{\boldsymbol{z}_i^T}{b_i}\right)^{-1}\left(\sum_{i \in A_{VAL}} \frac{\boldsymbol{z}_i}{b_i}\frac{log\,(\theta_i)}{b_i}\right),\ \sigma_v^2\left(\sum_{i \in A_{VAL}} \frac{\boldsymbol{z}_i}{b_i}\frac{\boldsymbol{z}_i^T}{b_i}\right)^{-1}\right)$$

$$\left[\,\sigma_v^2 \mid \hat{\boldsymbol{\theta}}^{DIR}, \hat{\boldsymbol{\psi}}^{DIR}, \boldsymbol{\theta}, \boldsymbol{\psi}, \boldsymbol{\beta}\,\right] \sim IG\left(\frac{m-1}{2}, \frac{1}{2}\sum_{i \in A_{VAL}}\left(\frac{log(\theta_i) - \boldsymbol{z}_i^T\boldsymbol{\beta}}{b_i}\right)^2\right)$$

$$\left[\,\psi_i \mid \hat{\boldsymbol{\theta}}^{DIR}, \hat{\boldsymbol{\psi}}^{DIR}, \boldsymbol{\theta}, \boldsymbol{\beta}, \sigma_v^2\,\right] \sim IG\left(\frac{n_i - 1}{2}, \frac{1}{2}\left(\,(\hat{\theta}_i^{DIR} - \theta_i)^2 + d_i\,\hat{\psi}_i\,\right)\right) \quad \text{for } \ i \in A_{VAL}$$

Processing steps for the HB_UM_LL_EV method

Step 0 Determine if the HB_UM_LL_EV method can be applied based on the specified inputs. Continue only if it is possible to apply the method with the given specifications.

Step 1 Use Table 1 (on page 10) to partition the small areas into the 3 area groups (A_{EXC} , A_{VAL} and A_{NUL}) for the parameter.

Step 2 If $m < (p+2)$ then we cannot apply the HB_UM_LL_EV method of estimation and we stop. If $m \geq (p+2)$ continue by computing the following starting values for the Gibbs sampler.

$$\theta_i^{(0)} = \hat{\theta}_i^{DIR} \quad \text{for } i \in A_{VAL}$$

$$\boldsymbol{\beta}^{(0)} = \left(\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i}{b_i} \frac{\mathbf{z}_i^T}{b_i} \right)^{-1} \left(\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i}{b_i} \frac{log\,(\theta_i^{(0)})}{b_i} \right)$$

$$\sigma_v^{2(0)} \sim IG\left(\frac{m-1}{2}, \frac{1}{2} \sum_{i \in A_{VAL}} \left(\frac{log\,(\theta_i^{(0)}) - \mathbf{z}_i^T \boldsymbol{\beta}^{(0)}}{b_i} \right)^2 \right)$$

$$\psi_i^{(0)} = \hat{\psi}_i \quad \text{for } i \in A_{VAL}$$

Process Notes:

- To apply this process, we must have $m \geq (p+2)$ so we can compute a proper regression coefficient. This means that m, the number of areas in A_{VAL}, must be at least 2 more than p, the number of auxiliary variables in $\mathbf{z}_i$.
- There is no problem calculating $log\,(\theta_i^{(0)})$ in $\boldsymbol{\beta}^{(0)}$ and $\sigma_v^{2(0)}$ because $log\,(\theta_i^{(0)}) = log\,(\hat{\theta}_i^{DIR})$ and $\hat{\theta}_i^{DIR} > 0$ for $i \in A_{VAL}$.
- We can also calculate $\boldsymbol{\beta}^{(g)}$ and $\sigma_v^{2(g)}$ in subsequent iterations ($g = 1, \ldots G$) because the generated values of $\theta_i^{(g)}$ in step 3 are always positive since they come from a Log-Normal distribution.

Methodology Notes:

- The value of $\sigma_v^{2(0)}$ is obtained from an Inverse Gamma distribution $IG(\alpha,\beta)$. To do this, we use the following relationship between the Gamma and Inverse Gamma distributions.

  If $x \sim Gamma(\alpha,\beta)$ with pdf $\dfrac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1}e^{-\beta x}$ for $x > 0$

  (x is Gamma with iscale parameter β)

  Then $x^{-1} \sim IG(\alpha,\beta)$ with pdf $\dfrac{\beta^\alpha}{\Gamma(\alpha)} \dfrac{e^{-\frac{\beta}{x}}}{x^{\alpha+1}}$ for $x > 0$

  (x^{-1} is Inverse Gamma with scale parameter β)

- Therefore, we obtain $\sigma_v^{2(0)}$ as follows.
  - First, we generate a random value from a Gamma distribution with shape parameter $\alpha = \dfrac{m-1}{2}$ and iscale (inverse scale) parameter $\beta = \dfrac{1}{2}\sum_{i \in A_{VAL}} \left(\dfrac{log(\theta_i^{(0)}) - \mathbf{z}_i^T \boldsymbol{\beta}^{(0)}}{b_i} \right)^2$.
  - Then we take the inverse of this random value as the required value of $\sigma_v^{2(0)}$.

- Note that $\sigma_v^{2(0)}$ is a positive scalar value.

Implementation Notes:

- The function RANGAM in SAS can be used to generate a pseudo-random variable from a Gamma distribution with shape parameter *alpha* (α) and iscale (inverse scale) parameter *beta* (β) as follows.

$$x = \frac{1}{\beta} \times \text{RANGAM}(0, alpha)$$

- The required value of $\sigma_v^{2(0)}$ is then obtained as x^{-1} which is equivalent to $\dfrac{\beta}{\text{RANGAM}(0, alpha)}$.

Step 3 Beginning with the starting values $\theta_i^{(0)}$, $\boldsymbol{\beta}^{(0)}$, $\sigma_v^{2\,(0)}$ and $\psi_i^{(0)}$, iterate from $g=1$ to $g=G$, computing these quantities at each iteration g in the following order.

1. $\theta_i^{(g)} \sim log\, N\left(\mathbf{z}_i^T \boldsymbol{\beta}^{(g-1)}, b_i^2\, \sigma_v^{2(g-1)} \right)$ for $i \in A_{VAL}$

2. $\alpha\,(\theta_i^{(g-1)}, \theta_i^{(g)}) = min\left\{ \dfrac{h(\theta_i^{(g)})}{h(\theta_i^{(g-1)})},\ 1 \right\}$ for $i \in A_{VAL}$

   where $h(\theta_i) = exp\left[-\dfrac{(\hat{\theta}_i^{DIR} - \theta_i)^2}{2\,\psi_i} \right]$

3. If $\xi < \alpha\,(\theta_i^{(g-1)}, \theta_i^{(g)})$ then $\theta_i^{(g)} = \theta_i^{(g)}$, else $\theta_i^{(g)} = \theta_i^{(g-1)}$ for $i \in A_{VAL}$

   where $\xi \sim U\left(0,1\right)$ is a Uniform distribution on the (0,1) interval

4. $\boldsymbol{\beta}^{(g)} \sim N\left(\left(\sum_{i\in A_{VAL}} \dfrac{\mathbf{z}_i}{b_i} \dfrac{\mathbf{z}_i^T}{b_i} \right)^{-1} \left(\sum_{i\in A_{VAL}} \dfrac{\mathbf{z}_i}{b_i} \dfrac{log\,(\theta_i^{(g)})}{b_i} \right),\ \sigma_v^{2(g-1)} \left(\sum_{i\in A_{VAL}} \dfrac{\mathbf{z}_i}{b_i} \dfrac{\mathbf{z}_i^T}{b_i} \right)^{-1} \right)$

5. $\sigma_v^{2(g)} \sim IG\left(\dfrac{m-1}{2}, \dfrac{1}{2} \sum_{i\in A_{VAL}} \left(\dfrac{log\,(\theta_i^{(g)}) - \mathbf{z}_i^T \boldsymbol{\beta}^{(g)}}{b_i} \right)^2 \right)$

6. $\psi_i^{(g)} \sim IG\left(\dfrac{n_i - 1}{2}, \dfrac{1}{2}\left((\hat{\theta}_i^{DIR} - \theta_i^{(g)})^2 + d_i\, \hat{\psi}_i \right) \right)$

7. For any areas $i \in A_{NUL}$ compute $\theta_i^{(g)} = exp(u_i)$ where $u_i \sim N\left(\mathbf{z}_i^T \boldsymbol{\beta}^{(g)}, b_i^2\, \sigma_v^{2(g)} \right)$

Methodology Notes:

- Here we use $log\, N\left(\mathbf{z}_i^T \boldsymbol{\beta}^{(g-1)}, b_i^2\, \sigma_v^{2(g-1)} \right)$ as our proposal distribution in the Metropolis-Hastings algorithm.
- The value of $\theta_i^{(g)}$ (for $i \in A_{VAL}$) is obtained by generating a random value from a univariate Log-Normal distribution with mean $\mathbf{z}_i^T \boldsymbol{\beta}^{(g-1)}$ and variance $b_i^2\, \sigma_v^{2(g-1)}$ (proposal step) . Each $\theta_i^{(g)}$ is a scalar value and we generate m independent values corresponding to $i \in A_{VAL}$.
- To obtain a value $\theta_i^{(g)}$ from a Log-Normal distribution we first generate a random value u_i from a Normal distribution with mean $\mathbf{z}_i^T \boldsymbol{\beta}^{(g-1)}$ and variance $b_i^2\, \sigma_v^{2(g-1)}$ and then take $\theta_i^{(g)} = exp(u_i)$.
- The quantity $\alpha(\theta_i^{(g-1)}, \theta_i^{(g)})$ is known as the acceptance probability. We generate a Uniform random variable ξ on the interval (0,1). If $\xi < \alpha(\theta_i^{(g-1)}, \theta_i^{(g)})$ then we accept the current value of $\theta_i^{(g)}$, else we reset it to the previous value ($\theta_i^{(g)} = \theta_i^{(g-1)}$) (acceptance step).
- The value of $\boldsymbol{\beta}^{(g)}$ is obtained by generating a random value from a multivariate Normal distribution with mean $\left(\sum_{i \in A_{VAL}} \mathbf{z}_i\, \mathbf{z}_i^T / b_i^2\right)^{-1} \left(\sum_{i \in A_{VAL}} \mathbf{z}_i\, log(\theta_i^{(g)}) / b_i^2\right)$ and variance-covariance matrix $\sigma_v^{2(g-1)} \left(\sum_{i \in A_{VAL}} \mathbf{z}_i\, \mathbf{z}_i^T / b_i^2\right)^{-1}$. Note that $\boldsymbol{\beta}^{(g)}$ is a $p \times 1$ vector.
- The values of $\sigma_v^{2(g)}$ and $\psi_i^{(g)}$ are obtained from the Inverse Gamma distribution as described earlier.
- The values $\theta_i^{(g)} = exp(\, u_i\,)$ where $u_i \sim N\left(\mathbf{z}_i^T \boldsymbol{\beta}^{(g)}, b_i^2\, \sigma_v^{2(g)} \right)$ for $i \in A_{NUL}$ are obtained from the linking model. These values are used to produce the HB estimates for $i \in A_{NUL}$.

Implementation Note:

- The quantity $\left(\sum_{i \in A_{VAL}} \mathbf{z}_i\, \mathbf{z}_i^T / b_i^2\right)^{-1}$ is constant over all iterations so we only need to compute it once.

Step 4 We discard the generated values for $g = 1, 2, \ldots, B$ where $B < G$. For the remaining samples, we define the index set $E_\delta = \{ B+\delta, B+2\delta, B+3\delta, \ldots\ , G \}$ of size $E = (G-B)\ / \delta$, by selecting every δ^{th} sample, where $4 \le \delta \le 10$. We then use the samples in this set to produce the following HB estimates of the small areas.

$$\hat{\theta}_i^{HB} = \begin{cases} \hat{\theta}_i^{DIR} & \text{for } i \in A_{EXC} \\ \dfrac{1}{E} \sum\limits_{g \in E_\delta} \theta_i^{(g)} & \text{for } i \in A_{VAL} \\ \dfrac{1}{E} \sum\limits_{g \in E_\delta} \theta_i^{(g)} & \text{for } i \in A_{NUL} \end{cases}$$

$$v\hat{a}r(\hat{\theta}_i^{HB}) = \begin{cases} \hat{\psi}_i & \text{for } i \in A_{EXC} \\ \dfrac{1}{E-1} \sum\limits_{g \in E_\delta} (\theta_i^{(g)} - \hat{\theta}_i^{HB})^2 & \text{for } i \in A_{VAL} \\ \dfrac{1}{E-1} \sum\limits_{g \in E_\delta} (\theta_i^{(g)} - \hat{\theta}_i^{HB})^2 & \text{for } i \in A_{NUL} \end{cases}$$

We also compute the following estimate of the regression coefficient and its estimated variance.

$$\hat{\boldsymbol{\beta}}^{HB} = \frac{1}{G-B} \sum_{g \in E_\delta} \boldsymbol{\beta}^{(g)}$$

$$v\hat{a}r(\hat{\boldsymbol{\beta}}^{HB}) = \frac{1}{G-B-1} \sum_{g \in E_\delta} (\boldsymbol{\beta}^{(g)} - \hat{\boldsymbol{\beta}}^{HB})(\boldsymbol{\beta}^{(g)} - \hat{\boldsymbol{\beta}}^{HB})^T$$

Methodology Note:

- The value of B is the length of the burn-in period. We do not use the values of $\theta_i^{(g)}$, $\boldsymbol{\beta}^{(g)}$, $\sigma_v^{2(g)}$ and $\psi_i^{(g)}$ in the burn-in period because they are less stable than the values obtained for $g > B$.

Process Notes:

- Since $E = (G-B)/\delta$ must be divisible by δ, it is better to define B, δ and E as inputs and set G as $G = B + \delta E$.
- The default values are $B = 2000$, $\delta = 4$ and $E = 10000$ so $G = 42000$.

Implementation Notes:

- We do not need to store the values of $\theta_i^{(g)}$, $\boldsymbol{\beta}^{(g)}$, $\sigma_v^{2(g)}$ and $\psi_i^{(g)}$ for $g = 1, 2 \ldots B$. We just need to generate them from one iteration to another.
- However, we need to store the values of $\theta_i^{(g)}$ and $\boldsymbol{\beta}^{(g)}$ for $g > B$ to calculate $\hat{\theta}_i^{HB}$, $v\hat{a}r(\hat{\theta}_i^{HB})$, $\hat{\boldsymbol{\beta}}^{HB}$ and $v\hat{a}r(\hat{\boldsymbol{\beta}}^{HB})$.
- If we are benchmarking the estimates $\hat{\theta}_i^{HB}$ in step 5 then we also need to store $\sigma_v^{2(g)}$ and $\psi_i^{(g)}$ for $g > B$ to compute $\hat{\sigma}_v^{2\,HB} = \frac{1}{E} \sum_{g \in E_\delta} \sigma_v^{2(g)}$ and $\hat{\psi}_i^{HB} = \frac{1}{E} \sum_{g \in E_\delta} \psi_i^{(g)}$, the HB estimates of σ_v^2 and ψ_i respectively.

Step 5 If benchmarking is required and the parameter corresponds to a mean or a total then we attempt (see Process Note below) to apply the method of difference adjustment to adjust the estimates $\hat{\theta}_i^{HB}$ and produce benchmark estimates $\hat{\theta}_i^{HB,b}$ and variance estimates $v\hat{a}r(\hat{\theta}_i^{HB,b})$ as follows:

$$\hat{\theta}_i^{HB,b} = \begin{cases} \hat{\theta}_i^{HB} & \text{for } i \in A_{EXC} \\ \hat{\theta}_i^{HB} + \alpha_i (\hat{\theta}^* - \sum_{d \in A} \omega_d \, \hat{\theta}_d^{HB}) & \text{for } i \in A_{VAL} \\ \hat{\theta}_i^{HB} & \text{for } i \in A_{NUL} \end{cases}$$

$$\text{with } \alpha_i = \frac{\omega_i \, (\hat{\psi}_i^{HB} + b_i^2 \, \hat{\sigma}_v^{2\,HB})}{\sum_{i \in A_{VAL}} \omega_i^2 \, (\hat{\psi}_i^{HB} + b_i^2 \, \hat{\sigma}_v^{2\,HB})} \qquad \text{for } i \in A_{VAL}$$

$$\text{and } \hat{\sigma}_v^{2\,HB} = \frac{1}{E} \sum_{g \in E_\delta} \sigma_v^{2(g)}$$

$$\text{and } \hat{\psi}_i^{HB} = \frac{1}{E} \sum_{g \in E_\delta} \psi_i^{(g)}$$

$$v\hat{a}r(\hat{\theta}_i^{HB,b}) = \begin{cases} v\hat{a}r(\hat{\theta}_i^{HB}) & \text{for } i \in A_{EXC} \\ v\hat{a}r(\hat{\theta}_i^{HB}) + (\hat{\theta}_i^{HB,b} - \hat{\theta}_i^{HB})^2 & \text{for } i \in A_{VAL} \\ v\hat{a}r(\hat{\theta}_i^{HB}) & \text{for } i \in A_{NUL} \end{cases}$$

where ω_i is defined in the table below

parameter type	**definition of ω_i for $i \in A$**
TOTAL	$\omega_i = 1$
MEAN	$\omega_i = N_i / N$ where $N = \sum_{i \in A} N_i$

Process Notes:

- The term $\hat{\theta}^*$ is either $\hat{\theta}^{DIR}$ (a population value provided by the user) or a value calculated from the parameter estimates as $\sum_{i \in A} \omega_i \hat{\theta}_i^{DIR}$ with $\omega_i = 1$ for a total and $\omega_i = N_i / N$ for a mean. Either way $\hat{\theta}^*$ is treated as a constant in the above calculation.
- If the user provides $\hat{\theta}^{DIR}$ then $\hat{\theta}^* = \hat{\theta}^{DIR}$ and this value is used for benchmarking.
- If the user does not provide $\hat{\theta}^{DIR}$ then we attempt to calculate $\hat{\theta}^*$ as follows.
  - If the parameter is a total and $\hat{\theta}_i^{DIR} \neq \cdot$ for all $i \in A$ then we compute $\hat{\theta}^*$ as $\sum_{i \in A} \hat{\theta}_i^{DIR}$.
  - If the parameter is a mean and we have $\hat{\theta}_i^{DIR} \neq \cdot$ and N_i known for all $i \in A$ then we compute $\hat{\theta}^*$ as $\sum_{i \in A} N_i \hat{\theta}_i^{DIR} \Big/ \sum_{i \in A} N_i$.
  - Otherwise, benchmarking cannot be applied.

Methodology Notes:

- We only adjust the values of $\hat{\theta}_i^{HB}$ for $i \in A_{VAL}$. We do not adjust $\hat{\theta}_i^{HB}$ for $i \in A_{EXC}$ or $i \in A_{NUL}$.
- Note that by definition $\hat{\theta}_i^{HB,b} = \hat{\theta}_i^{HB} = \hat{\theta}_i^{DIR}$ and $v\hat{a}r(\hat{\theta}_i^{HB,b}) = v\hat{a}r(\hat{\theta}_i^{HB}) = \psi_i$ for $i \in A_{EXC}$.
- The reference to the method of benchmarking by difference adjustment is the November 2007 notes from Fuller.
- The formula for the benchmark variance $v\hat{a}r(\hat{\theta}_i^{HB,b})$ for $i \in A_{VAL}$ is given on page 635 of You, Rao and Dick (2004).

Benchmarking

Benchmarking is used to ensure that the resulting small-area estimates add up to a specified population value $\hat{\theta}^{DIR}$ or the weighted sum of the direct estimates $\sum_{i \in A} \omega_i \hat{\theta}_i^{DIR}$ where $\omega_i = 1$ for a total and $\omega_i = N_i / \sum_{i \in A} N_i$ for a mean. Benchmarking can only be applied when the parameter of interest is a total or a mean. No benchmarking is possible for other parameters.

Logic for benchmarking of the HB methods

The following table shows the logic used in determining the method for benchmarking when this option is specified by the user.

Table 2 Logic for benchmarking.

parameter type	conditions		benchmark method	benchmark result
TOTAL	$\hat{\theta}^{DIR}$ is provided		difference adjustment	$\sum_{i \in A} \hat{\theta}_i^{HB} = \hat{\theta}^{DIR}$
	$\hat{\theta}^{DIR}$ is not provided	$\hat{\theta}_i^{DIR}$ known for all $i \in A$	difference adjustment	$\sum_{i \in A} \hat{\theta}_i^{HB} = \sum_{i \in A} \hat{\theta}_i^{DIR}$
		$\hat{\theta}_i^{DIR}$ not known for all $i \in A$	none	none
MEAN	$\hat{\theta}^{DIR}$ is provided	N_i known for all $i \in A$	difference adjustment	$\sum_{i \in A} \omega_i \hat{\theta}_i^{HB} = \hat{\theta}^{DIR}$ with $\omega_i = N_i / \sum_{i \in A} N_i$
		N_i not known for all $i \in A$	none	none
	$\hat{\theta}^{DIR}$ is not provided	N_i and $\hat{\theta}_i^{DIR}$ known for all $i \in A$	difference adjustment	$\sum_{i \in A} \omega_i \hat{\theta}_i^{HB} = \sum_{i \in A} \omega_i \hat{\theta}_i^{DIR}$ with $\omega_i = N_i / \sum_{i \in A} N_i$
		N_i and $\hat{\theta}_i^{DIR}$ not known for all $i \in A$	none	none
Other	none		none	none

If benchmarking is required for a total or a mean, then we determine if it is possible by checking the conditions in the above table. The benchmark result is shown in the last column of the table.

Notes on benchmarking for the HB methods

The following points should be kept in mind regarding benchmarking. They provide a summary of the above logic and important notes on the required inputs.

- Benchmarking can only be applied when the parameter of interest is a total or a mean.
- If you provide a value $\hat{\theta}^{DIR}$ then this quantity must represent the population total or the population mean across all areas including the exclusion areas.

- If the parameter is a total then $\hat{\theta}^{DIR}$ must correspond to the population total.
- If the parameter is a mean then $\hat{\theta}^{DIR}$ must correspond to the population mean.

- If $\hat{\theta}^{DIR}$ is provided then we attempt to benchmark the estimates to this value using the method of difference adjustment.
- If $\hat{\theta}^{DIR}$ is not provided then we attempt to benchmark the estimates to the weighted sum of the direct estimates $\sum_{i \in A} \omega_i \hat{\theta}_i^{DIR}$ where $\omega_i = 1$ for a total and $\omega_i = N_i / \sum_{i \in A} N_i$ for a mean. If this is not possible then no benchmarking is done.
- To benchmark a mean, we must know $\hat{\theta}^{DIR}$ and N_i for all $i \in A$ so we can produce $\omega_i = N_i / \sum_{i \in A} N_i$ for $i \in A$ and $\sum_{i \in A} \omega_i \hat{\theta}_i^{DIR}$.
- To benchmark a total, we must know $\hat{\theta}^{DIR}$ for all $i \in A$ so we can produce $\sum_{i \in A} \hat{\theta}_i^{DIR}$. We do not need the N_i values to benchmark totals.

The proof of the benchmark result for the method of difference adjustment is shown below.

Proof of benchmarking based on difference adjustment

Under the method of difference adjustment, the benchmarked estimates $\hat{\theta}_i^{HB,b}$ are defined as follows.

$$\hat{\theta}_i^{HB,b} = \begin{cases} \hat{\theta}_i^{HB} & \text{for } i \in A_{EXC} \\ \hat{\theta}_i^{HB} + \alpha_i (\hat{\theta}^* - \sum_{d \in A} \omega_d \hat{\theta}_d^{HB}) & \text{for } i \in A_{VAL} \\ \hat{\theta}_i^{HB} & \text{for } i \in A_{NUL} \end{cases}$$

$$\text{where } \alpha_i = \frac{\omega_i (\dot{\psi}_i + b_i^2 \hat{\sigma}_v^{2HB})}{\sum_{d \in A_{VAL}} \omega_d^2 (\dot{\psi}_d + b_d^2 \hat{\sigma}_v^{2HB})} \quad \text{for } i \in A_{VAL}$$

The term $\hat{\theta}^*$ is either $\hat{\theta}^{DIR}$ (a population value provided by the user) or a value calculated from the parameter estimates as $\sum_{i \in A} \omega_i \hat{\theta}_i^{DIR}$ where $\omega_i = 1$ for a total and $\omega_i = N_i / \sum_{i \in A} N_i$ for a mean.

The $\dot{\psi}_i$ represent either the known values ψ_i (for methods HB_MM_KV, HB_UM_LL_KV and HB_UM_LCU_KV) or the estimated values $\hat{\psi}_i^{HB}$ (for methods HB_MM_EV and HB_UM_LL_EV).

Prove: We need to show that under the above definition, $\sum_{i\in A}\omega_i\,\hat{\theta}_i^{HB,b}=\hat{\theta}^*$.

Proof: Substituting the value of $\hat{\theta}_i^{HB,b}$ into the left-hand side of the equation leads to the following derivation.

$$
\begin{aligned}
&\sum_{i\in A}\omega_i\,\hat{\theta}_i^{HB,b}\\
&=\sum_{i\in A_{EXC}}\omega_i\,\hat{\theta}_i^{HB,b}+\sum_{i\in A_{VAL}}\omega_i\,\hat{\theta}_i^{HB,b}+\sum_{i\in A_{NUL}}\omega_i\,\hat{\theta}_i^{HB,b}\\
&=\sum_{i\in A_{EXC}}\omega_i\,\hat{\theta}_i^{HB}+\sum_{i\in A_{VAL}}\omega_i\left(\hat{\theta}_i^{HB}+\alpha_i(\hat{\theta}^*-\sum_{d\in A}\omega_d\,\hat{\theta}_d^{HB})\right)+\sum_{i\in A_{NUL}}\omega_i\,\hat{\theta}_i^{HB}\\
&=\sum_{i\in A_{EXC}}\omega_i\,\hat{\theta}_i^{HB}+\sum_{i\in A_{VAL}}\omega_i\,\hat{\theta}_i^{HB}+\sum_{i\in A_{VAL}}\omega_i\left(\alpha_i(\hat{\theta}^*-\sum_{d\in A}\omega_d\,\hat{\theta}_d^{HB})\right)+\sum_{i\in A_{NUL}}\omega_i\,\hat{\theta}_i^{HB}\\
&=\sum_{i\in A_{EXC}}\omega_i\,\hat{\theta}_i^{HB}+\sum_{i\in A_{VAL}}\omega_i\,\hat{\theta}_i^{HB}+\sum_{i\in A_{NUL}}\omega_i\,\hat{\theta}_i^{HB}+\sum_{i\in A_{VAL}}\omega_i\left(\alpha_i(\hat{\theta}^*-\sum_{d\in A}\omega_d\,\hat{\theta}_d^{HB})\right)\\
&=\sum_{i\in A}\omega_i\,\hat{\theta}_i^{HB}+\sum_{i\in A_{VAL}}\omega_i\left(\alpha_i(\hat{\theta}^*-\sum_{d\in A}\omega_d\,\hat{\theta}_d^{HB})\right)\\
&=\sum_{i\in A}\omega_i\,\hat{\theta}_i^{HB}+\sum_{i\in A_{VAL}}\omega_i\left(\frac{\omega_i(\dot{\psi}_i+b_i^2\,\hat{\sigma}_v^{2\,HB})}{\sum_{d\in A_{VAL}}\omega_d^2(\dot{\psi}_d+b_d^2\,\hat{\sigma}_v^{2\,HB})}(\hat{\theta}^*-\sum_{d\in A}\omega_d\,\hat{\theta}_d^{HB})\right)\\
&=\sum_{i\in A}\omega_i\,\hat{\theta}_i^{HB}+\left(\frac{\sum_{i\in A_{VAL}}\omega_i^2(\dot{\psi}_i+b_i^2\,\hat{\sigma}_v^{2\,HB})}{\sum_{d\in A_{VAL}}\omega_d^2(\dot{\psi}_d+b_d^2\,\hat{\sigma}_v^{2\,HB})}(\hat{\theta}^*-\sum_{d\in A}\omega_d\,\hat{\theta}_d^{HB})\right)\\
&=\sum_{i\in A}\omega_i\,\hat{\theta}_i^{HB}+(\hat{\theta}^*-\sum_{d\in A}\omega_d\,\hat{\theta}_d^{HB})\\
&=\hat{\theta}^*
\end{aligned}
$$

This shows that the adjusted values satisfy the benchmark constraint and completes the proof.

Diagnostics

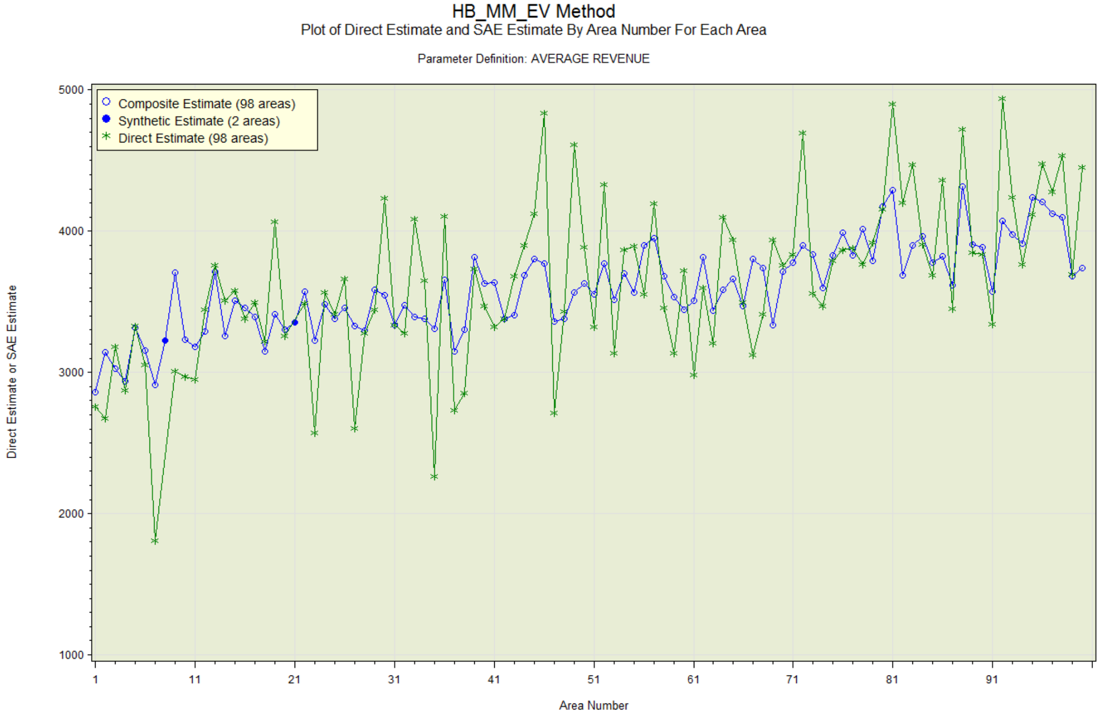
This section describes the diagnostic plots and tests used to test the fitness of the small-area model. These are based only on the areas in $A_{VAL} \cup A_{NUL}$. They do not include areas in the exclusion group A_{EXC} because these areas are not used to build the model.

Plots on the direct estimates and the SAE estimates

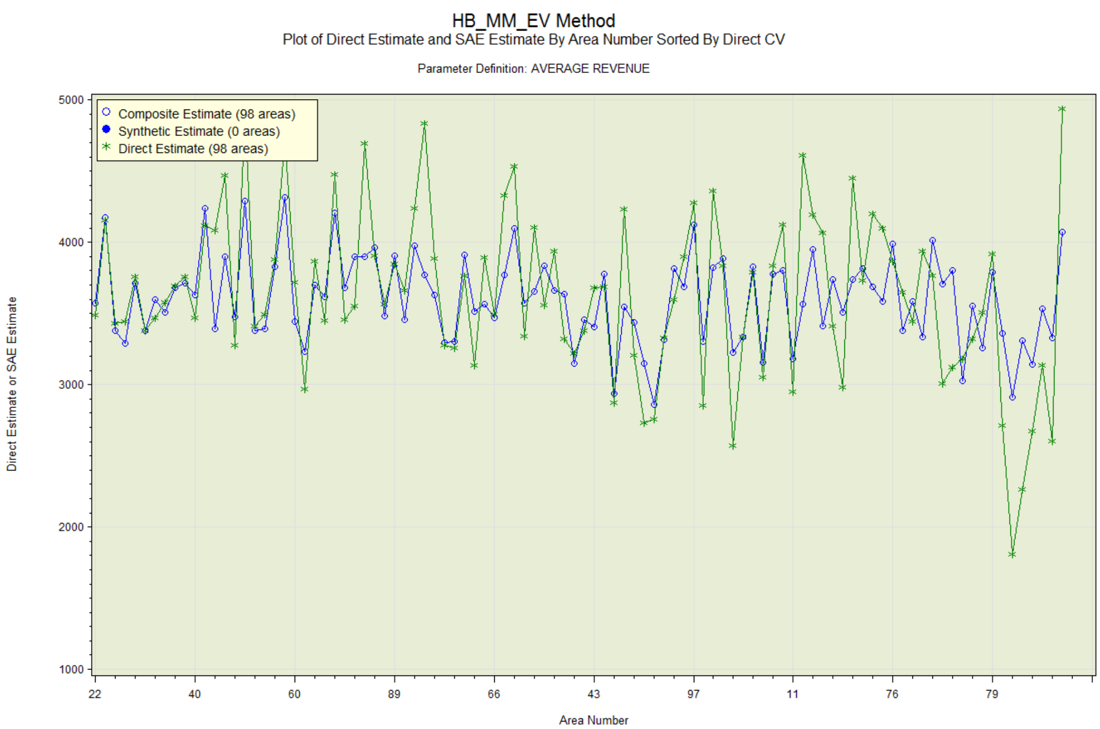
These plots provide a comparison between the direct estimates and the small-area estimation (SAE) estimates. The direct estimates are shown as asterisks (*) connected by a green line. The SAE estimates are shown as blue circles (for areas in area group A_{VAL}) or blue dots (for areas in area group A_{NUL}). They are all connected by a blue line. The areas in area group A_{VAL} have composite estimates while the areas in area group A_{NUL} have synthetic estimates.

- For an area in area group A_{VAL}, we expect the SAE estimate to be close to the direct estimate if the following conditions apply:
  - The small-area model provides a good fit.
  - The direct estimate has been produced through unbiased or approximately unbiased design-based estimation methods such as calibration estimation with sufficient number of sample units in the calibration groups and the small area.
- If the direct estimate for an area has poor precision then the corresponding SAE estimate may be quite different. We prefer the SAE estimate if the small-area model provides a good fit.
- For an area in area group A_{NUL} to be displayed on these plots, the direct estimate must be known but its direct variance is missing (for classification to area group A_{NUL}). The SAE estimate may not be close to the direct estimate since the missing direct variance may indicate a lack of precision and high variability in the direct estimate. If the small-area model fits well then the SAE estimate is preferred over the direct estimate.

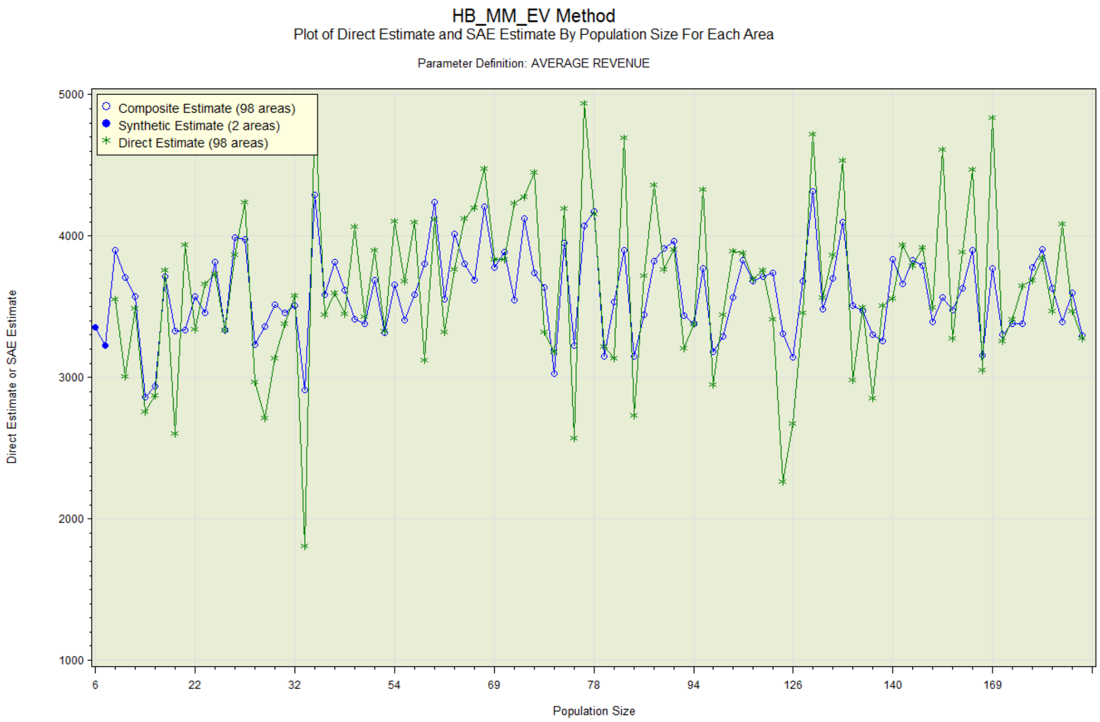
Direct Estimate and SAE Estimate By Area Number For Each Area



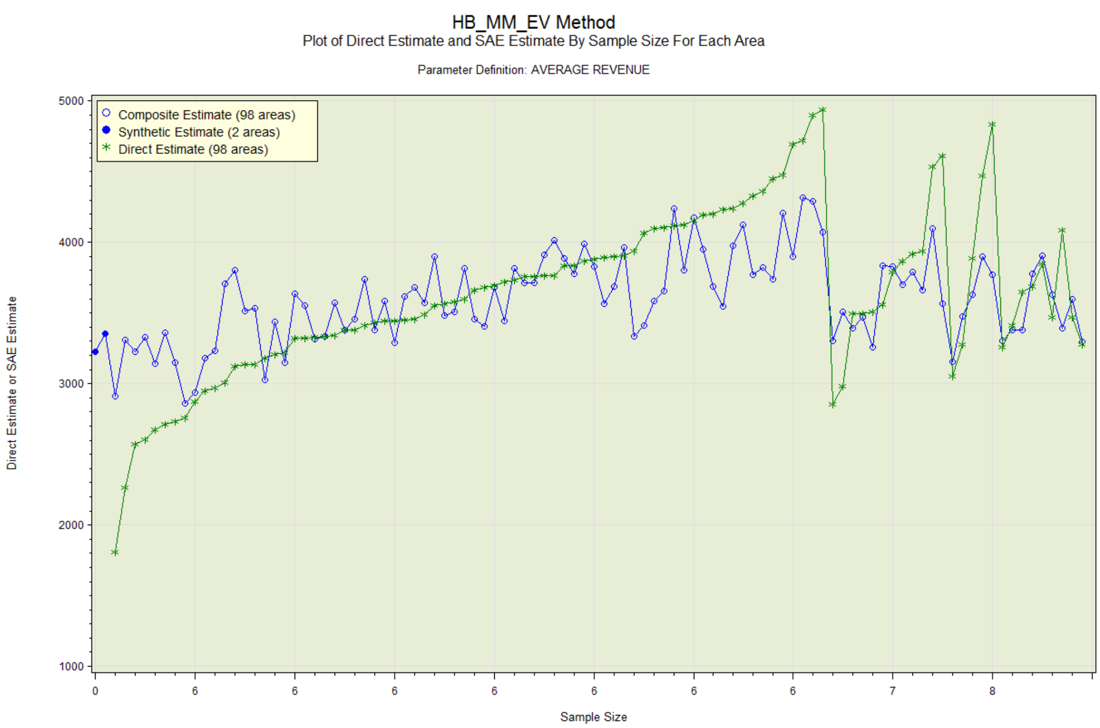
Direct Estimate and SAE Estimate By Area Number Sorted By Direct CV



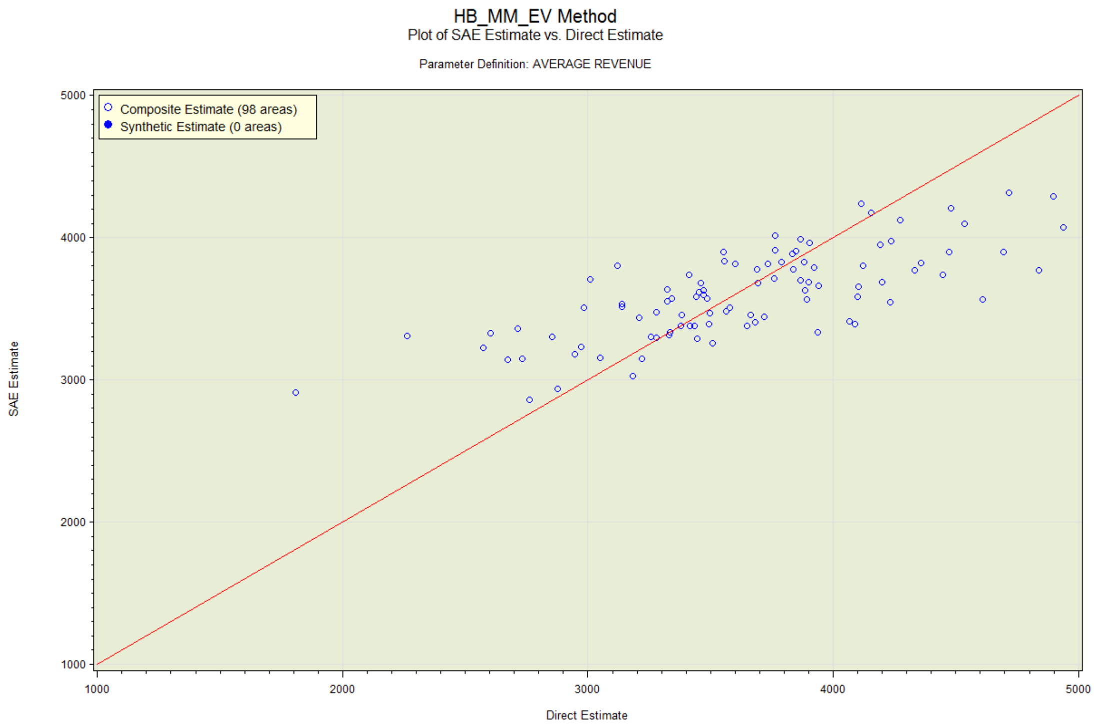
Direct Estimate and SAE Estimate By Population Size for Each Area



Direct Estimate and SAE Estimate By Sample Size for Each Area

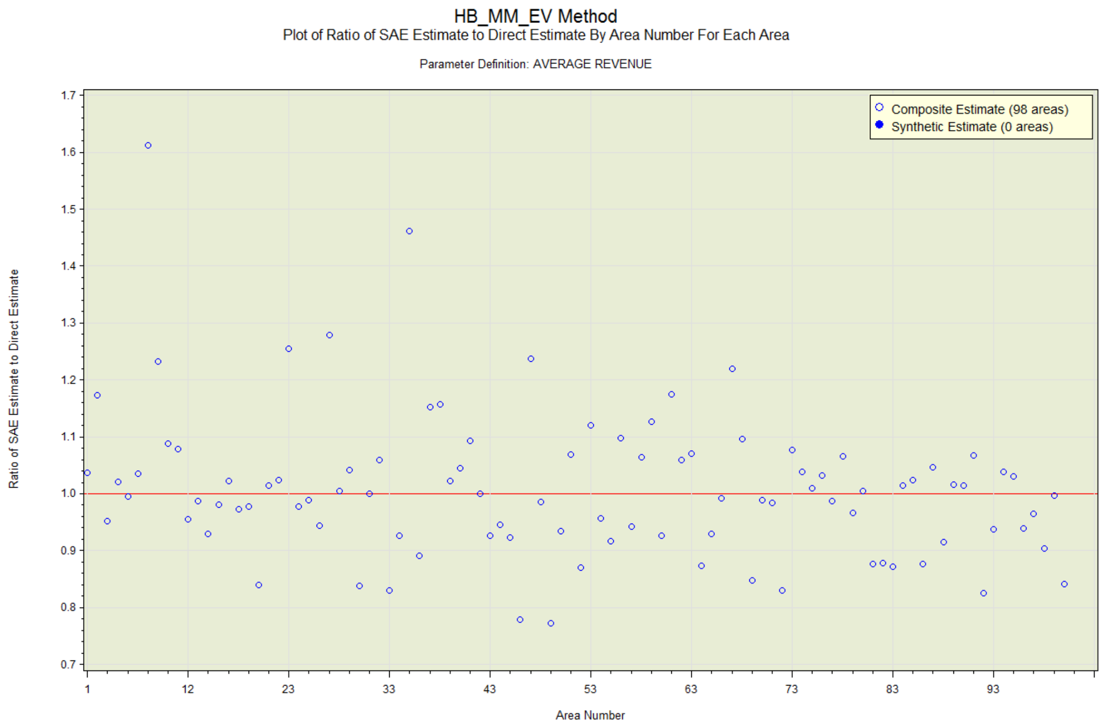


SAE Estimate vs. Direct Estimate

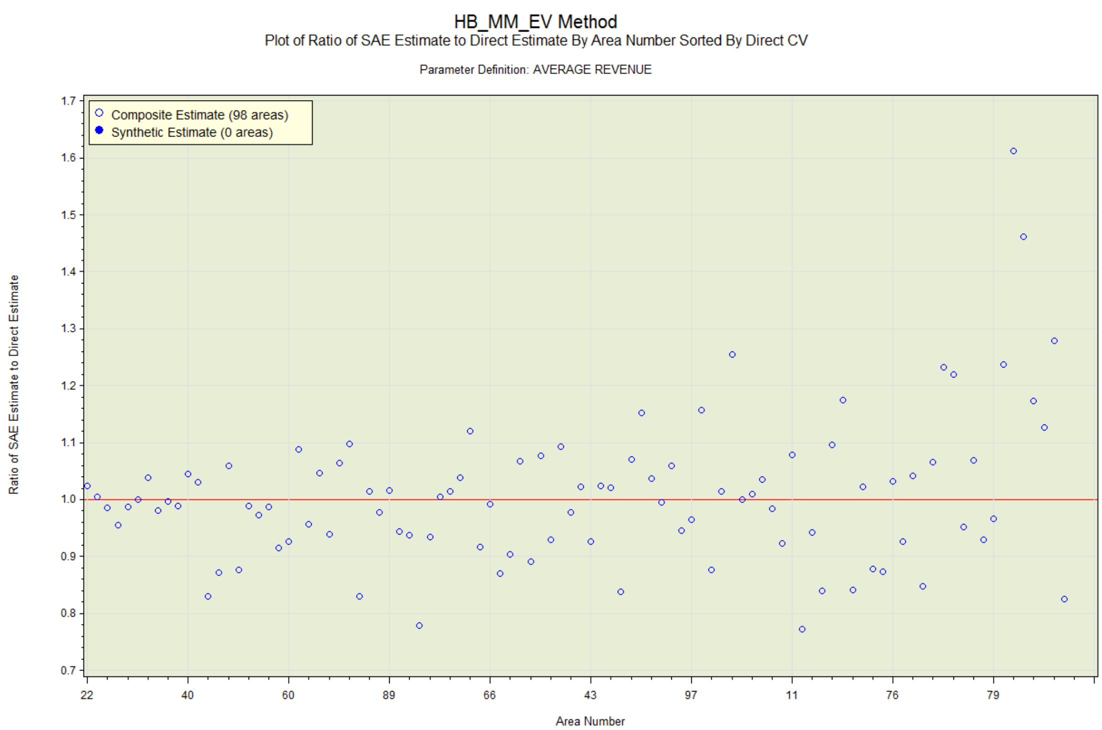


The red reference line on the plot has an intercept of 0 and a slope of 1. If the model fits well, then the two axes should have similar scale and the points should be close to the line. However, areas in area group A_{NUL} may have synthetic estimates which are considerably different from the direct estimates.

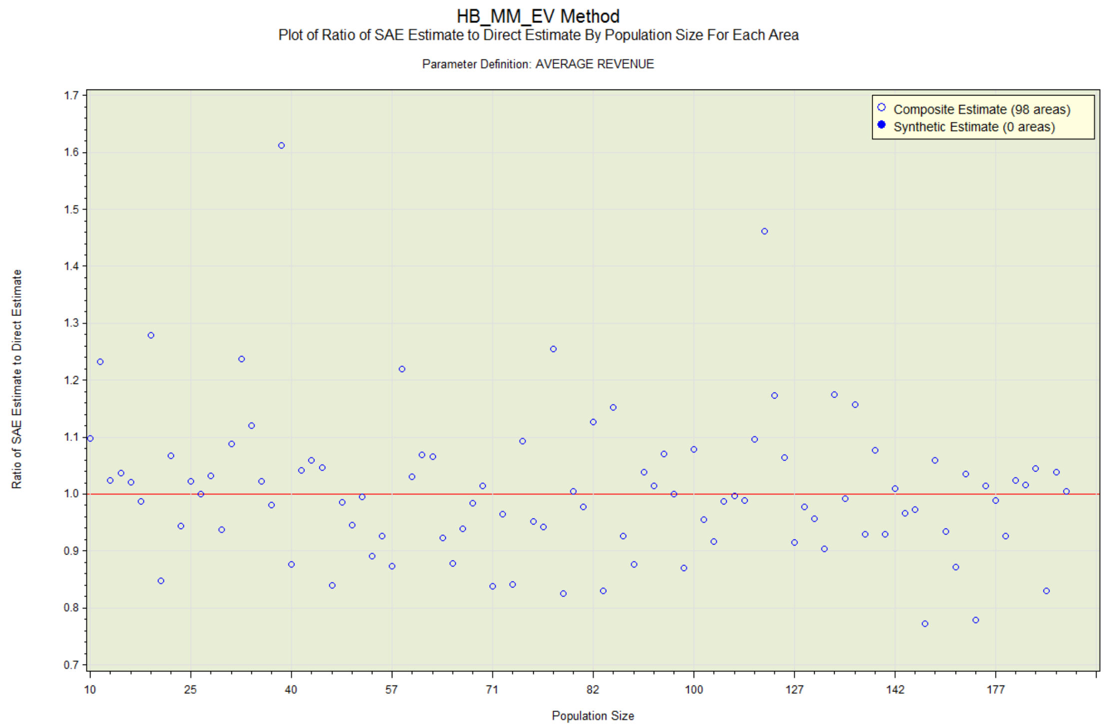
Ratio of SAE Estimate to Direct Estimate By Area Number For Each Area



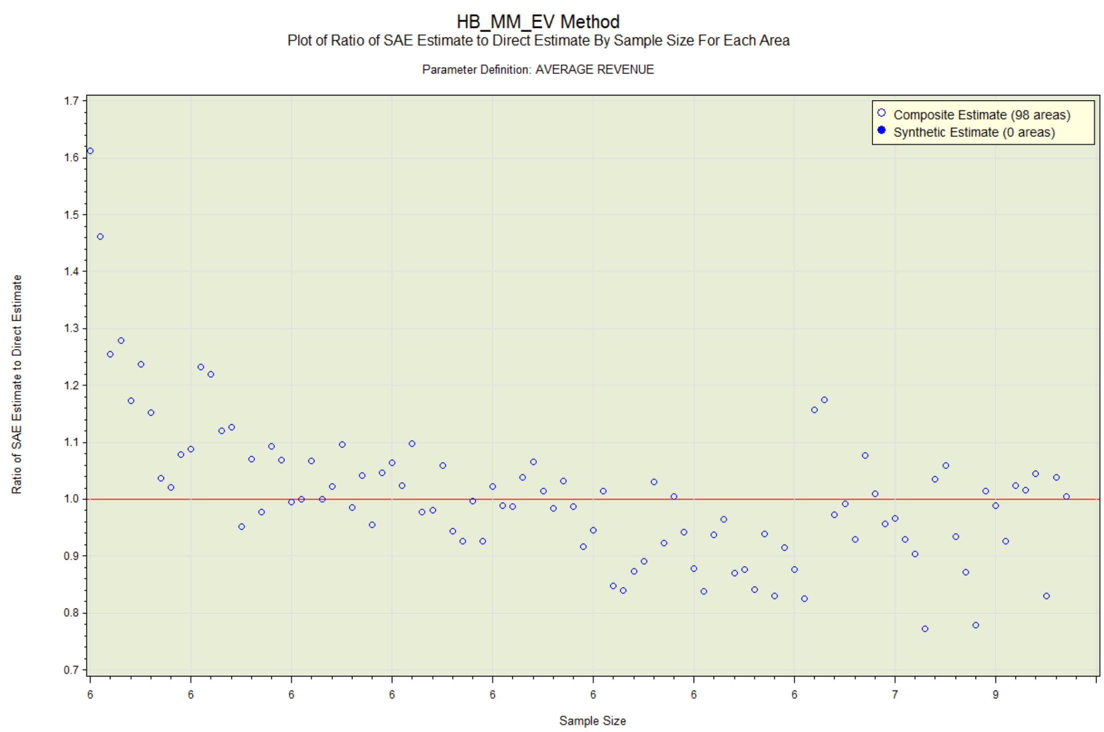
Ratio of SAE Estimate to Direct Estimate By Area Number Sorted By Direct CV



Ratio of SAE Estimate to Direct Estimate By Population Size for Each Area



Ratio of SAE Estimate to Direct Estimate By Sample Size for Each Area

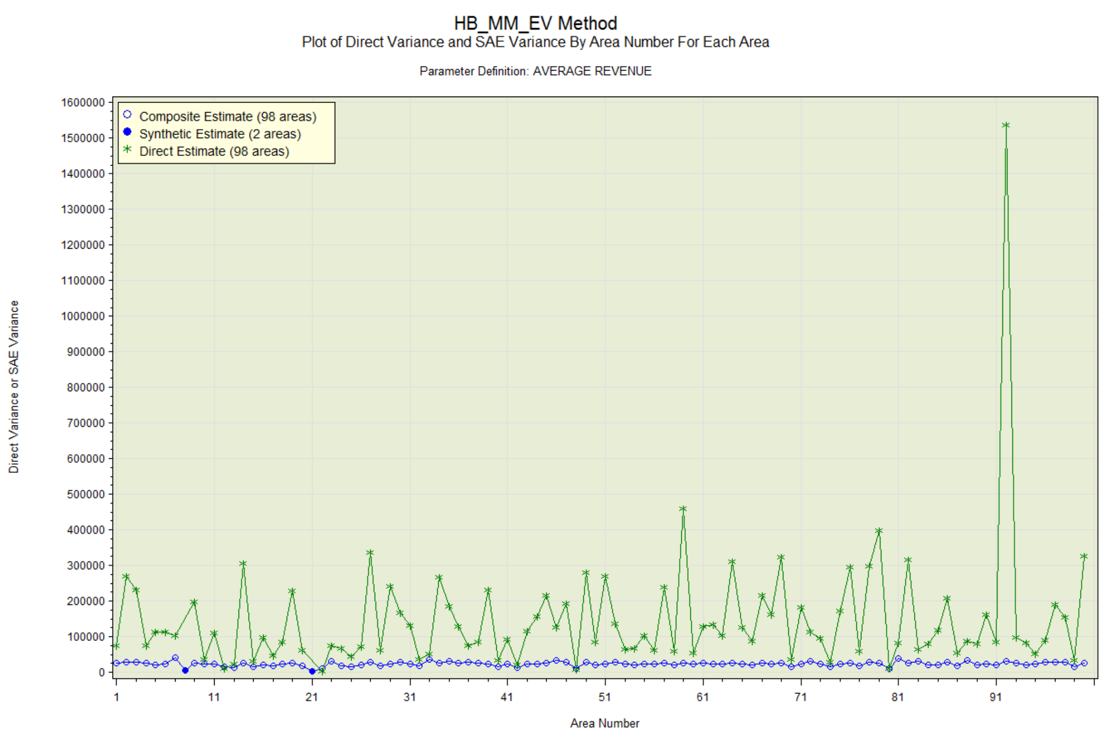


Plots on the direct variances and the SAE variances

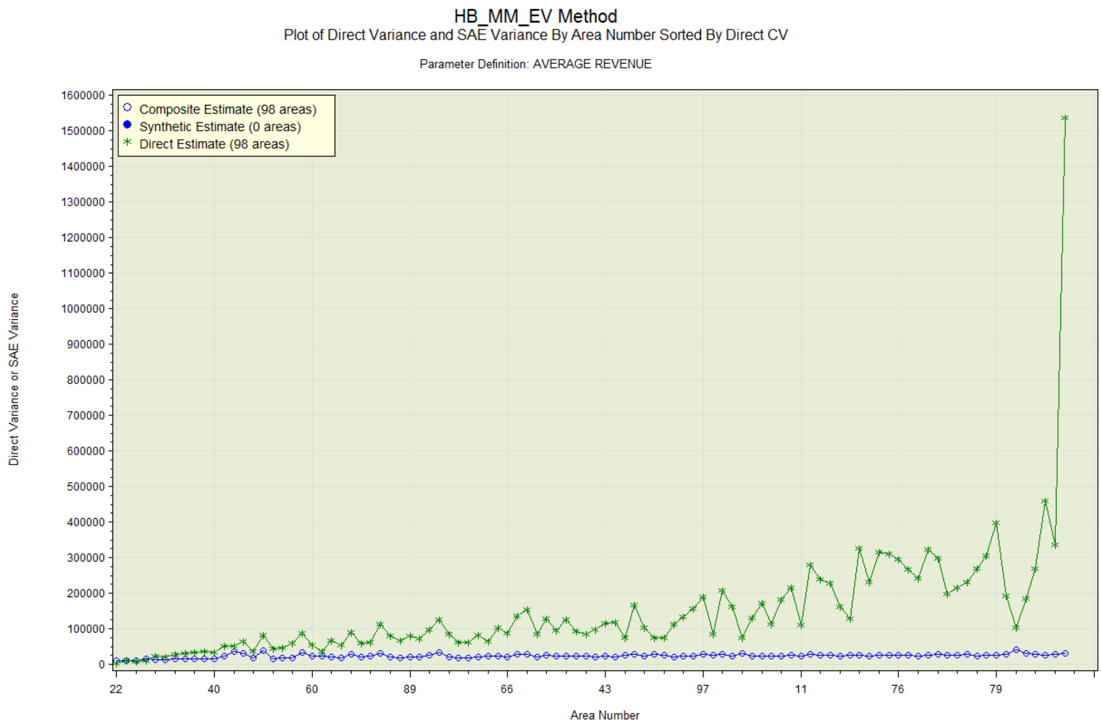
These plots provide a comparison between the direct variances and the SAE variances. The direct variances are shown as asterisks (*) connected by a green line. The SAE variances are shown as blue circles (for areas in area group A_{VAL}) or blue dots (for areas in area group A_{NUL}). They are all connected by a blue line. The areas in area group A_{VAL} have composite estimates while the areas in area group A_{NUL} have synthetic estimates.

- For an area in area group A_{VAL}, we expect the SAE variance to be generally smaller than the direct variance if the small-area model provides a good fit. However, if the direct estimate is based on calibration estimation with a sufficiently large number of sample units in the area then the direct variance should be comparable to the SAE variance.
- For an area in area group A_{NUL} to be displayed on these plots, the direct variance must be known but its direct estimate is missing (for classification to area group A_{NUL}). This is an unlikely scenario. If the model fits well then the SAE estimate and its variance are valid estimates for the area.

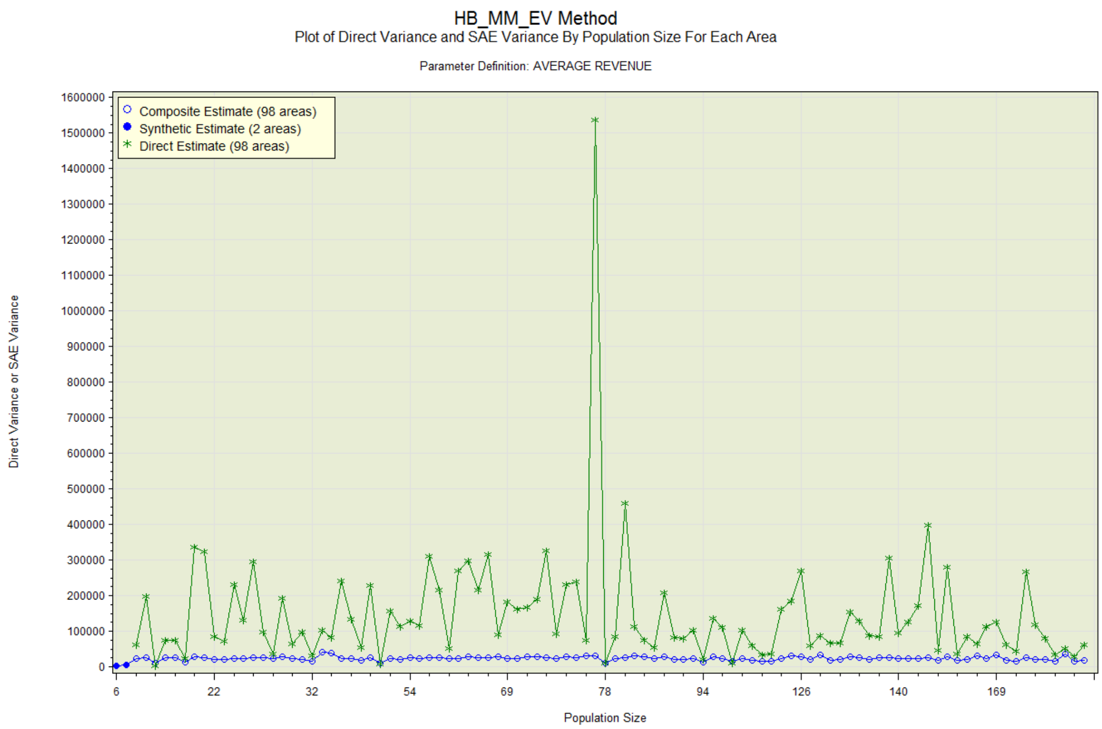
Direct Variance and SAE Variance By Area Number For Each Area



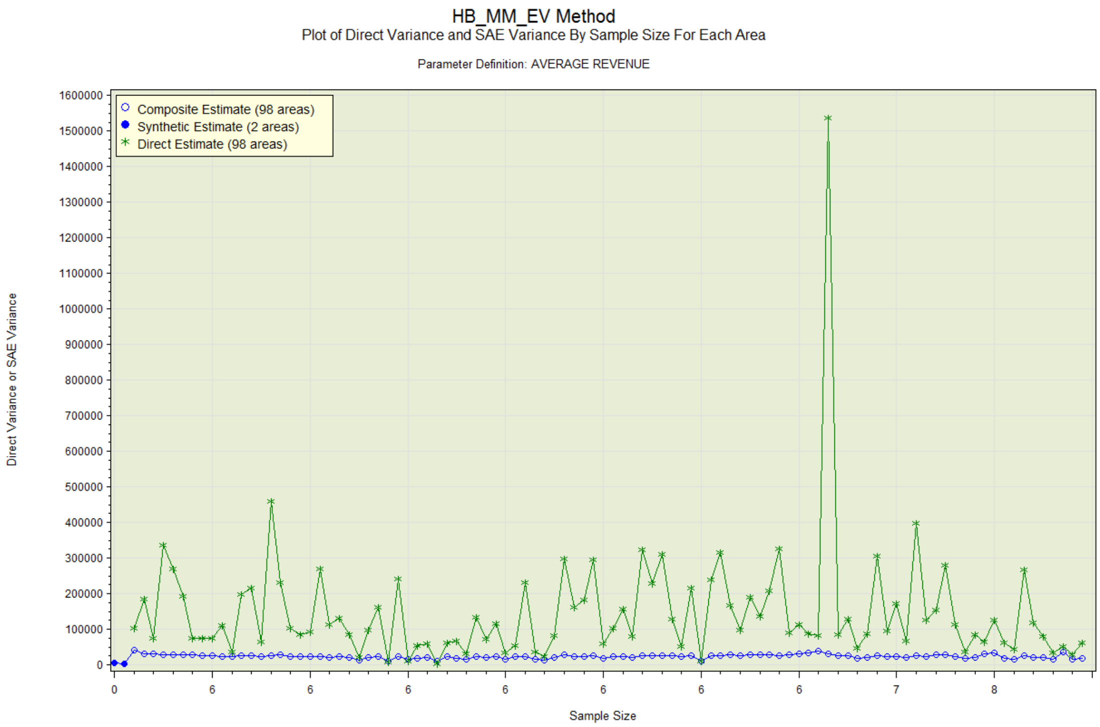
Direct Variance and SAE Variance By Area Number Sorted By Direct CV



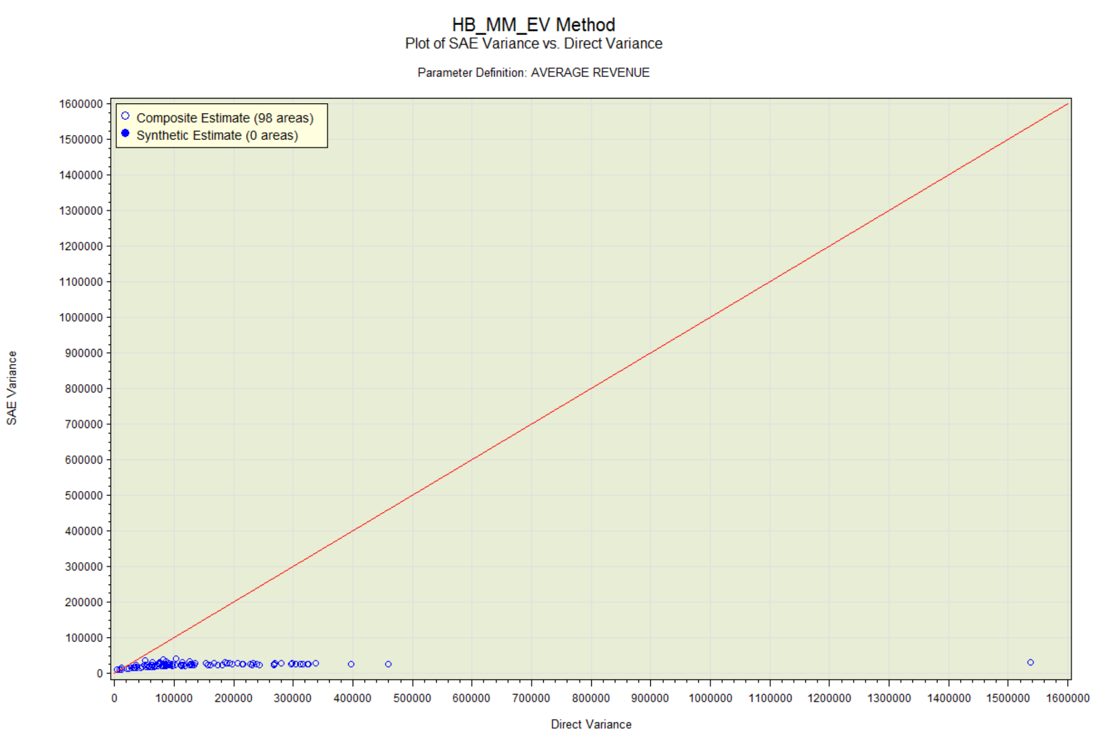
Direct Variance and SAE Variance By Population Size for Each Area



Direct Variance and SAE Variance By Sample Size for Each Area

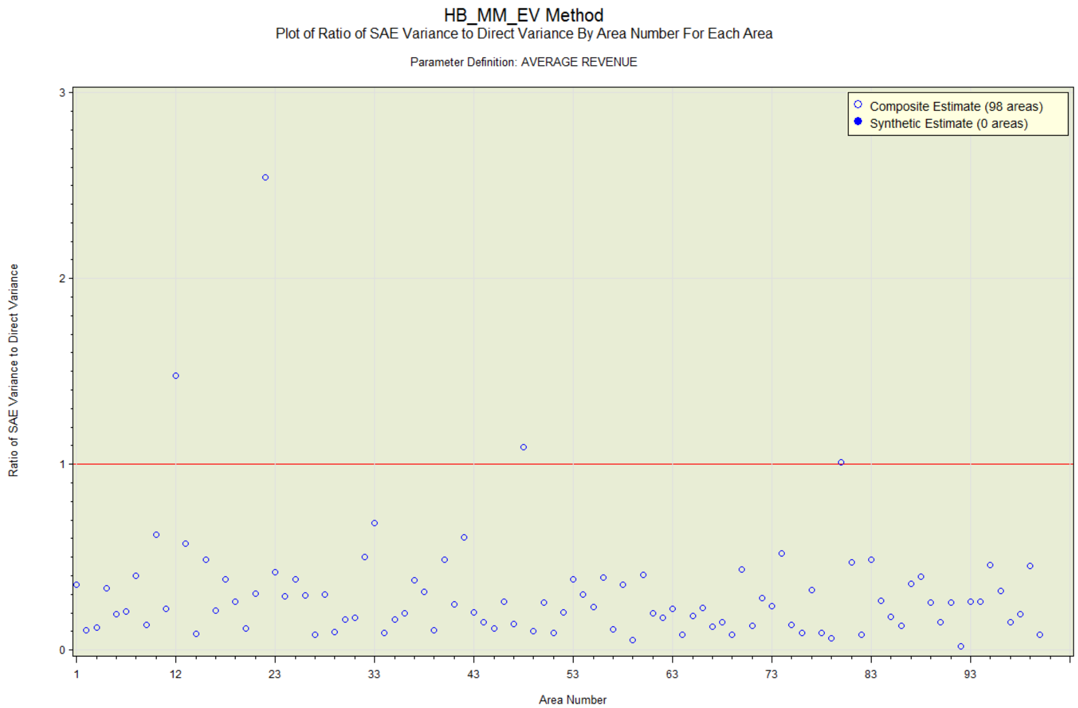


SAE Variance vs. Direct Variance

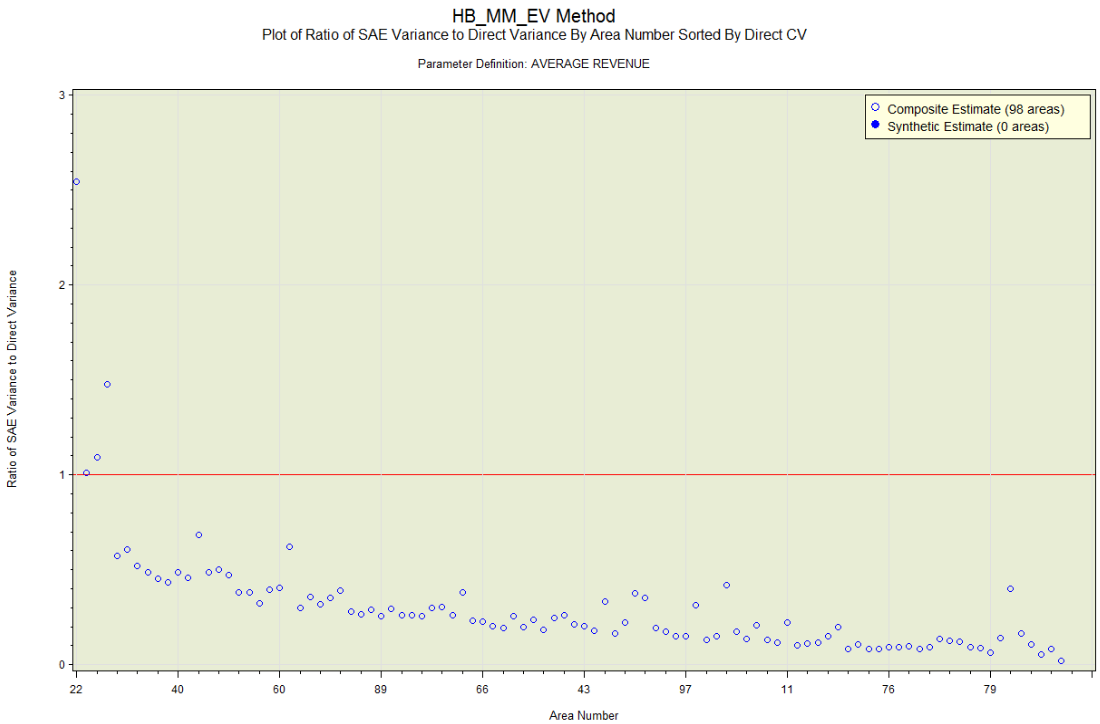


The red reference line on the plot has an intercept of 0 and a slope of 1. If the model fits well, then the SAE variance estimates should be generally smaller than the direct variances. This means most of points on the plot should be below the red line.

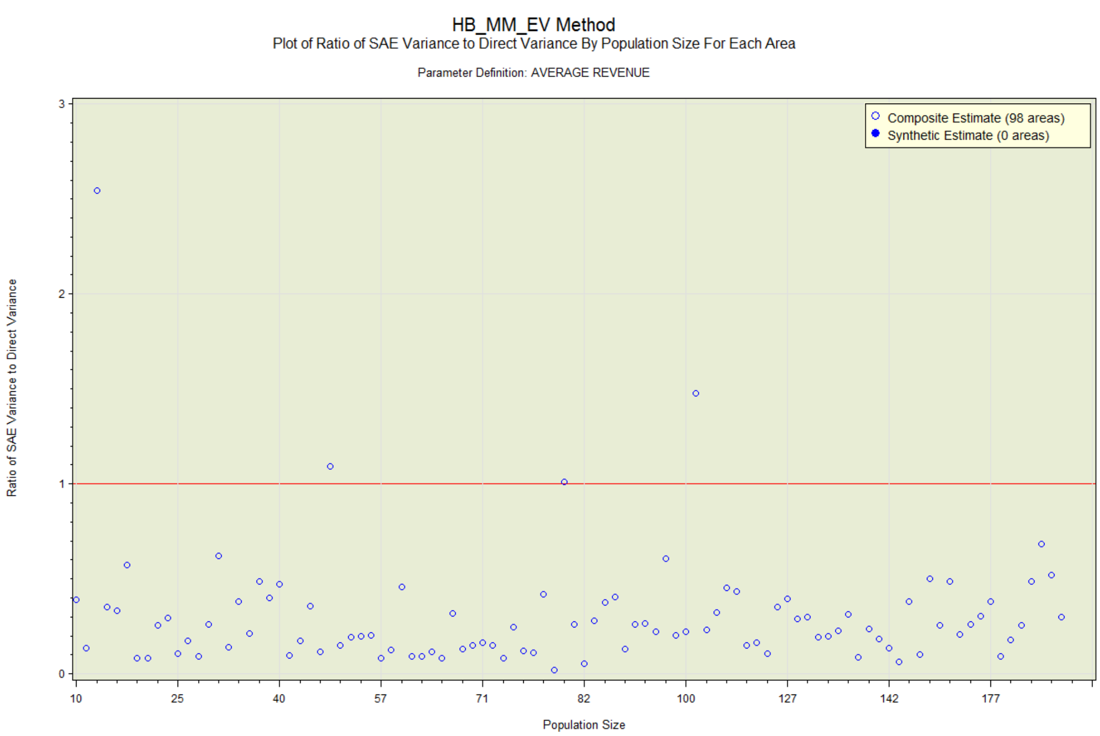
Ratio of SAE Variance to Direct Variance By Area Number For Each Area



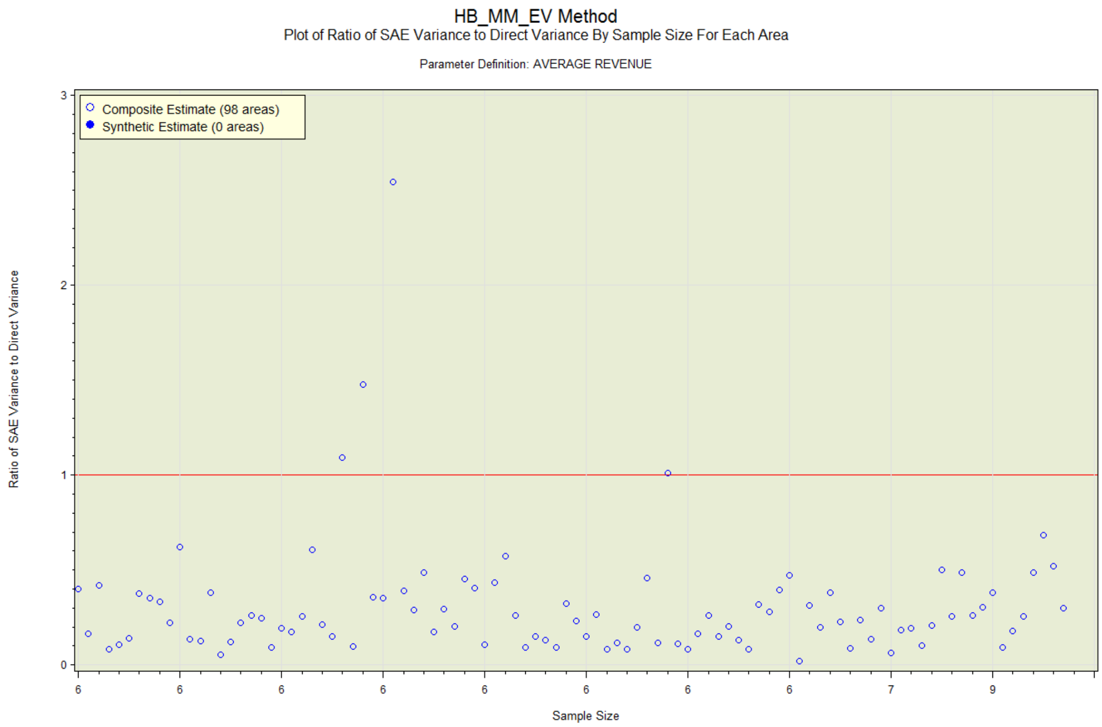
Ratio of SAE Variance to Direct Variance By Area Number Sorted By Direct CV



Ratio of SAE Variance to Direct Variance By Population Size for Each Area



Ratio of SAE Variance to Direct Variance By Sample Size for Each Area

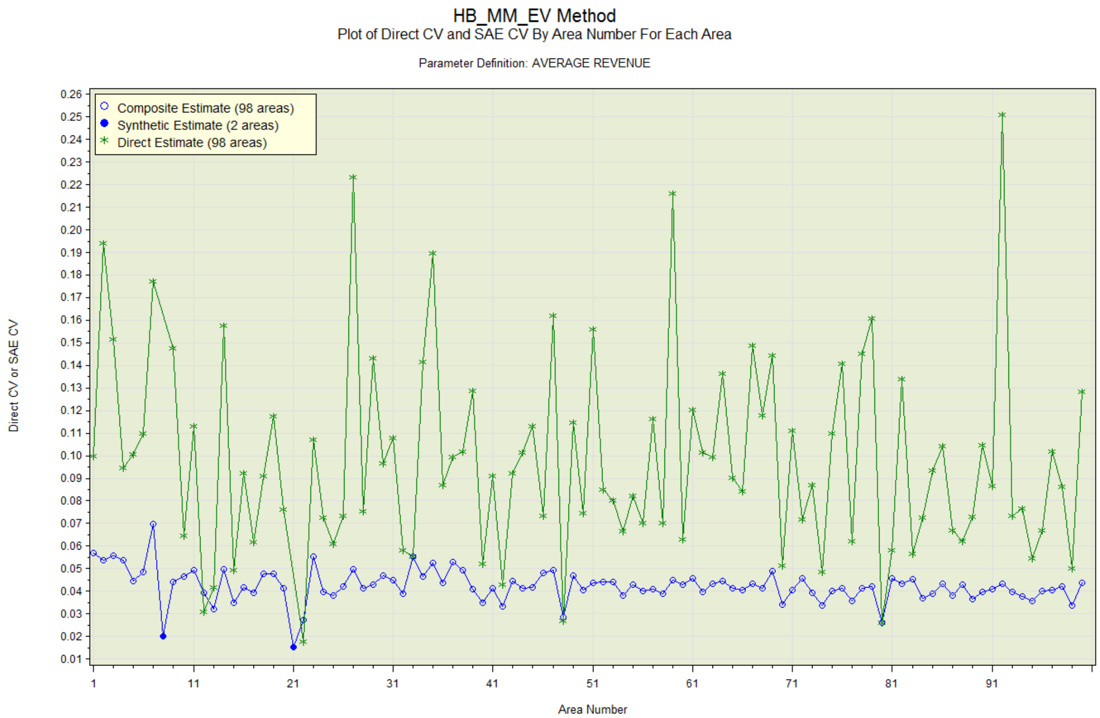


Plots on the direct CVs and the SAE CVs

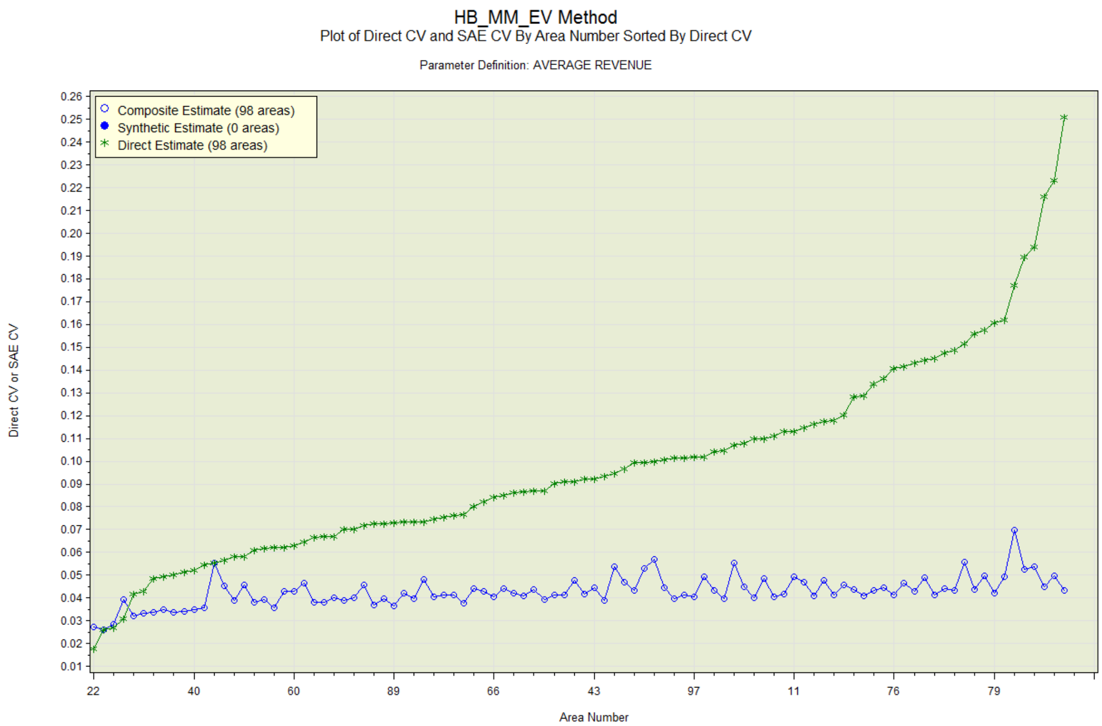
These plots provide a comparison between the direct and SAE coefficients of variation. The SAE CVs are shown as blue circles (for the areas in area group A_{VAL}) connected by a blue line. There are no blue dots corresponding to areas in area group A_{NUL}. This is because the direct estimate and the direct variance cannot be missing to produce and display a CV.

For each area in area group A_{VAL}, we expect the SAE CV to be generally smaller than the direct CV if the small-area model provides a good fit. However, if the direct estimate is based on calibration estimation with a sufficiently large number of sample units in the area then the direct CV should be comparable to the SAE CV.

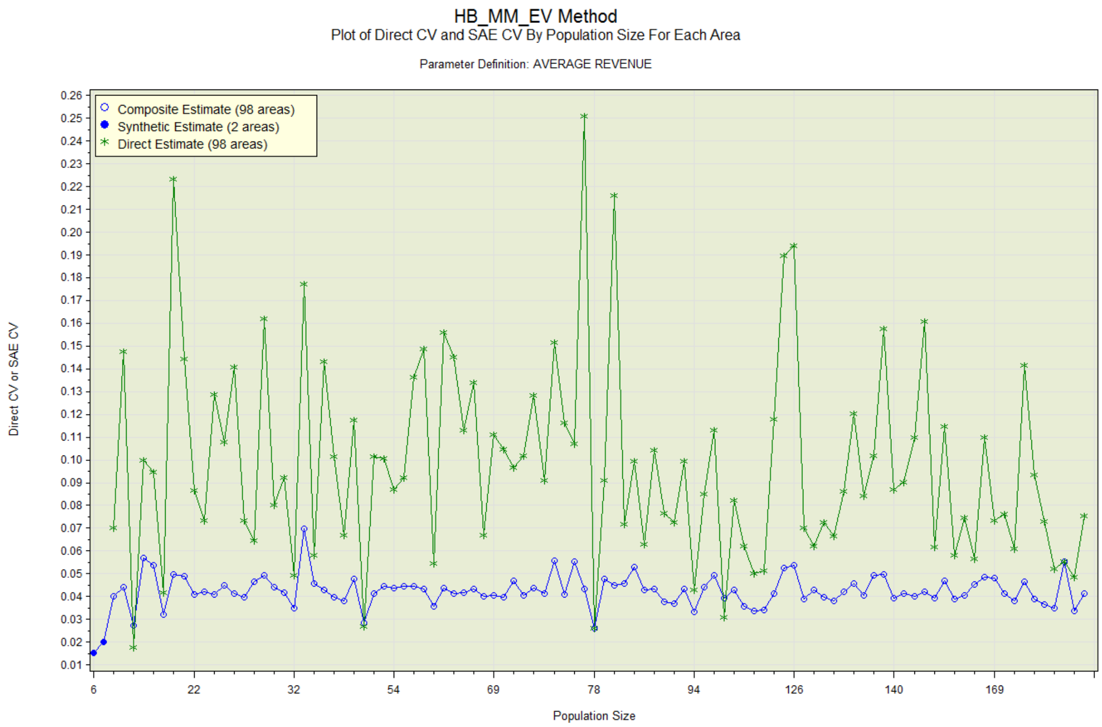
Direct CV and SAE CV By Area Number For Each Area



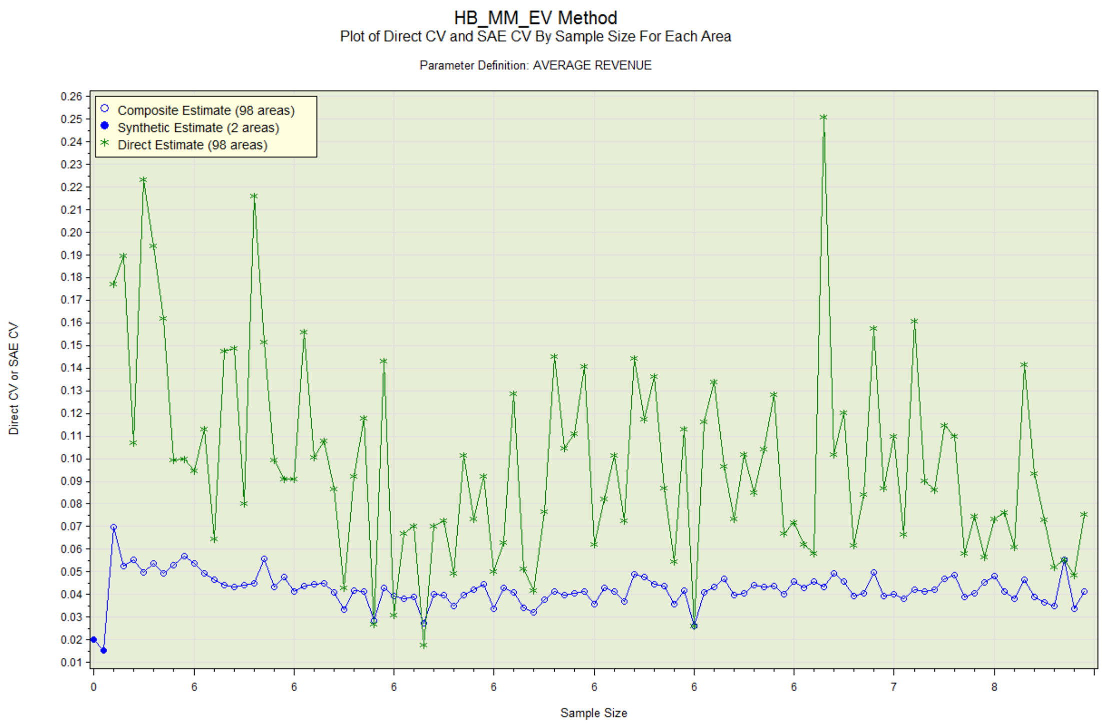
Direct CV and SAE CV By Area Number Sorted By Direct CV



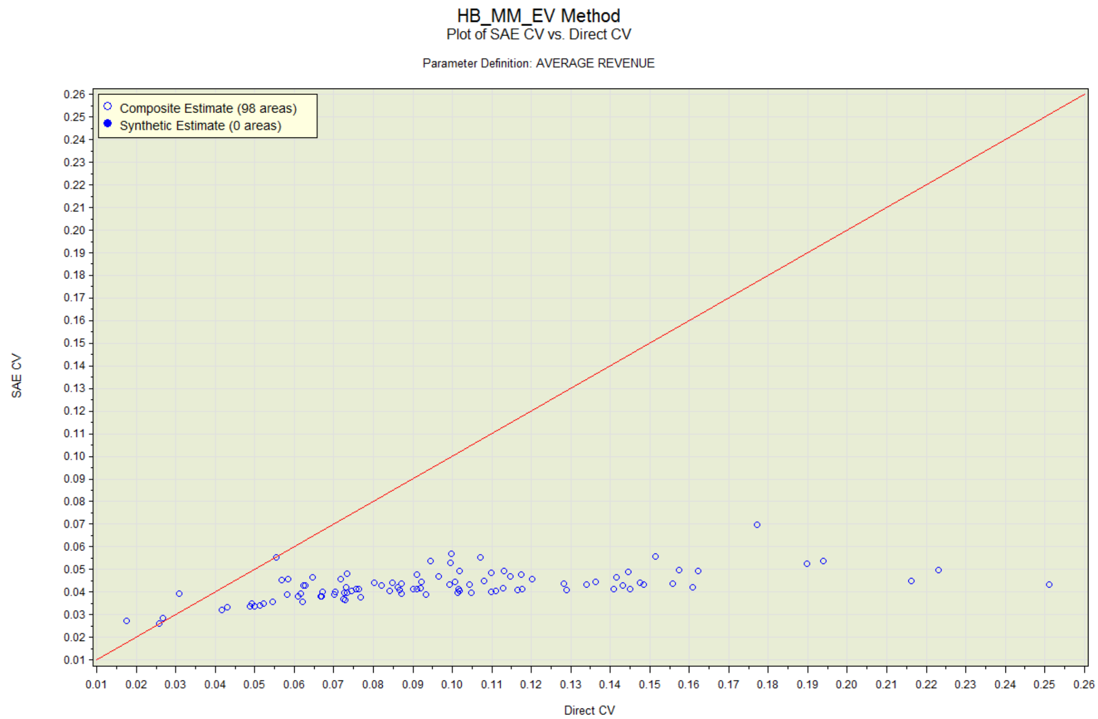
Direct CV and SAE CV By Population Size for Each Area



Direct CV and SAE CV By Sample Size for Each Area

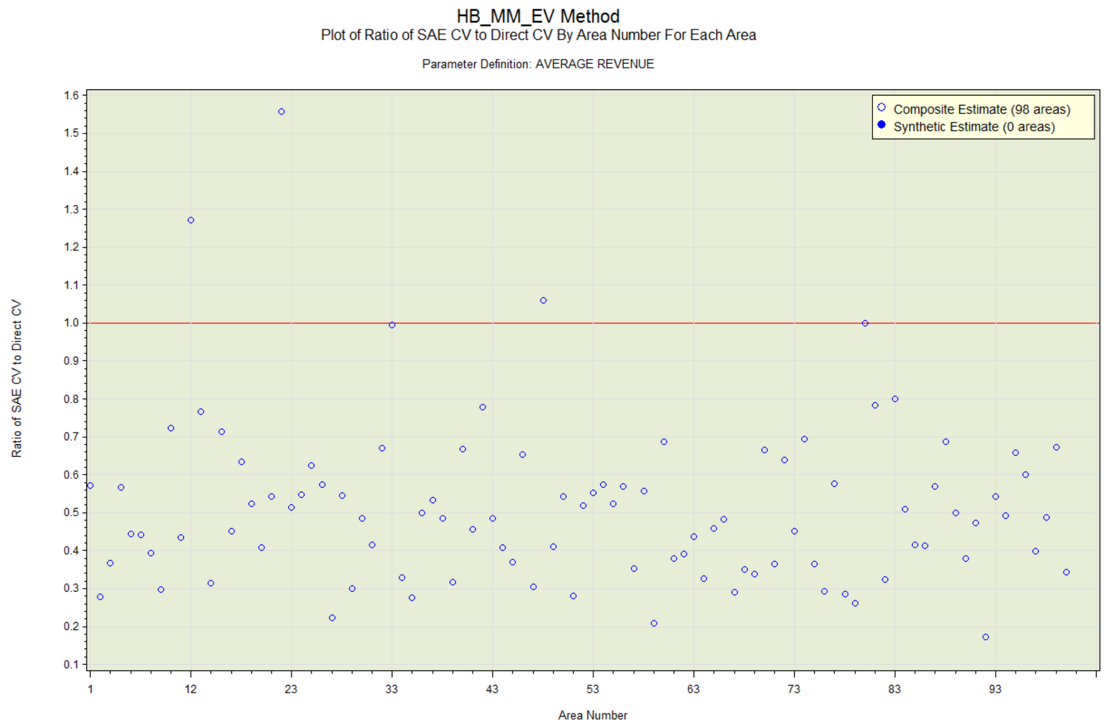


SAE CV vs. Direct CV

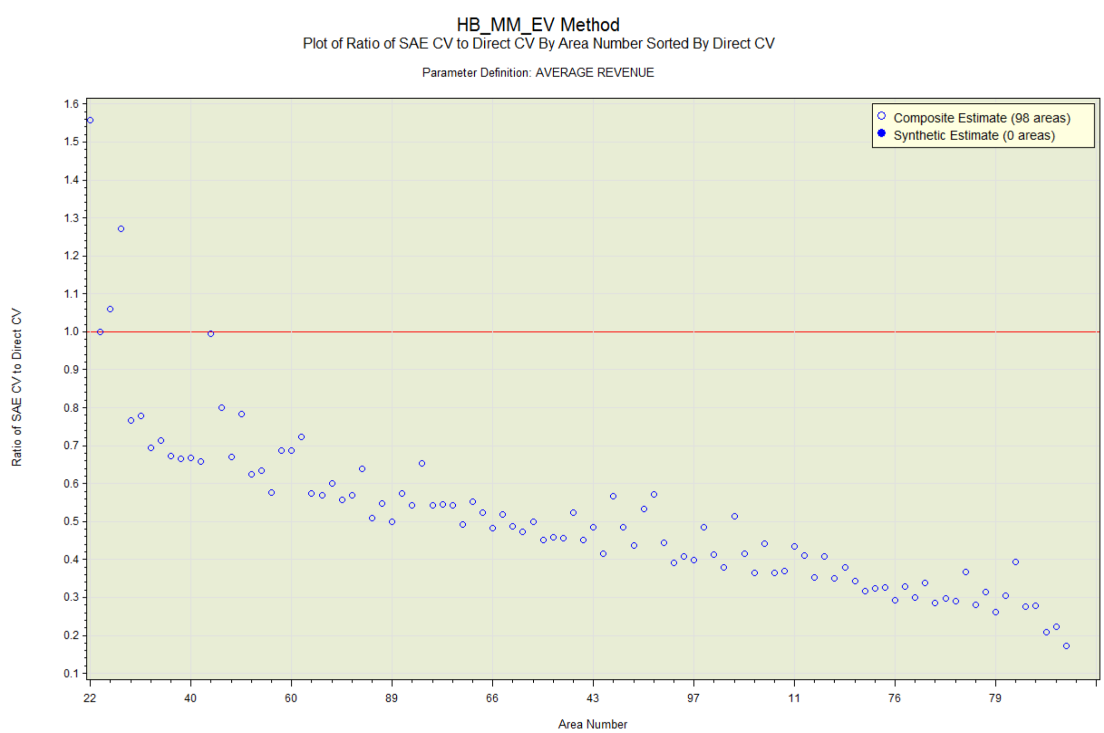


The red line on the plot has an intercept of 0 and a slope of 1. If the model fits well then the SAE RRMSE estimates should be generally smaller than the direct CVs. This means most of points on the plot should be below the red line.

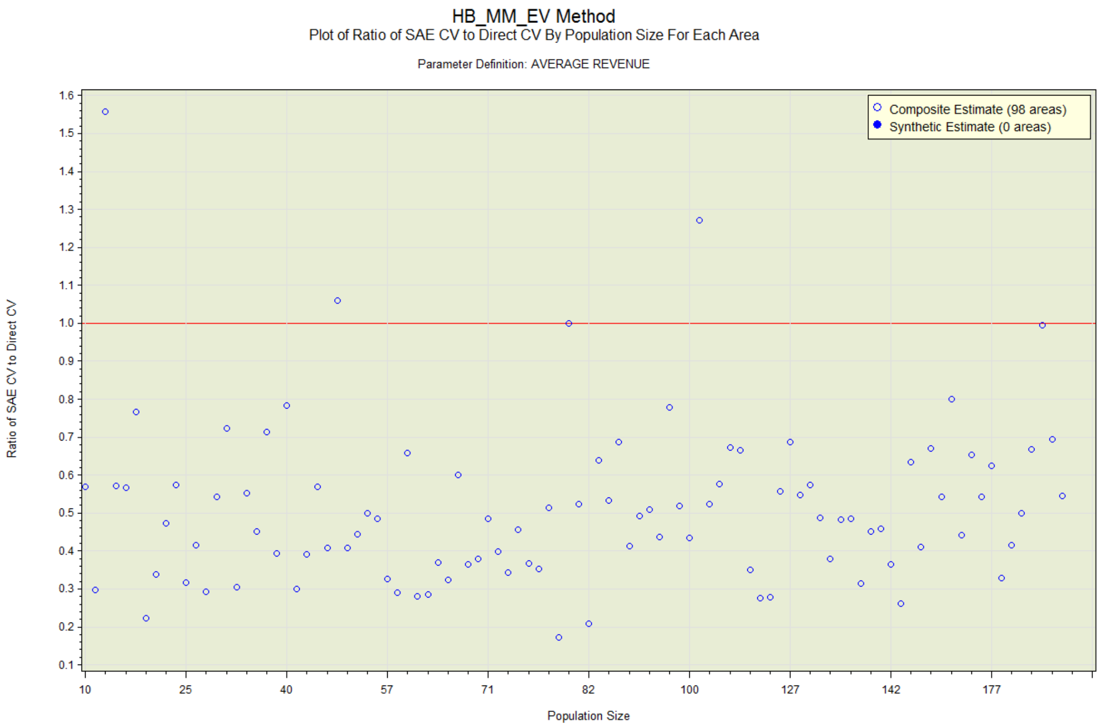
Ratio of SAE CV to Direct CV By Area Number For Each Area



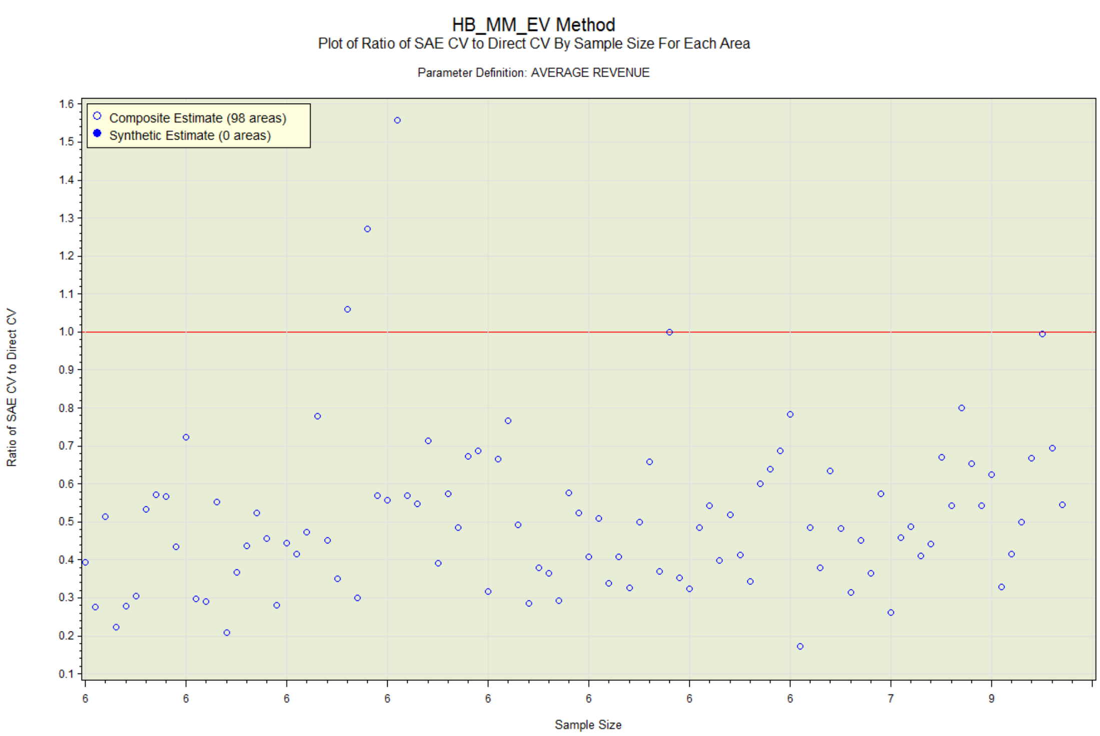
Ratio of SAE CV to Direct CV By Area Number Sorted By Direct CV



Ratio of SAE CV to Direct CV By Population Size for Each Area



Ratio of SAE CV to Direct CV By Sample Size for Each Area



Plots on the fit of the model

These plots are used to provide an evaluation of the fit of the small-area model.

Individual predictive probabilities

This plot provides information on the degree of overestimation or underestimation under the model. This is based on the individual predictive probabilities. They are calculated as follows.

1. Generate the predicted data $y_i^{(g)}$ under the model for all areas $i \in A_{VAL}$.

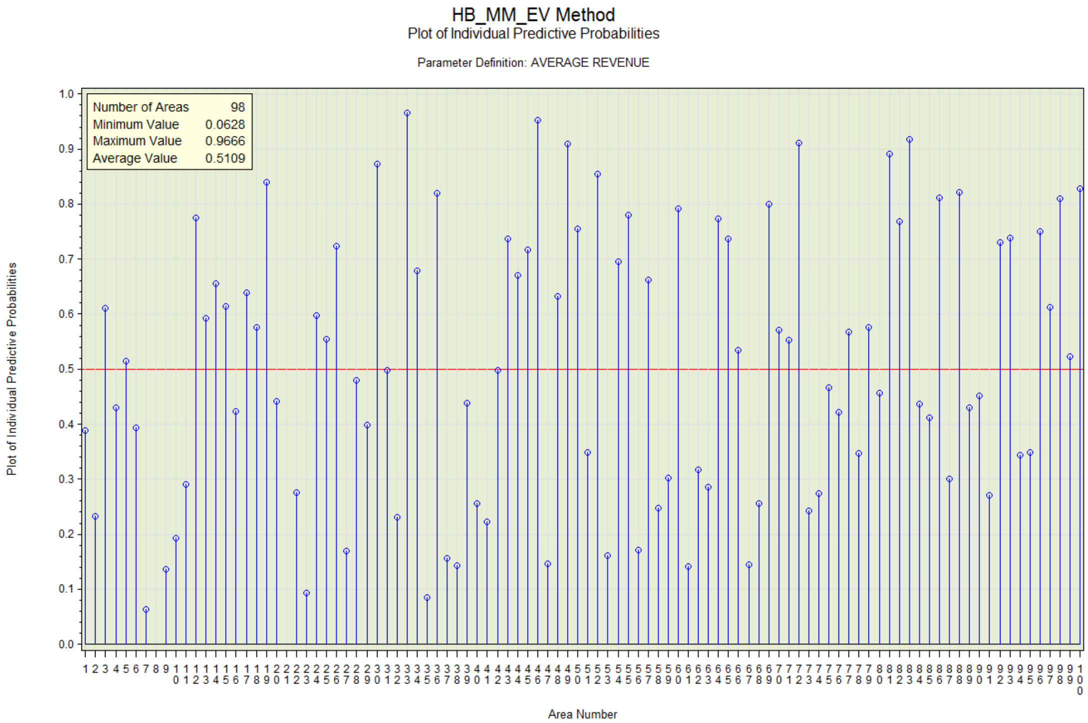
   - If the direct variances are assumed known, we generate the predicted data $y_i^{(g)}$ under the model as $y_i^{(g)} \sim N(\theta_i^{(g)}, \psi_i)$ for $i \in A_{VAL}$.

   - If the direct variance are estimated, we generate the predicted data $y_i^{(g)}$ under the model as $y_i^{(g)} \sim N(\theta_i^{(g)}, \psi_i^{(g)})$ for $i \in A_{VAL}$.

2. Estimate the individual predictive probability p_i^* as $\frac{1}{(G-B)/\Delta g} \sum_{\substack{g=B+\Delta g \\ \Delta g}}^{G} I\{y_i^{(g)} < \hat{\theta}_i^{DIR}\}$ for $i \in A_{VAL}$ where $I\{\}$ is the indicator function with value 1 if the expression is true and 0 if it is false.

3. Plot the individual predictive probabilities p_i^* and compute the average of the p_i^* over all areas in $i \in A_{VAL}$.

   We expect the average to be around 0.5 if there is no overestimation or underestimation under the model.



The plot shows the values of p_i^* over the areas $i \in A_{VAL}$. The red reference line on the plot has value 0.5. If the model fits well, then the average should be close to this value. The inset box shows the minimum, maximum and average values of the p_i^* along with the number of areas in A_{VAL}. Areas on the plot without p_i^* values belong to either A_{EXC} or A_{NUL}.

Posterior Predictive Standardized Residuals

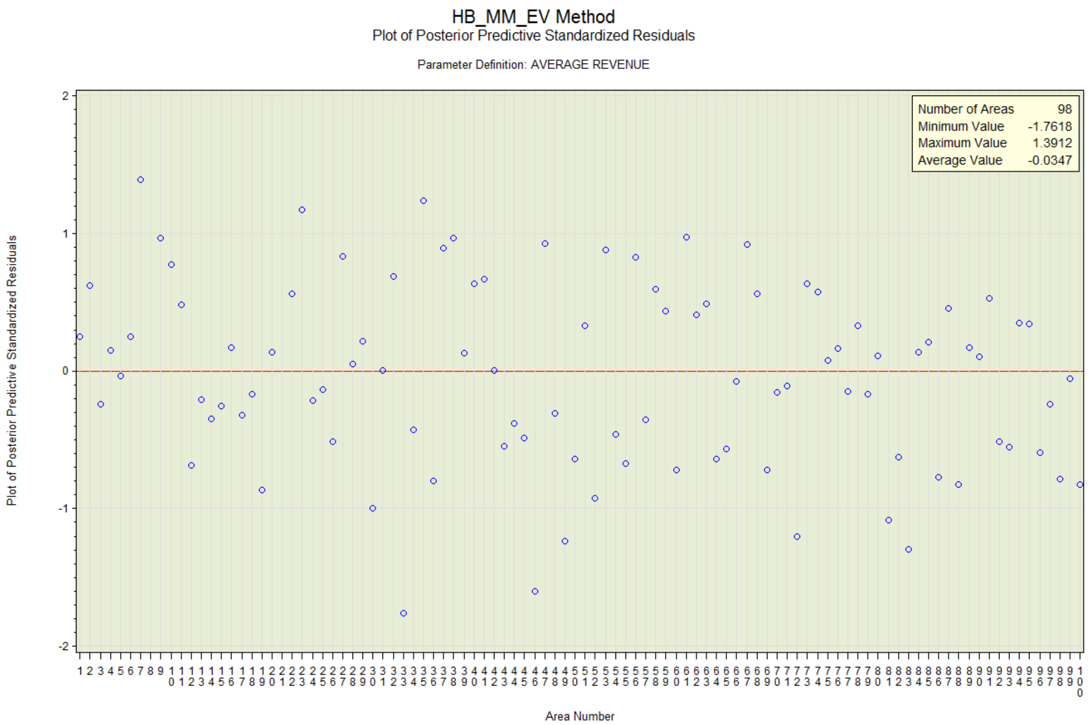
This plot displays the posterior predictive standardized residuals under the posterior predictive distribution. We use the predicted data $y_i^{(g)}$ under the model. We compute the predictive mean

$\bar{y}_i^* = \frac{1}{(G-B)/\Delta g} \sum_{\substack{g=B+\Delta g \\ \Delta g}}^{G} y_i^{(g)}$ and predictive variance

$var(y_i^*) = \frac{1}{(((G-B)/\Delta g)-1)} \sum_{\substack{g=B+\Delta g \\ \Delta g}}^{G} (y_i^{(g)} - \bar{y}_i^*)^2$ of the $y_i^{(g)}$ for each $i \in A_{VAL}$. Then we calculate the posterior predictive standardized residuals as follows.

$$d_i^* = \frac{\bar{y}_i^* - \hat{\theta}_i^{DIR}}{\sqrt{\operatorname{var}(y_i^*)}}$$

If the model fits well, then the d_i^* values should appear to be from an approximately Normal distribution with mean 0 and variance 1. Therefore, we expect most of the values of d_i^* to be between -2 and 2.



The red reference line on the plot has a value 0. The inset box shows the minimum, maximum and average values of the d_i^* along with the number of areas in A_{VAL}. Areas on the plot without d_i^* values belong to either A_{EXC} or A_{NUL}.

Derivation of full conditional distributions for the HB models

This section shows the derivation of the full conditional (posterior) distributions for the parameters in each of the models. This is done as follows. We use the sampling and linking models as well as the prior distributions for the parameters to derive their joint likelihood function. Then we isolate the expressions in the likelihood function involving each of the parameters. The resulting expressions provide us with the full conditional distributions for the parameters in the model.

Matched FH model with known sampling variances

The sampling and linking models are written in the Bayesian framework as follows.

$$\left[\, \hat{\theta}_i^{DIR} \,\middle|\, \theta_i \,\right] \sim N\left(\, \theta_i, \psi_i \,\right)$$
$$\left[\, \theta_i \,\middle|\, \boldsymbol{\beta}, \sigma_v^2 \,\right] \sim N\left(\, \mathbf{z}_i^T \boldsymbol{\beta}, b_i^2\, \sigma_v^2 \,\right)$$

The prior distributions $\pi(\,\cdot\,)$ for the unknown parameters $\boldsymbol{\beta}$ and σ_v^2 are given by the following specifications.

$$\pi(\boldsymbol{\beta}) \propto 1$$
$$\pi(\sigma_v^2) \propto (\sigma_v^2)^{-1/2}$$

The joint likelihood function $f(\boldsymbol{\theta},\boldsymbol{\beta},\sigma_v^2 \,|\, \hat{\boldsymbol{\theta}}^{DIR},\boldsymbol{\psi})$ for the unknown parameters in the model is obtained by multiplying the four density functions for $\hat{\boldsymbol{\theta}}^{DIR}$, $\boldsymbol{\theta}$, $\boldsymbol{\beta}$ and σ_v^2. It is given by the following expression.

$$\begin{aligned} f(\boldsymbol{\theta},\boldsymbol{\beta},\sigma_v^2 \,|\, \hat{\boldsymbol{\theta}}^{DIR},\boldsymbol{\psi}) &= \prod_{i \in A_{VAL}} \left\{ \frac{1}{\sqrt{2\pi\psi_i}}\, exp\left[-\frac{(\hat{\theta}_i^{DIR} - \theta_i)^2}{2\psi_i} \right] \frac{1}{\sqrt{2\pi\, b_i^2\, \sigma_v^2}}\ \ exp\left[-\frac{(\,\theta_i - \mathbf{z}_i^T \boldsymbol{\beta}\,)^2}{2\, b_i^2\, \sigma_v^2} \right] \right\} \pi(\boldsymbol{\beta})\, \pi(\sigma_v^2) \\ &\propto exp\left[-\sum\nolimits_{i \in A_{VAL}} \frac{(\hat{\theta}_i^{DIR} - \theta_i)^2}{2\psi_i} \right] \left(\sigma_v^2\right)^{-\frac{m}{2}}\ exp\left[-\sum\nolimits_{i \in A_{VAL}} \frac{(\,\theta_i - \mathbf{z}_i^T \boldsymbol{\beta}\,)^2}{2\, b_i^2\, \sigma_v^2} \right] \left(\sigma_v^2\right)^{-\frac{1}{2}} \\ &\propto exp\left[-\sum\nolimits_{i \in A_{VAL}} \frac{(\hat{\theta}_i^{DIR} - \theta_i)^2}{2\psi_i} + \frac{(\,\theta_i - \mathbf{z}_i^T \boldsymbol{\beta}\,)^2}{2\, b_i^2\, \sigma_v^2} \right] \left(\sigma_v^2\right)^{-(\frac{m+1}{2})} \end{aligned}$$

The full conditional distributions are obtained by isolating the expressions involving each of the parameters. From this product form, it is clear that the θ_i are independent. For each θ_i we obtain a density function which is proportional to the following expression.

$$
\begin{aligned}
f(\theta_i \mid \hat{\boldsymbol{\theta}}^{DIR}, \boldsymbol{\psi}, \boldsymbol{\beta}, \sigma_v^2) &\propto exp\left[-\frac{(\hat{\theta}_i^{DIR} - \theta_i)^2}{2\psi_i} - \frac{(\theta_i - \mathbf{z}_i^T \boldsymbol{\beta})^2}{2\, b_i^2\, \sigma_v^2} \right] \\
&\propto exp\left[-\frac{1}{2}\left\{ \left(\frac{1}{\psi_i} + \frac{1}{b_i^2\,\sigma_v^2} \right)\theta_i^2 - 2\left(\frac{\hat{\theta}_i^{DIR}}{\psi_i} + \frac{\mathbf{z}_i^T \boldsymbol{\beta}}{b_i^2\,\sigma_v^2} \right)\theta_i \right\} \right] \\
&\propto exp\left[-\frac{1}{2}\left\{ \left(\theta_i - \frac{b_i^2\,\sigma_v^2\,\hat{\theta}_i^{DIR} + \psi_i\,\mathbf{z}_i^T \boldsymbol{\beta}}{b_i^2\,\sigma_v^2 + \psi_i} \right)^2 \left(\frac{1}{\psi_i} + \frac{1}{b_i^2\,\sigma_v^2} \right) \right\} \right] \\
&\propto exp\left[-\frac{1}{2}\left(\theta_i - \left[\gamma_i\,\hat{\theta}_i^{DIR} + (1-\gamma_i)\,\mathbf{z}_i^T \boldsymbol{\beta} \right] \right)^2 \left(\frac{1}{\psi_i\,\gamma_i} \right) \right] \\
&\text{where} \quad \gamma_i = \frac{b_i^2\,\sigma_v^2}{b_i^2\,\sigma_v^2 + \psi_i}
\end{aligned}
$$

This shows that each θ_i has a Normal distribution with mean $\gamma_i\,\hat{\theta}_i^{DIR} + (1-\gamma_i)\,\mathbf{z}_i^T \boldsymbol{\beta}$ and variance $\psi_i\,\gamma_i$.

The full conditional distribution of $\boldsymbol{\beta}$ is derived from the only term in the joint likelihood function that involves $\boldsymbol{\beta}$.

$$
\begin{aligned}
f(\boldsymbol{\beta} \mid \hat{\boldsymbol{\theta}}^{DIR}, \boldsymbol{\psi}, \boldsymbol{\theta}, \sigma_v^2) &\propto exp\left[-\sum_{i \in A_{VAL}} \frac{(\theta_i - \mathbf{z}_i^T \boldsymbol{\beta})^2}{2\, b_i^2\, \sigma_v^2} \right] \\
&\propto exp\left[-\sum_{i \in A_{VAL}} \frac{\theta_i^2 - 2\,\theta_i\,\mathbf{z}_i^T \boldsymbol{\beta} + \boldsymbol{\beta}^T \mathbf{z}_i\,\mathbf{z}_i^T \boldsymbol{\beta}}{2\, b_i^2\, \sigma_v^2} \right] \\
&\propto exp\left[-\frac{1}{2}\left\{ \sum_{i \in A_{VAL}} \frac{-2\,\theta_i\,\mathbf{z}_i^T \boldsymbol{\beta}}{b_i^2\,\sigma_v^2} + \boldsymbol{\beta}^T \sum_{i \in A_{VAL}} \frac{\mathbf{z}_i\,\mathbf{z}_i^T}{b_i^2\,\sigma_v^2} \boldsymbol{\beta} \right\} \right] \\
&\propto exp\left[-\frac{1}{2}\left\{ \sum_{i \in A_{VAL}} \left(\boldsymbol{\beta} - \boldsymbol{A}\left(\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i\,\theta_i}{b_i^2\,\sigma_v^2} \right) \right)^T \boldsymbol{A}^{-1} \left(\boldsymbol{\beta} - \boldsymbol{A}\left(\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i\,\theta_i}{b_i^2\,\sigma_v^2} \right) \right) \right\} \right] \\
&\text{where} \quad \boldsymbol{A} = \left(\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i\,\mathbf{z}_i^T}{b_i^2\,\sigma_v^2} \right)^{-1} = \sigma_v^2 \left(\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i\,\mathbf{z}_i^T}{b_i^2} \right)^{-1}
\end{aligned}
$$

This is a multivariate Normal distribution with mean $\boldsymbol{A}\left(\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i\,\theta_i}{b_i^2\,\sigma_v^2} \right)$ and variance-covariance matrix $\boldsymbol{A}$. Thus, we have the following.

$$
f(\boldsymbol{\beta} \mid \hat{\boldsymbol{\theta}}^{DIR}, \boldsymbol{\psi}, \boldsymbol{\theta}, \sigma_v^2) \sim N\left(\left(\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i}{b_i} \frac{\mathbf{z}_i^T}{b_i} \right)^{-1} \left(\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i}{b_i} \frac{\theta_i}{b_i} \right),\ \sigma_v^2 \left(\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i}{b_i} \frac{\mathbf{z}_i^T}{b_i} \right)^{-1} \right)
$$

Finally, the full conditional distribution of σ_v^2 is determined from the terms in the joint likelihood function that only involve σ_v^2.

$$f(\sigma_v^2 \,\middle|\, \hat{\boldsymbol{\theta}}^{DIR}, \boldsymbol{\psi}, \boldsymbol{\theta}, \boldsymbol{\beta}) \propto exp\left[-\sum\nolimits_{i \in A_{VAL}} \frac{(\theta_i - \mathbf{z}_i^T \boldsymbol{\beta})^2}{2\, b_i^2\, \sigma_v^2}\right] \left(\sigma_v^2\right)^{-(\frac{m+1}{2})}$$

$$\propto \frac{1}{\left(\sigma_v^2\right)^{(\frac{m-1}{2})+1}} exp\left[-\frac{1}{2\sigma_v^2} \sum\nolimits_{i \in A_{VAL}} \left(\frac{\theta_i - \mathbf{z}_i^T \boldsymbol{\beta}}{b_i}\right)^2\right]$$

If we compare this expression with the pdf of the Inverse Gamma function $IG(\alpha, \beta)$, we see that

$f(\sigma_v^2 \,|\, \hat{\boldsymbol{\theta}}^{DIR}, \boldsymbol{\psi}, \boldsymbol{\theta}, \boldsymbol{\beta})$ is $IG(\alpha, \beta)$ with $\alpha = \dfrac{m-1}{2}$ and $\beta = \dfrac{1}{2} \sum\nolimits_{i \in A_{VAL}} \left(\dfrac{\theta_i - \mathbf{z}_i^T \boldsymbol{\beta}}{b_i}\right)^2$.

Unmatched log-linear model with known sampling variances

The sampling and linking models are written in the Bayesian framework as follows.

$$\left[\hat{\theta}_i^{DIR} \,\middle|\, \theta_i\right] \sim N\left(\theta_i, \psi_i\right)$$

$$\left[log(\theta_i) \,\middle|\, \boldsymbol{\beta}, \sigma_v^2\right] \sim N\left(\mathbf{z}_i^T \boldsymbol{\beta}, b_i^2\, \sigma_v^2\right)$$

The prior distributions $\pi(\cdot)$ for the unknown parameters $\boldsymbol{\beta}$ and σ_v^2 are given by the following specifications.

$$\pi(\boldsymbol{\beta}) \propto 1$$

$$\pi(\sigma_v^2) \propto (\sigma_v^2)^{-1/2}$$

The joint likelihood function $f(\boldsymbol{\theta}, \boldsymbol{\beta}, \sigma_v^2 \,|\, \hat{\boldsymbol{\theta}}^{DIR}, \boldsymbol{\psi})$ for the unknown parameters in the model is obtained by multiplying the density functions for $\hat{\boldsymbol{\theta}}^{DIR}$, $log(\boldsymbol{\theta})$, $\boldsymbol{\beta}$ and σ_v^2. It is given by the following expression.

$$f(\boldsymbol{\theta}, \boldsymbol{\beta}, \sigma_v^2 \,|\, \hat{\boldsymbol{\theta}}^{DIR}, \boldsymbol{\psi}) = \prod_{i \in A_{VAL}} \left\{\frac{1}{\sqrt{2\pi\psi_i}} exp\left[-\frac{(\hat{\theta}_i^{DIR} - \theta_i)^2}{2\psi_i}\right] \frac{1}{\sqrt{2\pi b_i^2\, \sigma_v^2}} \frac{1}{\theta_i} exp\left[-\frac{(log(\theta_i) - \mathbf{z}_i^T \boldsymbol{\beta})^2}{2\, b_i^2\, \sigma_v^2}\right]\right\} \pi(\boldsymbol{\beta})\, \pi(\sigma_v^2)$$

$$= \prod_{i \in A_{VAL}} \left\{\frac{1}{\sqrt{2\pi\psi_i}} exp\left[-\frac{(\hat{\theta}_i^{DIR} - \theta_i)^2}{2\psi_i}\right] \frac{1}{\sqrt{2\pi b_i^2\, \sigma_v^2}} \frac{1}{\theta_i} exp\left[-\frac{(log(\theta_i) - \mathbf{z}_i^T \boldsymbol{\beta})^2}{2\, b_i^2\, \sigma_v^2}\right]\right\} \left(\sigma_v^2\right)^{-1/2}$$

$$\propto \prod_{i \in A_{VAL}} \left\{exp\left[-\frac{(\hat{\theta}_i^{DIR} - \theta_i)^2}{2\psi_i}\right] \frac{1}{\theta_i} exp\left[-\frac{(log(\theta_i) - \mathbf{z}_i^T \boldsymbol{\beta})^2}{2\, b_i^2\, \sigma_v^2}\right]\right\} \left(\sigma_v^2\right)^{-(\frac{m+1}{2})}$$

The full conditional distributions are obtained by isolating the expressions involving each of the parameters. From this product form, it is clear that the θ_i are independent. For each θ_i we obtain a distribution which is proportional to the product of Normal and Log-Normal density functions.

$$f(\theta_i \mid \hat{\boldsymbol{\theta}}^{DIR}, \boldsymbol{\psi}, \boldsymbol{\beta}, \sigma_v^2) \propto exp\left[-\frac{(\hat{\theta}_i^{DIR} - \theta_i)^2}{2\psi_i}\right] \frac{1}{\theta_i} exp\left[-\frac{(log(\theta_i) - \mathbf{z}_i^T \boldsymbol{\beta})^2}{2\, b_i^2\, \sigma_v^2}\right]$$

This distribution does not have the closed form expression of a known probability density function. To use the Gibbs sampler on this distribution, we apply the Metropolis-Hastings algorithm with the proposal distribution defined as Log-Normal with density function

$$\frac{1}{\theta_i} \frac{1}{\sqrt{2\pi b_i^2 \sigma_v^2}} exp\left[-\frac{(log(\theta_i) - \mathbf{z}_i^T \boldsymbol{\beta})^2}{2\, b_i^2\, \sigma_v^2}\right].$$

The full conditional distribution of $\boldsymbol{\beta}$ is derived from the term in the joint likelihood function that only involves $\boldsymbol{\beta}$.

$$\begin{aligned}
f(\boldsymbol{\beta} \mid \hat{\boldsymbol{\theta}}^{DIR}, \boldsymbol{\psi}, \boldsymbol{\theta}, \sigma_v^2) &\propto exp\left[-\sum_{i \in A_{VAL}} \frac{(log(\theta_i) - \mathbf{z}_i^T \boldsymbol{\beta})^2}{2\, b_i^2\, \sigma_v^2}\right] \\
&\propto exp\left[-\sum_{i \in A_{VAL}} \frac{(log(\theta_i))^2 - 2\,log(\theta_i)\,\mathbf{z}_i^T \boldsymbol{\beta} + \boldsymbol{\beta}^T \mathbf{z}_i\, \mathbf{z}_i^T \boldsymbol{\beta}}{2\, b_i^2\, \sigma_v^2}\right] \\
&\propto exp\left[-\frac{1}{2}\left\{\sum_{i \in A_{VAL}} \frac{-2\,log(\theta_i)\,\mathbf{z}_i^T \boldsymbol{\beta}}{b_i^2\, \sigma_v^2} + \boldsymbol{\beta}^T \sum_{i \in A_{VAL}} \frac{\mathbf{z}_i\, \mathbf{z}_i^T}{b_i^2\, \sigma_v^2} \boldsymbol{\beta}\right\}\right] \\
&\propto exp\left[-\frac{1}{2}\left\{\sum_{i \in A_{VAL}} \left(\boldsymbol{\beta} - \boldsymbol{A}\left(\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i\, log(\theta_i)}{b_i^2\, \sigma_v^2}\right)\right)^T \boldsymbol{A}^{-1} \left(\boldsymbol{\beta} - \boldsymbol{A}\left(\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i\, log(\theta_i)}{b_i^2\, \sigma_v^2}\right)\right)\right\}\right] \\
&\text{where} \quad \boldsymbol{A} = \left(\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i\, \mathbf{z}_i^T}{b_i^2\, \sigma_v^2}\right)^{-1} = \sigma_v^2 \left(\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i\, \mathbf{z}_i^T}{b_i^2}\right)^{-1}
\end{aligned}$$

This is a multivariate Normal distribution with mean $\boldsymbol{A}\left(\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i\, log(\theta_i)}{b_i^2\, \sigma_v^2}\right)$ and variance-covariance matrix $\boldsymbol{A}$. Thus, we have the following.

$$f(\boldsymbol{\beta} \mid \hat{\boldsymbol{\theta}}^{DIR}, \boldsymbol{\psi}, \boldsymbol{\theta}, \sigma_v^2) \sim N\left(\left(\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i}{b_i} \frac{\mathbf{z}_i^T}{b_i}\right)^{-1} \left(\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i}{b_i} \frac{log(\theta_i)}{b_i}\right), \sigma_v^2 \left(\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i}{b_i} \frac{\mathbf{z}_i^T}{b_i}\right)^{-1}\right)$$

Finally, the full conditional distribution of σ_v^2 is determined from the last two terms in the joint likelihood function which only involve σ_v^2.

$$f(\sigma_v^2 \,\big|\, \hat{\boldsymbol{\theta}}^{DIR}, \boldsymbol{\psi}, \boldsymbol{\theta}, \boldsymbol{\beta}) \propto exp\left[-\sum_{i \in A_{VAL}} \frac{(log(\theta_i) - \mathbf{z}_i^T \boldsymbol{\beta})^2}{2\, b_i^2\, \sigma_v^2}\right] \left(\sigma_v^2\right)^{-(\frac{m+1}{2})}$$

$$\propto \frac{1}{\left(\sigma_v^2\right)^{(\frac{m-1}{2})+1}} exp\left[-\frac{1}{2\sigma_v^2} \sum_{i \in A_{VAL}} \left(\frac{log(\theta_i) - \mathbf{z}_i^T \boldsymbol{\beta}}{b_i}\right)^2\right]$$

If we compare this expression with the pdf of the Inverse Gamma function $IG(\alpha, \beta)$, we see that

$f(\sigma_v^2 \,\big|\, \hat{\boldsymbol{\theta}}^{DIR}, \boldsymbol{\psi}, \boldsymbol{\theta}, \boldsymbol{\beta})$ is $IG(\alpha, \beta)$ with $\alpha = \dfrac{m-1}{2}$ and $\beta = \dfrac{1}{2} \sum_{i \in A_{VAL}} \left(\dfrac{log(\theta_i) - \mathbf{z}_i^T \boldsymbol{\beta}}{b_i}\right)^2$.

Unmatched log census undercount model with known sampling variances

The sampling and linking models are written in the Bayesian framework as follows.

$$\left[\, \hat{\theta}_i^{DIR} \,\big|\, \theta_i \,\right] \sim N\left(\, \theta_i, \psi_i \,\right)$$

$$g(\theta_i) = \left[\, log\left(\frac{\theta_i}{\theta_i + C_i}\right) \,\bigg|\, \boldsymbol{\beta}, \sigma_v^2 \,\right] \sim N\left(\, \mathbf{z}_i^T \boldsymbol{\beta}, b_i^2\, \sigma_v^2 \,\right)$$

The prior distributions $\pi(\cdot)$ for the unknown parameters $\boldsymbol{\beta}$ and σ_v^2 are given by the following specifications.

$$\pi(\boldsymbol{\beta}) \propto 1$$
$$\pi(\sigma_v^2) \propto (\sigma_v^2)^{-1/2}$$

The joint likelihood function $f(\boldsymbol{\theta}, \boldsymbol{\beta}, \sigma_v^2)$ for the unknown parameters in the model is then given by the following expression.

$$f(\boldsymbol{\theta}, \boldsymbol{\beta}, \sigma_v^2) = \prod_{i \in A_{VAL}} \left\{ \frac{1}{\sqrt{2\pi\psi_i}} exp\left[-\frac{(\hat{\theta}_i^{DIR} - \theta_i)^2}{2\psi_i}\right] g'(\theta_i) \frac{1}{\sqrt{2\pi\, b_i^2\, \sigma_v^2}} exp\left[-\frac{(g(\theta_i) - \mathbf{z}_i^T \boldsymbol{\beta})^2}{2\, b_i^2\, \sigma_v^2}\right]\right\} \pi(\boldsymbol{\beta})\, \pi(\sigma_v^2)$$

$$= \prod_{i \in A_{VAL}} \left\{ \frac{1}{\sqrt{2\pi\psi_i}} exp\left[-\frac{(\hat{\theta}_i^{DIR} - \theta_i)^2}{2\psi_i}\right] g'(\theta_i) \frac{1}{\sqrt{2\pi\, b_i^2\, \sigma_v^2}} exp\left[-\frac{(g(\theta_i) - \mathbf{z}_i^T \boldsymbol{\beta})^2}{2\, b_i^2\, \sigma_v^2}\right]\right\} \left(\sigma_v^2\right)^{-1/2}$$

$$\propto \prod_{i \in A_{VAL}} \left\{ exp\left[-\frac{(\hat{\theta}_i^{DIR} - \theta_i)^2}{2\psi_i}\right] g'(\theta_i)\, exp\left[-\frac{(g(\theta_i) - \mathbf{z}_i^T \boldsymbol{\beta})^2}{2\, b_i^2\, \sigma_v^2}\right]\right\} \left(\sigma_v^2\right)^{-(\frac{m+1}{2})}$$

where $g'(\theta_i) = \dfrac{\partial}{\partial \theta_i} log(\dfrac{\theta_i}{\theta_i + C_i}) = \dfrac{C_i}{\theta_i\, (\theta_i + C_i)}$

The full conditional distributions are obtained by isolating the expressions involving each of the parameters. From this product form, it is clear that the θ_i are independent. For each θ_i we obtain a distribution which is proportional to the product of Normal density function and a density function proportional to $g'(\theta_i)\, exp\left[-\frac{(g(\theta_i)-\mathbf{z}_i^T\boldsymbol{\beta})^2}{2\,b_i^2\,\sigma_v^2}\right]$.

$$f(\theta_i \big| \boldsymbol{\beta},\sigma_v^2) \propto exp\left[-\frac{(\hat{\theta}_i^{DIR}-\theta_i)^2}{2\psi_i}\right] g'(\theta_i)\, exp\left[-\frac{(g(\theta_i)-\mathbf{z}_i^T\boldsymbol{\beta})^2}{2\,b_i^2\,\sigma_v^2}\right]$$

This distribution does not have the closed form expression of a known probability density function. To use the Gibbs sampler on this distribution, we apply the Metropolis-Hastings algorithm with the proposal distribution defined as the Normal distribution with density function proportional to $exp\left[-\frac{(\hat{\theta}_i^{DIR}-\theta_i)^2}{2\psi_i}\right]$.

The full conditional distribution of $\boldsymbol{\beta}$ is derived from the term in the joint likelihood function that only involves $\boldsymbol{\beta}$.

$$\begin{aligned}
f(\boldsymbol{\beta} \,|\, \boldsymbol{\theta},\sigma_v^2) &\propto exp\left[-\sum\nolimits_{i\in A_{VAL}} \frac{(g(\theta_i)-\mathbf{z}_i^T\boldsymbol{\beta})^2}{2\,b_i^2\,\sigma_v^2}\right] \\
&\propto exp\left[-\sum\nolimits_{i\in A_{VAL}} \frac{(g(\theta_i))^2-2\,g(\theta_i)\,\mathbf{z}_i^T\boldsymbol{\beta}+\boldsymbol{\beta}^T z_i\, z_i^T\boldsymbol{\beta}}{2\,b_i^2\,\sigma_v^2}\right] \\
&\propto exp\left[-\frac{1}{2}\left\{\sum\nolimits_{i\in A_{VAL}} \frac{-2\,g(\theta_i)\,\mathbf{z}_i^T\boldsymbol{\beta}}{b_i^2\,\sigma_v^2}+\boldsymbol{\beta}^T\sum\nolimits_{i\in A_{VAL}} \frac{\mathbf{z}_i\,\mathbf{z}_i^T}{b_i^2\,\sigma_v^2}\boldsymbol{\beta}\right\}\right] \\
&\propto exp\left[-\frac{1}{2}\left\{\sum\nolimits_{i\in A_{VAL}} \left(\boldsymbol{\beta}-\boldsymbol{A}\left(\sum\nolimits_{i\in A_{VAL}} \frac{\mathbf{z}_i\,g(\theta_i)}{b_i^2\,\sigma_v^2}\right)\right)^T \boldsymbol{A}^{-1}\left(\boldsymbol{\beta}-\boldsymbol{A}\left(\sum\nolimits_{i\in A_{VAL}} \frac{\mathbf{z}_i\,g(\theta_i)}{b_i^2\,\sigma_v^2}\right)\right)\right\}\right] \\
&\text{where}\quad \boldsymbol{A}=\left(\sum\nolimits_{i\in A_{VAL}} \frac{\mathbf{z}_i\,\mathbf{z}_i^T}{b_i^2\,\sigma_v^2}\right)^{-1}=\sigma_v^2\left(\sum\nolimits_{i\in A_{VAL}} \frac{\mathbf{z}_i\,\mathbf{z}_i^T}{b_i^2}\right)^{-1}
\end{aligned}$$

This is a multivariate Normal distribution with mean $\boldsymbol{A}\left(\sum\nolimits_{i\in A_{VAL}} \frac{\mathbf{z}_i\,g(\theta_i)}{b_i^2\,\sigma_v^2}\right)$ and variance-covariance matrix $\boldsymbol{A}$. Thus, we have the following.

$$f(\boldsymbol{\beta} \,|\, \boldsymbol{\theta},\sigma_v^2) \sim N\left(\left(\sum\nolimits_{i\in A_{VAL}} \frac{\mathbf{z}_i}{b_i}\frac{\mathbf{z}_i^T}{b_i}\right)^{-1}\left(\sum\nolimits_{i\in A_{VAL}} \frac{\mathbf{z}_i}{b_i}\frac{g(\theta_i)}{b_i}\right),\ \sigma_v^2\left(\sum\nolimits_{i\in A_{VAL}} \frac{\mathbf{z}_i}{b_i}\frac{\mathbf{z}_i^T}{b_i}\right)^{-1}\right)$$

Finally, the full conditional distribution of σ_v^2 is determined from the last two terms in the joint likelihood function which only involve σ_v^2.

$$f(\sigma_v^2 \mid \boldsymbol{\theta}, \boldsymbol{\beta}) \propto exp\left[-\sum_{i \in A_{VAL}} \frac{(g(\theta_i) - \boldsymbol{z}_i^T \boldsymbol{\beta})^2}{2\, b_i^2\, \sigma_v^2} \right] \left(\sigma_v^2\right)^{-(\frac{m+1}{2})}$$

$$\propto \frac{1}{\left(\sigma_v^2\right)^{(\frac{m-1}{2})+1}} exp\left[-\frac{1}{2\sigma_v^2} \sum_{i \in A_{VAL}} \left(\frac{g(\theta_i) - \boldsymbol{z}_i^T \boldsymbol{\beta}}{b_i} \right)^2 \right]$$

If we compare this expression with the pdf of the Inverse Gamma function $IG(\alpha, \beta)$, we see that

$f(\sigma_v^2 \mid \boldsymbol{\theta}, \boldsymbol{\beta})$ is $IG(\alpha, \beta)$ with $\alpha = \dfrac{m-1}{2}$ and $\beta = \dfrac{1}{2} \sum_{i \in A_{VAL}} \left(\dfrac{g(\theta_i) - \boldsymbol{z}_i^T \boldsymbol{\beta}}{b_i} \right)^2$.

Matched FH model with estimated sampling variances

The Bayesian model for this method includes the following distributional assumptions.

$$\left[\, \hat{\theta}_i^{DIR} \mid \theta_i \,\right] \sim N\left(\, \theta_i, \psi_i \,\right)$$
$$\left[\, \theta_i \mid \boldsymbol{\beta}, \sigma_v^2 \,\right] \sim N\left(\, \boldsymbol{z}_i^T \boldsymbol{\beta}, b_i^2\, \sigma_v^2 \,\right)$$
$$\left[\, d_i\, \hat{\psi}_i \mid \psi_i \,\right] \sim \psi_i\, \chi_{d_i}^2 \quad \text{where } d_i = (n_i - 1)$$

The prior distributions $\pi(\cdot)$ for the unknown parameters $\boldsymbol{\psi}$, $\boldsymbol{\beta}$ and σ_v^2 are given by the following specifications.

$$\pi(\boldsymbol{\beta}) \propto 1$$
$$\pi(\sigma_v^2) \propto (\sigma_v^2)^{-1/2}$$
$$\pi(\psi_i) \propto (\psi_i)^{-1/2}$$

The unknown parameters are $\boldsymbol{\theta}$, $\boldsymbol{\psi}$, $\boldsymbol{\beta}$ and σ_v^2. The joint likelihood function $f(\boldsymbol{\theta}, \boldsymbol{\psi}, \boldsymbol{\beta}, \sigma_v^2 \mid \hat{\boldsymbol{\theta}}^{DIR}, \hat{\boldsymbol{\psi}})$ for the unknown parameters is obtained by multiplying the density functions of the above six distributions. Before we do this, we first consider the probability density function (pdf) of the distribution of $\psi_i\, \chi_{d_i}^2$.

The pdf of $\psi_i\, \chi_{d_i}^2$ can be obtained by the usual method of transformation of variables. We first let $y = \psi_i\, x$ where x has a $\chi_{d_i}^2$ distribution with pdf $\dfrac{1}{2^{d_i/2}\, \Gamma(d_i/2)} x^{\frac{d_i}{2}-1} exp(-\dfrac{x}{2})$. The pdf of y is given by $g(y) = f(y) \left| \dfrac{\partial x}{\partial y} \right|$ with $\left| \dfrac{\partial x}{\partial y} \right| = \dfrac{1}{\psi_i}$. It follows that $g(y) = \dfrac{1}{2^{d_i/2}\, \Gamma(d_i/2)} \left(\dfrac{y}{\psi_i} \right)^{\frac{d_i}{2}-1} exp(-\dfrac{y}{2\, \psi_i}) \dfrac{1}{\psi_i}$. Since y is really $d_i\, \hat{\psi}_i$, we then obtain the pdf based on the unknown parameters in the model. Thus, we have the following expression.

$$\frac{1}{2^{d_i/2}\,\Gamma\left(d_i/2\right)}\left(\frac{d_i\,\hat{\psi}_i}{\psi_i}\right)^{\frac{d_i}{2}-1} exp(-\frac{d_i\,\hat{\psi}_i}{2\,\psi_i})\,\frac{1}{\psi_i}\ \propto\ \left(\psi_i\right)^{-\frac{d_i}{2}} exp(-\frac{d_i\,\hat{\psi}_i}{2\,\psi_i})$$

We can now write the joint likelihood function $f(\boldsymbol{\theta}, \boldsymbol{\psi}, \boldsymbol{\beta}, \sigma_v^2 \,|\, \hat{\boldsymbol{\theta}}^{DIR}, \hat{\boldsymbol{\psi}})$ for the unknown parameters as follows.

$$f(\boldsymbol{\theta}, \boldsymbol{\psi}, \boldsymbol{\beta}, \sigma_v^2 \,|\, \hat{\boldsymbol{\theta}}^{DIR}, \hat{\boldsymbol{\psi}})$$

$$\propto \prod_{i\in A_{VAL}} \left\{ \frac{1}{\sqrt{\psi_i}} exp\left[-\frac{(\hat{\theta}_i^{DIR}-\theta_i)^2}{2\psi_i}\right] \frac{1}{\sqrt{\sigma_v^2}} exp\left[-\frac{(\theta_i-\mathbf{z}_i^T\boldsymbol{\beta})^2}{2\,b_i^2\,\sigma_v^2}\right] \left(\psi_i\right)^{-\frac{d_i}{2}} exp(-\frac{d_i\,\hat{\psi}_i}{2\,\psi_i})\,\pi(\psi_i) \right\} \pi(\boldsymbol{\beta})\,\pi(\sigma_v^2)$$

$$\propto \prod_{i\in A_{VAL}} \left\{ (\psi_i)^{-\frac{1}{2}} exp\left[-\frac{(\hat{\theta}_i^{DIR}-\theta_i)^2}{2\psi_i}\right] \left(\sigma_v^2\right)^{-\frac{1}{2}} exp\left[-\frac{(\theta_i-\mathbf{z}_i^T\boldsymbol{\beta})^2}{2\,b_i^2\,\sigma_v^2}\right] \left(\psi_i\right)^{-\frac{d_i}{2}} exp(-\frac{d_i\,\hat{\psi}_i}{2\,\psi_i})\,(\psi_i)^{-\frac{1}{2}} \right\} \left(\sigma_v^2\right)^{-\frac{1}{2}}$$

$$\propto \prod_{i\in A_{VAL}} \left\{ (\psi_i)^{-\frac{(d_i+2)}{2}} \right\} exp\left[-\sum_{i\in A_{VAL}} \frac{(\hat{\theta}_i^{DIR}-\theta_i)^2}{2\psi_i} + \frac{(\theta_i-\mathbf{z}_i^T\boldsymbol{\beta})^2}{2\,b_i^2\,\sigma_v^2} + \frac{d_i\,\hat{\psi}_i}{2\,\psi_i}\right] \left(\sigma_v^2\right)^{-(\frac{m+1}{2})}$$

The full conditional distributions are obtained by isolating the expressions involving each of the parameters. From this form, it is clear that the θ_i are independent. For each θ_i we obtain a density function which is proportional to the following expression.

$$f(\theta_i \,|\, \hat{\boldsymbol{\theta}}^{DIR}, \hat{\boldsymbol{\psi}}, \boldsymbol{\psi}, \boldsymbol{\beta}, \sigma_v^2) \propto exp\left[-\frac{(\hat{\theta}_i^{DIR}-\theta_i)^2}{2\psi_i} - \frac{(\theta_i-\mathbf{z}_i^T\boldsymbol{\beta})^2}{2\,b_i^2\,\sigma_v^2}\right]$$

$$\propto exp\left[-\frac{1}{2}\left\{\left(\frac{1}{\psi_i}+\frac{1}{b_i^2\,\sigma_v^2}\right)\theta_i^2 - 2\left(\frac{\hat{\theta}_i^{DIR}}{\psi_i}+\frac{\mathbf{z}_i^T\boldsymbol{\beta}}{b_i^2\,\sigma_v^2}\right)\theta_i\right\}\right]$$

$$\propto exp\left[-\frac{1}{2}\left\{\left(\theta_i - \frac{b_i^2\,\sigma_v^2\,\hat{\theta}_i^{DIR} + \psi_i\,\mathbf{z}_i^T\boldsymbol{\beta}}{b_i^2\,\sigma_v^2+\psi_i}\right)^2 \left(\frac{1}{\psi_i}+\frac{1}{b_i^2\,\sigma_v^2}\right)\right\}\right]$$

$$\propto exp\left[-\frac{1}{2}\left(\theta_i - \left[\gamma_i\,\hat{\theta}_i^{DIR} + (1-\gamma_i)\,\mathbf{z}_i^T\boldsymbol{\beta}\right]\right)^2 \left(\frac{1}{\gamma_i\,\psi_i}\right)\right]$$

$$\text{where}\quad \gamma_i = \frac{b_i^2\,\sigma_v^2}{b_i^2\,\sigma_v^2+\psi_i}$$

This shows that each θ_i has a Normal distribution with mean $\gamma_i\,\hat{\theta}_i^{DIR} + (1-\gamma_i)\,\mathbf{z}_i^T\boldsymbol{\beta}$ and variance $\gamma_i\,\psi_i$.

The full conditional distribution of $\boldsymbol{\beta}$ is derived from the only term in the joint likelihood function that involves $\boldsymbol{\beta}$.

$$f(\boldsymbol{\beta} \mid \hat{\boldsymbol{\theta}}^{DIR}, \hat{\boldsymbol{\psi}}, \boldsymbol{\theta}, \boldsymbol{\psi}, \sigma_v^2) \propto exp\left[-\sum_{i \in A_{VAL}} \frac{(\theta_i - \mathbf{z}_i^T \boldsymbol{\beta})^2}{2\, b_i^2\, \sigma_v^2}\right]$$

$$\propto exp\left[-\sum_{i \in A_{VAL}} \frac{\theta_i^2 - 2\theta_i\, \mathbf{z}_i^T \boldsymbol{\beta} + \boldsymbol{\beta}^T \mathbf{z}_i\, \mathbf{z}_i^T \boldsymbol{\beta}}{2\, b_i^2\, \sigma_v^2}\right]$$

$$\propto exp\left[-\frac{1}{2}\left\{\sum_{i \in A_{VAL}} \frac{-2\theta_i\, \mathbf{z}_i^T \boldsymbol{\beta}}{b_i^2\, \sigma_v^2} + \boldsymbol{\beta}^T \sum_{i \in A_{VAL}} \frac{\mathbf{z}_i\, \mathbf{z}_i^T}{b_i^2\, \sigma_v^2} \boldsymbol{\beta}\right\}\right]$$

$$\propto exp\left[-\frac{1}{2}\left\{\sum_{i \in A_{VAL}} \left(\boldsymbol{\beta} - \boldsymbol{A}\left(\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i\, \theta_i}{b_i^2\, \sigma_v^2}\right)\right)^T \boldsymbol{A}^{-1}\left(\boldsymbol{\beta} - \boldsymbol{A}\left(\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i\, \theta_i}{b_i^2\, \sigma_v^2}\right)\right)\right\}\right]$$

$$\text{where} \quad \boldsymbol{A} = \left(\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i\, \mathbf{z}_i^T}{b_i^2\, \sigma_v^2}\right)^{-1} = \sigma_v^2 \left(\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i\, \mathbf{z}_i^T}{b_i^2}\right)^{-1}$$

This is a multivariate Normal distribution with mean $\boldsymbol{A}\left(\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i\, \theta_i}{b_i^2\, \sigma_v^2}\right)$ and variance-covariance matrix $\boldsymbol{A}$. Thus, we have the following.

$$f(\boldsymbol{\beta} \mid \hat{\boldsymbol{\theta}}^{DIR}, \hat{\boldsymbol{\psi}}, \boldsymbol{\psi}, \boldsymbol{\beta}, \sigma_v^2) \sim N\left(\left(\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i}{b_i} \frac{\mathbf{z}_i^T}{b_i}\right)^{-1} \left(\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i}{b_i} \frac{\theta_i}{b_i}\right),\ \sigma_v^2 \left(\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i}{b_i} \frac{\mathbf{z}_i^T}{b_i}\right)^{-1}\right)$$

The full conditional distribution of σ_v^2 is determined from the terms in the joint likelihood function that only involve σ_v^2.

$$f(\sigma_v^2 \mid \hat{\boldsymbol{\theta}}^{DIR}, \hat{\boldsymbol{\psi}}, \boldsymbol{\theta}, \boldsymbol{\psi}, \boldsymbol{\beta}) \propto \left(\sigma_v^2\right)^{-(\frac{m+1}{2})} exp\left[-\sum_{i \in A_{VAL}} \frac{(\theta_i - \mathbf{z}_i^T \boldsymbol{\beta})^2}{2\, b_i^2\, \sigma_v^2}\right]$$

$$\propto \frac{1}{\left(\sigma_v^2\right)^{(\frac{m-1}{2})+1}} exp\left[-\frac{1}{2\sigma_v^2} \sum_{i \in A_{VAL}} \left(\frac{\theta_i - \mathbf{z}_i^T \boldsymbol{\beta}}{b_i}\right)^2\right]$$

If we compare this expression with the pdf of the Inverse Gamma function $IG(\alpha, \beta)$, we see that $f(\sigma_v^2 \mid \hat{\boldsymbol{\theta}}^{DIR}, \hat{\boldsymbol{\psi}}, \boldsymbol{\theta}, \boldsymbol{\psi}, \boldsymbol{\beta})$ is $IG(\alpha, \beta)$ with $\alpha = \frac{m-1}{2}$ and $\beta = \frac{1}{2} \sum_{i \in A_{VAL}} \left(\frac{\theta_i - \mathbf{z}_i^T \boldsymbol{\beta}}{b_i}\right)^2$.

Finally, the full conditional distribution of each ψ_i is obtained from the joint likelihood function by isolating the terms involving ψ_i. Thus, we obtain the following.

$$f(\psi_i \mid \hat{\boldsymbol{\theta}}^{DIR}, \hat{\boldsymbol{\psi}}, \boldsymbol{\theta}, \boldsymbol{\beta}, \sigma_v^2) \propto \frac{1}{(\psi_i)^{\frac{d_i}{2}+1}} exp\left[-\frac{(\hat{\theta}_i^{DIR} - \theta_i)^2}{2\psi_i} - \frac{d_i\, \hat{\psi}_i}{2\, \psi_i}\right]$$

Comparing this form with that of the Inverse Gamma distribution, we see that $f(\psi_i \,|\, \hat{\boldsymbol{\theta}}^{DIR}, \hat{\boldsymbol{\psi}}, \boldsymbol{\theta}, \boldsymbol{\beta}, \sigma_v^2\,)$ is $IG(\alpha, \beta)$ with $\alpha = \dfrac{d_i}{2} = \dfrac{n_i - 1}{2}$ and $\beta = \dfrac{1}{2}\left\{\, (\hat{\theta}_i^{DIR} - \theta_i)^2 + d_i\, \hat{\psi}_i \,\right\}$.

Unmatched log-linear model with estimated sampling variances

The Bayesian model for this method includes the following distributional assumptions.

$$\begin{aligned}
\left[\, \hat{\theta}_i^{DIR} \,\middle|\, \theta_i \,\right] &\sim N\left(\, \theta_i, \psi_i \,\right)\\
\left[\, log(\theta_i) \,\middle|\, \boldsymbol{\beta}, \sigma_v^2 \,\right] &\sim N\left(\, \boldsymbol{z}_i^T \boldsymbol{\beta}, b_i^2\, \sigma_v^2 \,\right)\\
\left[\, d_i\, \hat{\psi}_i \,\middle|\, \psi_i \,\right] &\sim \psi_i\, \chi_{d_i}^2 \quad \text{where } d_i = (n_i - 1)
\end{aligned}$$

The Bayesian prior distributions $\pi(\,\cdot\,)$ for the unknown parameters $\boldsymbol{\psi}$, $\boldsymbol{\beta}$ and σ_v^2 are given by the following specifications.

$$\begin{aligned}
&\pi(\boldsymbol{\beta}) \propto 1\\
&\pi(\sigma_v^2) \propto (\sigma_v^2)^{-1/2}\\
&\pi(\,\psi_i\,) \propto (\,\psi_i\,)^{-1/2}
\end{aligned}$$

The unknown parameters in this model are $\boldsymbol{\theta} = \left\{\, \theta_i \right\}$, $\boldsymbol{\psi} = \left\{\, \psi_i \right\}$, $\boldsymbol{\beta}$ and σ_v^2 . The joint likelihood function $f(\boldsymbol{\theta}, \boldsymbol{\psi}, \boldsymbol{\beta}, \sigma_v^2 \,|\, \hat{\boldsymbol{\theta}}^{DIR}, \hat{\boldsymbol{\psi}})$ for the unknown parameters is obtained by multiplying the density functions of the above six distributions. Before we do this, we first consider the probability density function (pdf) of the distribution of $\psi_i\, \chi_{d_i}^2$.

The pdf of $\psi_i\, \chi_{d_i}^2$ can be obtained by the usual method of transformation of variables. We first let $y = \psi_i\, x$ where x has a $\chi_{d_i}^2$ distribution with pdf $f(x) \dfrac{1}{2^{d_i/2}\, \Gamma\left(d_i / 2\right)} x^{\frac{d_i}{2} - 1} exp(-\dfrac{x}{2})$. The pdf of y is given by $g(y) = f(y) \left| \dfrac{\partial x}{\partial y} \right|$ with $\left| \dfrac{\partial x}{\partial y} \right| = \dfrac{1}{\psi_i}$. It follows that $g(y) = \dfrac{1}{2^{d_i/2}\, \Gamma\left(d_i / 2\right)} \left(\dfrac{y}{\psi_i}\right)^{\frac{d_i}{2} - 1} exp(-\dfrac{y}{2\,\psi_i}) \dfrac{1}{\psi_i}$.

Since y is really $d_i\, \hat{\psi}_i$, we then obtain the pdf based on the unknown parameters in the model. Thus, we have the following expression.

$$\frac{1}{2^{d_i/2}\, \Gamma\left(d_i / 2\right)} \left(\frac{d_i\, \hat{\psi}_i}{\psi_i}\right)^{\frac{d_i}{2} - 1} exp(-\frac{d_i\, \hat{\psi}_i}{2\,\psi_i}) \frac{1}{\psi_i} \;\propto\; \left(\psi_i\right)^{-\frac{d_i}{2}} exp(-\frac{d_i\, \hat{\psi}_i}{2\,\psi_i})$$

We can now write the joint likelihood function $f(\boldsymbol{\theta}, \boldsymbol{\psi}, \boldsymbol{\beta}, \sigma_v^2 \,|\, \hat{\boldsymbol{\theta}}^{DIR}, \hat{\boldsymbol{\psi}})$ for the unknown parameters as follows.

$$f(\boldsymbol{\theta}, \boldsymbol{\psi}, \boldsymbol{\beta}, \sigma_v^2 \mid \hat{\boldsymbol{\theta}}^{DIR}, \hat{\boldsymbol{\psi}})$$

$$\propto \prod_{i \in A_{VAL}} \left\{ \frac{1}{\sqrt{\psi_i}} exp\left[-\frac{(\hat{\theta}_i^{DIR} - \theta_i)^2}{2\psi_i} \right] \frac{1}{\sqrt{\sigma_v^2}} \frac{1}{\theta_i} exp\left[-\frac{(\,log(\theta_i) - \boldsymbol{z}_i^T \boldsymbol{\beta}\,)^2}{2\, b_i^2\, \sigma_v^2} \right] (\psi_i)^{-\frac{d_i}{2}} exp(-\frac{d_i\, \hat{\psi}_i}{2\, \psi_i})\, \pi(\,\psi_i) \right\} \pi(\boldsymbol{\beta})\, \pi(\sigma_v^2)$$

$$\propto \prod_{i \in A_{VAL}} \left\{ (\,\psi_i\,)^{-\frac{1}{2}} exp\left[-\frac{(\hat{\theta}_i^{DIR} - \theta_i)^2}{2\psi_i} \right] \left(\sigma_v^2\right)^{-\frac{1}{2}} \frac{1}{\theta_i} exp\left[-\frac{(\,log(\theta_i) - \boldsymbol{z}_i^T \boldsymbol{\beta}\,)^2}{2\, b_i^2\, \sigma_v^2} \right] (\,\psi_i\,)^{-\frac{d_i}{2}} exp(-\frac{d_i\, \hat{\psi}_i}{2\, \psi_i})\,(\,\psi_i\,)^{-\frac{1}{2}} \right\} \left(\sigma_v^2\right)^{-\frac{1}{2}}$$

$$\propto \prod_{i \in A_{VAL}} \left\{ (\,\psi_i\,)^{-\frac{(d_i+2)}{2}} exp\left[-\frac{(\hat{\theta}_i^{DIR} - \theta_i)^2}{2\psi_i} \right] \frac{1}{\theta_i} exp\left[-\frac{(\,log(\theta_i) - \boldsymbol{z}_i^T \boldsymbol{\beta}\,)^2}{2\, b_i^2\, \sigma_v^2} \right] exp(-\frac{d_i\, \hat{\psi}_i}{2\, \psi_i}) \right\} \times \left(\sigma_v^2\right)^{-(\frac{m+1}{2})}$$

The full conditional distributions are obtained by isolating the expressions involving each of the parameters. From this form, it is clear that the θ_i are independent. For each θ_i we obtain a distribution which is proportional to the product of Normal and Log-Normal density functions.

$$f(\theta_i \mid \hat{\boldsymbol{\theta}}^{DIR}, \hat{\boldsymbol{\psi}}, \boldsymbol{\psi}, \boldsymbol{\beta}, \sigma_v^2\,) \propto exp\left[-\frac{(\hat{\theta}_i^{DIR} - \theta_i)^2}{2\psi_i} \right] \frac{1}{\theta_i} exp\left[-\frac{(\,log(\theta_i) - \boldsymbol{z}_i^T \boldsymbol{\beta}\,)^2}{2\, b_i^2\, \sigma_v^2} \right]$$

This distribution does not have the closed form expression of a known probability density function. To use the Gibbs sampler on this distribution, we apply the Metropolis-Hastings algorithm with the proposal distribution defined as Log-Normal with density function proportional to

$$\frac{1}{\theta_i} exp\left[-\frac{(log(\theta_i) - \boldsymbol{z}_i^T \boldsymbol{\beta})^2}{2\, b_i^2\, \sigma_v^2} \right].$$

The full conditional distribution of $\boldsymbol{\beta}$ is derived from the only term in the joint likelihood function that involves $\boldsymbol{\beta}$.

$$f(\boldsymbol{\beta} \mid \hat{\boldsymbol{\theta}}^{DIR}, \hat{\boldsymbol{\psi}}, \boldsymbol{\theta}, \boldsymbol{\psi}, \sigma_v^2\,) \propto exp\left[-\sum_{i \in A_{VAL}} \frac{(log(\theta_i) - \boldsymbol{z}_i^T \boldsymbol{\beta})^2}{2\, b_i^2\, \sigma_v^2} \right]$$

$$\propto exp\left[-\sum_{i \in A_{VAL}} \frac{(log(\theta_i))^2 - 2\,log(\theta_i)\,\boldsymbol{z}_i^T \boldsymbol{\beta} + \boldsymbol{\beta}^T \boldsymbol{z}_i\, \boldsymbol{z}_i^T \boldsymbol{\beta}}{2\, b_i^2\, \sigma_v^2} \right]$$

$$\propto exp\left[-\frac{1}{2} \left\{ \sum_{i \in A_{VAL}} \frac{-2\,log(\theta_i)\,\boldsymbol{z}_i^T \boldsymbol{\beta}}{b_i^2\, \sigma_v^2} + \boldsymbol{\beta}^T \sum_{i \in A_{VAL}} \frac{\boldsymbol{z}_i\, \boldsymbol{z}_i^T}{b_i^2\, \sigma_v^2} \boldsymbol{\beta} \right\} \right]$$

$$\propto exp\left[-\frac{1}{2} \left\{ \sum_{i \in A_{VAL}} \left(\boldsymbol{\beta} - \boldsymbol{A} \left(\sum_{i \in A_{VAL}} \frac{\boldsymbol{z}_i\, log(\theta_i)}{b_i^2\, \sigma_v^2} \right) \right)^T \boldsymbol{A}^{-1} \left(\boldsymbol{\beta} - \boldsymbol{A} \left(\sum_{i \in A_{VAL}} \frac{\boldsymbol{z}_i\, log(\theta_i)}{b_i^2\, \sigma_v^2} \right) \right) \right\} \right]$$

$$\text{where} \quad \boldsymbol{A} = \left(\sum_{i \in A_{VAL}} \frac{\boldsymbol{z}_i\, \boldsymbol{z}_i^T}{b_i^2\, \sigma_v^2} \right)^{-1} = \sigma_v^2 \left(\sum_{i \in A_{VAL}} \frac{\boldsymbol{z}_i\, \boldsymbol{z}_i^T}{b_i^2} \right)^{-1}$$

This is a multivariate Normal distribution with mean $\boldsymbol{A}\left(\sum_{i\in A_{VAL}} \frac{\boldsymbol{z}_i\, log(\theta_i)}{b_i^2\,\sigma_v^2}\right)$ and variance-covariance matrix $\boldsymbol{A}$. Thus, we have the following.

$$f(\boldsymbol{\beta}\,|\,\hat{\boldsymbol{\theta}}^{DIR}, \hat{\boldsymbol{\psi}}, \boldsymbol{\theta}, \boldsymbol{\psi}, \sigma_v^2\,) \sim N\left(\left(\sum_{i\in A_{VAL}} \frac{\boldsymbol{z}_i}{b_i}\frac{\boldsymbol{z}_i^T}{b_i}\right)^{-1}\left(\sum_{i\in A_{VAL}} \frac{\boldsymbol{z}_i}{b_i}\frac{log\,(\theta_i)}{b_i}\right),\ \sigma_v^2\left(\sum_{i\in A_{VAL}} \frac{\boldsymbol{z}_i}{b_i}\frac{\boldsymbol{z}_i^T}{b_i}\right)^{-1}\right)$$

The full conditional distribution of σ_v^2 is determined from the terms in the joint likelihood function that only involve σ_v^2.

$$\begin{aligned} f(\sigma_v^2\,\left|\,\hat{\boldsymbol{\theta}}^{DIR}, \hat{\boldsymbol{\psi}}, \boldsymbol{\theta}, \boldsymbol{\psi}, \boldsymbol{\beta}\,\right.) &\propto \left(\sigma_v^2\right)^{-(\frac{m+1}{2})} exp\left[-\sum_{i\in A_{VAL}} \frac{(log(\theta_i)-\boldsymbol{z}_i^T\boldsymbol{\beta})^2}{2\,b_i^2\,\sigma_v^2}\right] \\ &\propto \frac{1}{\left(\sigma_v^2\right)^{(\frac{m-1}{2})+1}}\, exp\left[-\frac{1}{2\sigma_v^2}\sum_{i\in A_{VAL}}\left(\frac{log(\theta_i)-\boldsymbol{z}_i^T\boldsymbol{\beta}}{b_i}\right)^2\right] \end{aligned}$$

If we compare this expression with the pdf of the Inverse Gamma function $IG(\alpha,\beta)$, we see that

$f(\sigma_v^2\,\left|\,\hat{\boldsymbol{\theta}}^{DIR}, \hat{\boldsymbol{\psi}}, \boldsymbol{\theta}, \boldsymbol{\psi}, \boldsymbol{\beta}\right.)$ is $IG(\alpha,\beta)$ with $\alpha=\frac{m-1}{2}$ and $\beta=\frac{1}{2}\sum_{i\in A_{VAL}}\left(\frac{log(\theta_i)-\boldsymbol{z}_i^T\boldsymbol{\beta}}{b_i}\right)^2$.

The full conditional distribution of each ψ_i is obtained from the joint likelihood function by isolating the terms involving ψ_i. Thus we obtain the following.

$$f(\psi_i\,|\,\hat{\boldsymbol{\theta}}^{DIR}, \hat{\boldsymbol{\psi}}, \boldsymbol{\theta}, \boldsymbol{\beta}, \sigma_v^2\,) \propto \frac{1}{(\psi_i)^{\frac{d_i}{2}+1}}\, exp\left[-\frac{(\hat{\theta}_i^{DIR}-\theta_i)^2}{2\psi_i}-\frac{d_i\,\hat{\psi}_i}{2\,\psi_i}\right]$$

Comparing this form with that of the Inverse Gamma distribution, we see that $f(\psi_i\,|\,\hat{\boldsymbol{\theta}}^{DIR}, \hat{\boldsymbol{\psi}}, \boldsymbol{\theta}, \boldsymbol{\beta}, \sigma_v^2\,)$ is $IG(\alpha,\beta)$ with $\alpha=\frac{d_i}{2}=\frac{n_i-1}{2}$ and $\beta=\frac{1}{2}\left\{\,(\hat{\theta}_i^{DIR}-\theta_i)^2+d_i\,\hat{\psi}_i\,\right\}$.

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